Ambetronics Engineers Private Ltd

User Manual
Smart Gas Detector/ Transmitter

Model No: GT-2500-FLP
(ENC: JB-FLP-IIC/ JB-90/ WP-P/ WP)

For Oxygen/ Toxic/ Combustible/ PID/ NDIR (CO_2/ CH_4/ C_3H_8):

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1 SAFETY INFORMATION

Before installing / operating / Maintaining the instrument ensure that this ‘Operating & Installation Manual’ is read. Give particular attention to warning & cautions. All warnings are listed here and repeated in appropriate places of relevant subjects in the ‘Operating & Installation Manual’. Cautions will appear in section / subsection of the ‘Operating & Installation Manual’ where require.

CONDITION OF SAFE USE

- Smart Gas Detector/ Transmitter (GT-2500) is for use in an Ambient Temperature range of -15°C ≤ Ta ≤ +55°C.
- Users must follow the warnings and cautions as mentioned in the next section before use.

2 WARNINGS / CAUTIONS

- Access to the interior of the instrument when carrying out any work, must be conducted by qualified & trained personnel only.
- To reduce the risk of ignition of hazardous atmosphere, de-classify the area or disconnect the instrument supply before opening the instrument enclosure. Keep enclosure tightly closed during operating.
- Do not open Instrument enclosure or replace / refit the sensor in potentially hazardous atmospheres while power is still connected with the instrument.
- The body of Instrument enclosure must be earthen for electrical safety and to limit the effects of radio frequency interference. Earth points are provided outside of instrument enclosure.
- All screen / instruments earth / wiring is earthen at a single point to prevent false readings or always that may occur due to potential earth loops.
- At the end of their working life, replacement of sensor must be disposed in an environmentally safe manner. Disposal should be according to local waste management requirements & environment legislation. Alternatively, old replaceable sensors may be securely packaged & returned to ‘Ambetronics’.

3 INTRODUCTION

3.1 OVERVIEW

The GT-2500 is a microprocessor based Smart Gas Detector/ Transmitter with an easy to read LED display. The Detector/ Transmitter provides the industry standard Analog 4-20 mA output as well as optional RS-485 digital output. The Detector/ Transmitter utilizes smart sensor technology, using pluggable Sensor Modules, where all of the sensor information & its calibration data, Alarm setting, offset setting, output current setting is stored in Sensor Module, which is easy maintenance, servicing.

The reader of this manual should ensure that it is appropriate in all details for the exact instrument to be installed and or operated. If any doubt, contact ‘Ambetronics’ for advice. If any information is not covered in this manual, or if any comments / corrections are required in this manual, please contact Ambetronics using contact details given on the last page.

FOR FLAMEPROOF INSTRUMENTS

Explosion proof certified sealing device, such as a conduit seal with setting compound, Suitable for the conditions of use and correctly installed, shall be provided immediately to the entrance of the housing. Unused openings shall be closed with suitable explosion proof certified closing elements. The unit must be protected from extreme vibration and direct sunlight in hot environments as this may cause the temperature of the detector to rise above its specified limits and cause premature failure.

FOR WEATHERPROOF INSTRUMENTS:

In order to minimize fire or electric shock hazard, the unit must be protected against atmospheric precipitation and excessive humidity. Do not use the unit in areas threatened with excessive shocks, vibrations, dust, humidity, corrosive gases and oils. Do not use the unit in explosion hazard areas. Make sure that the ambient temperature does not exceed the recommended values. In such cases forced cooling of the unit must be considered.

3.2 FEATURES

- Provides a fast reliable output for detection a smallest leak of Oxygen, Toxic Gases, Combustible Gases and Volatile Organic Compound (VOC).
- Provision for smart-pluggable gas sensor module for Oxygen, Toxic, Combustible, PID sensor & NDIR sensor.
- Calibration data, Alarm data, Offset data, Output current data is saved in the particular sensor module, for easy maintenance & servicing.
- Highly resistant to poisoning and etching.
- Capable of detecting down to PPM, %V/V, %LEL.
- Digital display of Gas Concentration on 7 segment LED Display.
- Indication for ‘Sensor Open’, ‘Over Range’, ‘Sensor Replace’, ‘Cal Due’, ‘Cal Fail’, mA Loop Open.
- Standard 4-20mA signal output with configurable range.
- Standard 0-10V/0-5V/ 0-1V signal output with configurable range.
- Optional STEL and TWA set points can be configured for Toxic & VOC Gases.
- Optional RS-485 Communication Port with MODBUS RTU PROTOCOL.
- Non-Intrusive programming for Flameproof Model & Weatherproof Model (ENC: JB-90) using Magnetic Wand.
- Non-Intrusive programming for Weatherproof Model (ENC: WP-P) using Front Keys.
- Password protected programming with Password Changing Facility.
- Alarm Acknowledgement Facility from front as well as rear terminal (Only for FLP-IIC model).
- ‘Test Mode’ provided to check the electronics as Alarm LEDs, Relays, 4-20mA Output current with or without sensor module.
- Optional Alarm Relay contacts on board with two configurable Alarm Levels & One Fail Safe Relay.
- In WP model (ENC: JB-90) there are RGB LED’s used to indicate the status of the instrument. If the side LED’s color is green then it’s all ok condition, Blue color indicate the sensor open condition & user settable color for both alarms (AL1 & AL2).

3.3 APPLICATIONS

- Refineries
- Fertilizers Plants
- Gas Metering skid
- Pulp & Paper Plants
- Gas Metering Station
- Control Atmosphere
- Bullet Yard / Storage Yard
- Chemical & Petrochemical Plants
- Offshore Drilling & Processing Platforms
- Cold Storage
- Stack Monitoring
- Gas Cylinder Bank
- Oil & Gas Industries
- Heat Treatment Plants
- Burner / Furnace Areas
- Sewage Plants
- Chemical Processing Plant
- Acid Alkalizes & Dyes Mfg. Plants
- Chlorination Plant
- Ambient Monitoring
- Gas Pipeline Project
- Chemical Storage Area
- Power & Industrial Plants
- Coal Mine and Confined Area
- Automotive Industries / Paint Shops

3.4 TECHNICAL SPECIFICATIONS

Table 1

GENERAL		
Sensor Technology	:	Electrochemical / Catalytic / Pellistor/ NDIR/ PID
Detection Method	:	Diffusion
Gases Detected	:	(Kindly select Gas as specified in the table)
Range & Resolution	:	(Kindly select Range & Resolution as specified in the table)
Display	:	2-line, 4-digit 8mm (H) seven segment LED display & 8 LEDs to indicate status of instrument.
Control Action	:	I. Two independent alarm set points with Latch & Non-Latch Facility. II. User selectable Hysteresis and Logic option. III. Configurable STEL and TWA set points for Toxic & VOC gases.
Setting	:	1) Programming using Magnetic wand without opening enclosure cover for Flameproof & Weatherproof model (ENC: JB-90). 2) Programming using front keys in Weatherproof model (ENC: WP-P).

PERFORMANCE			
SR.NO	SENSOR TECHNOLOGY		CALIBRATION ACCURACY
1	Electrochemical		±2 % F.S
2	Catalytic / Pellistor		±2 % F.S
3	NDIR - CH_4 / CO_2 /C_3H_8		±10 % of Applied Gas/ F.S.
4	PID	0 to 4000 PPM	±10 % of Applied Gas/ F.S.
		0 to 1000 PPM	±5 % of Applied Gas/ F.S.
		0 to 40 PPM	±3 % of Applied Gas/ F.S.
Response Time	:	Less than 15 Seconds.	

ELECTRICAL		
Supply Voltage	:	18 to 36 VDC , typically 24VDC
Power Consumption	:	Less than 3.6 Watts.
Connector cross section	:	2.5 mm^2 for Flexible or Armoured Shielded Cable.
OUTPUT SIGNAL		
Standard Current Output	:	4-20mA / 0-10V/0-5V/0-1V Output with configurable range selection.
Current Output Accuracy	:	Current Output Accuracy : ±0.125% F.S
Voltage Output Accuracy	:	Voltage Output Accuracy : ±0.25% F.S
Load Driving Capacity	:	1) 560 ohm at 18VDC to 36VDC 2) 820 ohm load driving capacity at 22VDC to 36VDC.
Optional Relay	:	Three SPDT Relay (one for Failsafe and Two for Alarm indication) of rating NO: 10A 250VAC / 5A 30VDC & NC: 3A 125VAC / 3A 30VDC.
Optional Communication	:	Isolated RS-485 Communication Port with MODBUS RTU protocol.

ERROR MONITORING

- During Sensor Break/ Open the Display Shows 'SNSR OPEN' & Output current will be as per Upscale/ Downscale
- During Over Range the Display Shows 'OVER RNGE' & Output current will be as per Upscale/ Downscale
- During Output Current Open condition the display indicates "LOOP OPEN" & the message disappears after Acknowledgement
- mA Loop Open setting can be Enabled or Disabled
- Upscale current: 21mA/ 22mA Selectable
- Downscale current 1mA / 3.7mA Selectable
- Upscale voltage for 0-10V: 10V
- Upscale voltage for 0-5V: 5V
- Upscale voltage for 0-1V: 1V
- Downscale voltage for 0-10V/0-5V/0-1V: 0V
- Inhibit mode current is adjustable & user selective
  Oxygen for 25% V/V range : 3.8mA / 17.4mA
  Oxygen for 100% V/V range : 3.8mA / 7.34mA
  Nitrogen for 100% V/V range : 3.8mA/ 16.656mA
  Toxic / combustible / PID /NDIR : 2mA / 3.8mA / 4mA
- Inhibit mode for 0-10V voltage output is adjustable & user selective
  Oxygen for 25% V/V range : 0V / 8.36V
  Oxygen/N2 for 100% V/V range : 0V/ 2.09V
  Nitrogen for 100% V/V range : 0V/7.91V

Toxic / Combustible / PID /NDIR : 0V

- Inhibit mode for 0-5V voltage output is adjustable & user selective
  Oxygen for 25% V/V range : 0V / 4.18V
  Oxygen/N2 for 100% V/V range : 0V/ 1.045V
  Nitrogen for 100% V/V range : 0V/3.955V
  Toxic / combustible / PID /NDIR : 0V
- Inhibit mode for 0-1V voltage output is adjustable & user selective
  Oxygen for 25% V/V range : 0V / 836mV
  Oxygen/N2 for 100% V/V range : 0V/ 209mV
  Nitrogen for 100% V/V range : 0V/791mV
  Toxic / combustible / PID /NDIR : 0mV

ENVIRONMENTAL

Operating Temp	:	-15 to +50 °C
Storage Temp	:	-10 to +60 °C
Humidity	:	Less than 95% Non – Condensing.

ACCESSORIES (OPTIONAL)

• CE Certified 24 VDC Power Supply.	• Gas Sampling & Conditioning System.
• Canopy & Stand Mounting.	• Hooter cum Flasher.
• Gas Calibration Kit (0.5, 1, 3, 10) Litre.	• RS-485 to USB OR RS-232 Convertor
• PC Based SCADA Software, Modem.	• Ethernet converter

COMMON DELIVERABLE

• Test calibration certificate	• Reference calibration gas certificate
• User manual	• Standard mounting hardware

FLAMEPROOF HOUSING GT-2500-FLP (ENC: JB-FLP-IIC)

Protection Class	:	IP-66
Approval	:	CMRI approved for IIA & IIB or IIC gas group
Cabinet Material	:	Cast aluminium alloy,LM-6
Cable Entry	:	Double compression cable gland (EX-proof, M20 x 1.5mm (P))
Dimension	:	225mm(H) with sensor holder x 225mm(W) with cable gland & stopping plug x 122mm(D)
	:	225mm(H) with sensor holder x 266mm(W) with Hooter Cum Flasher & stopping plug x 122mm(D)
Mounting	:	Wall Mounting / Stand Mounting / Pipe Mounting
Weight	:	Approx. 2.2 kg
	:	Approx. 2.7 kg with Hooter Cum Flasher

FLAMEPROOF HOUSING GT-2500-FLP (ENC: JB90)

Protection Class	:	IP-66
Approval	:	CMRI approved for IIA & IIB or IIC gas group
Cabinet Material	:	Cast aluminium alloy,LM-6
Cable Entry	:	Double compression cable gland (EX-proof, M20 x 1.5mm (P))

Dimension	:	195mm(H) with sensor holder x 197mm(W) with cable gland & stopping plug x 97mm(D)
	:	195mm(H) with sensor holder x 238mm(W) with Hooter Cum Flasher & stopping plug x 97mm(D)
Mounting	:	Wall Mounting / Stand Mounting / Pipe Mounting
Weight	:	Approx. 1.6 kg
	:	Approx. 2.1kg with hooter cum flasher
WEATHERPROOF HOUSING GT-2500-WP (ENC: WP-P)		
Protection Class	:	IP-65
Cabinet Material	:	Polycarbonate, Grey Colour
Mounting	:	Wall Mounting / Stand Mounting / Pipe Mounting
WITH SINGLE CABLE GLAND & HOOTER CUM FLASHER		
Dimension	:	192mm(H) with Sensor Holder × 145mm(W) with × 55mm(D)
Weight	:	Approx. 565grams
WITH TWO SIDE CABLE GLAND		
Dimension	:	192mm(H) with Sensor Holder × 125mm(W) with × 55mm(D)
Weight	:	Approx. 504grams
WITH SINGLE CABLE GLAND		
Dimension	:	192mm(H) with Sensor Holder × 102mm(W) × 55mm(D)
Weight	:	Approx. 500grams
WEATHERPROOF HOUSING GT-2500-WP (ENC: JB-90)		
Protection Class	:	IP-65
Cabinet Material	:	Polycarbonate, Grey Colour (RAL7032)
Mounting	:	Wall Mounting / Stand Mounting / Pipe Mounting
WITH SINGLE CABLE GLAND & HOOTER CUM FLASHER		
Dimension	:	180mm(H) with Sensor Holder × 180mm(W) with × 70mm(D)
Weight	:	Approx. 594.5grams
WITH SINGLE CABLE GLAND & WITHOUT HOOTER CUM FLASHER		
Dimension	:	180mm(H) with Sensor Holder × 130mm(W) × 70mm(D)
Weight	:	Approx. 531grams

3.5 GAS WITH RANGE & RESOLUTION

Table 2

ELECROCHEMICAL SENSOR TECHNOLOGY					
SR. NO	**GASES**	**RANGE**	**UNIT**	**RES.**	**ORDER CODE**
O1	Oxygen (O_2)	25	% Vol.	0.01	SSM-FLP-O2-EC-08-03
O2	Oxygen (O_2)	0 to 100	% Vol.	0.01	SSM-FLP-O2-EC-14-03
NT1	Nitrogen (N_2)	0 to 100	% Vol.	0.01	SSM-FLP-N2-EC-14-03

TOXIC GASES					
T1	Ammonia (NH_3)	100	PPM	1	SSM-FLP-NH3-EC-14-01
T2	Ammonia (NH_3)	1000	PPM	1	SSM-FLP-NH3-EC-24-01
T3	Bromine (Br_2)	10	PPM	0.1	SSM-FLP-BR2-EC-05-01
T4	Carbon Monoxide (CO)	1000	PPM	1	SSM-FLP-CO-EC-24-01
T5	Carbon Monoxide (CO)	2000	PPM	1	SSM-FLP-CO-EC-28-01
T6	Chlorine (CL_2)	10	PPM	0.1	SSM-FLP-CL2-EC-05-01
T7	Ethylene Oxide (ETO)	100	PPM	1	SSM-FLP-ETO-EC-14-01
T8	Hydrogen (H_2)	2000	PPM	1	SSM-FLP-H2-EC-26-01
T9	Hydrogen Bromide (HBr)	100	PPM	1	SSM-FLP-HBR-EC-14-01
T10	Hydrogen Chloride (HCL)	100	PPM	1	SSM-FLP-HCL-EC-14-01
T11	Hydrogen Cyanide(HCN)	100	PPM	1	SSM-FLP-HCN-EC-14-01
T12	Hydrogen Fluoride (HF)	10	PPM	0.1	SSM-FLP-HF-EC-05-01
T13	Hydrogen Fluoride (HF)	100	PPM	1	SSM-FLP-HF-EC-14-01
T14	Hydrogen Sulfide (H_2S)	100	PPM	1	SSM-FLP-H2S-EC-14-01
T15	Ozone (O_3)	20	PPM	0.1	SSM-FLP-O3-EC-06-01
T16	Phosphine (PH_3)	10	PPM	0.1	SSM-FLP-PH3-EC-05-01
T17	Nitrogen Dioxide (NO_2)	20	PPM	0.1	SSM-FLP-NO2-EC-06-01
T18	Nitric Oxide (NO)	250	PPM	1	SSM-FLP-NO-EC-17-01
T19	Sulphur Dioxide (SO_2)	50	PPM	0.1	SSM-FLP-SO2-EC-11-01
T20	Sulphur Dioxide (SO_2)	2000	PPM	1	SSM-FLP-SO2-EC-26-01
CATALYTIC/ PELLISTOR SENSOR TECHNOLOGY					
COMBUSTIBLE GASES					
SR.NO	**GASES**	**RANGE**	**UNIT**	**RES.**	**ORDER CODE**
C1	Acetone $(CH_3)_2CO$	100	%LEL	1	SSM-FLP-ACETONE-CT-14-02
C2	Acetylene (C_2H_2)	100	%LEL	1	SSM-FLP-ACETYLENE-CT-14-02
C3	Ammonia (NH_3)	100	%LEL	1	SSM-FLP-NH3-CT-14-02
C4	Butane/n-Butane (C_4H_{10})	100	%LEL	1	SSM-FLP-BUTANE-CT-14-02
C5	Carbon Monoxide(CO)	100	%LEL	1	SSM-FLP-CO-CT-14-02
C6	Ethanol (C_2H_5OH)	100	%LEL	1	SSM-FLP-ETHANOL-CT-14-02
C7	Ethyl Acetate ($C_4H_8O_2$)	100	%LEL	1	SSM-FLP-ETHYL ACETATECT-14-02
C8	Ethylene (C_2H_4)	100	%LEL	1	SSM-FLP-ETHYLENE-CT-14-02
C9	Hexane/n-Hexane (C_6H_{14})	100	%LEL	1	SSM-FLP-HEXANE-CT-14-02
C10	Hydrogen (H_2)	100	%LEL	1	SSM-FLP-H2-CT-14-02
C11	Isopropanol ($CH_3C_2H_4OH$)	100	%LEL	1	SSM-FLP-IPA-CT-14-02

C12	Methane (CH_4)/HC	100	%LEL	1	SSM-FLP-METHANE-CT-14-02
C13	Methyl Ethyl Ketone (C_4H_8O)	100	%LEL	1	SSM-FLP-METHYL ETHYL KETONE-CT-14-02
C14	Methanol (CH_3OH)	100	%LEL	1	SSM-FLP-METHANOL-CT-14-02
C15	N-Heptane (C_7H_{16})	100	%LEL	1	SSM-FLP-N-HEPTANE-CT-14-02
C16	N-Pentane (C_5H_{12})	100	%LEL	1	SSM-FLP-N-PENTANE-CT-14-02
C17	Pentane/n-Pentane (C_5H_{12})	100	%LEL	1	SSM-FLP-PENTANE-CT-14-02
C18	Propane/n-Propane (C_3H_8)	100	%LEL	1	SSM-FLP-PROPANE-CT-14-02
C19	Toluene ($C_6H_5CH_3$)	100	%LEL	1	SSM-FLP-TOLUENE-CT-14-02
C20	Unleaded Petrol	100	%LEL	1	SSM-FLP-UNLEADED PETROL-CT-14-02
C21	CNG/LNG/LPG/Natural Gas/Flammable Gas	100	%LEL	1	SSM-FLP-CNG/LNG-CT-14-02

NDIR SENSOR TECHNOLOGY

SR.NO	GASES	RANGE	UNIT	RES.	ORDER CODE
N1	Carbon Dioxide (CO_2)	5000	PPM	1	SSM-FLP-CO2-IR-28-01
N2	Carbon Dioxide (CO_2)	5	%V/V	0.1	SSM-FLP-CO2-IR-04-03
N3	Carbon Dioxide (CO_2)	100	%V/V	0.1	SSM-FLP-CO2-IR-14-03
N4	Methane(CH_4)	100	%LEL	0.1	SSM-FLP-METHANE -IR-14-03
N5	Methane(CH_4)	5	%V/V	0.1	SSM-FLP-METHANE -IR-14-02
N6	Methane(CH_4)	100	%V/V	1	SSM-FLP-METHANE-IR-04-03
N7	Propane / LPG (C_3H_8)	100	%LEL	1	SSM-FLP-PROPANE-IR-14-02
N8	Propane / LPG (C_3H_8)	5	%V/V	0.1	SSM-FLP-PROPANE-IR-04-03
N9	Propane / LPG (C_3H_8)	100	%V/V	1	SSM-FLP-PROPANE-IR-14-03

PID SENSOR TECHNOLOGY

SR.NO	GASES	RANGE	UNIT	RES	ORDER CODE
P1	Isobutylene(C_4H_8) / other VOC	40	PPM	0.1	SSM-FLP-ISOBUTYLENE-PD-10-01
P2	Isobutylene(C_4H_8)/ other VOC	1000	PPM	1	SSM-FLP-ISOBUTYLENE-PD-24-01
P3	Isobutylene(SPAN C_4H_8)/ other VOC	4000	PPM	1	SSM-FLP-ISOBUTYLENE-PD-27-01

NOTE:

- In above Table, Range of gases start from zero.
- **Gases, which are not listed, are available on request & for other details contact factory.**
- All VOCs are available in PID detection principle in PPM ranges.
- PID detector will be provided by calibration with Isobutylene gas.
- In PID detector, VOC other than Isobutylene is calibrated with Isobutylene gas by setting VOC correction factor, In Calibration Report; VOC factor with respect to Isobutylene gas will be mentioned.
  Detection value of VOC = Isobutylene gas concentration value x factor

4 HARDWARE DETAILS

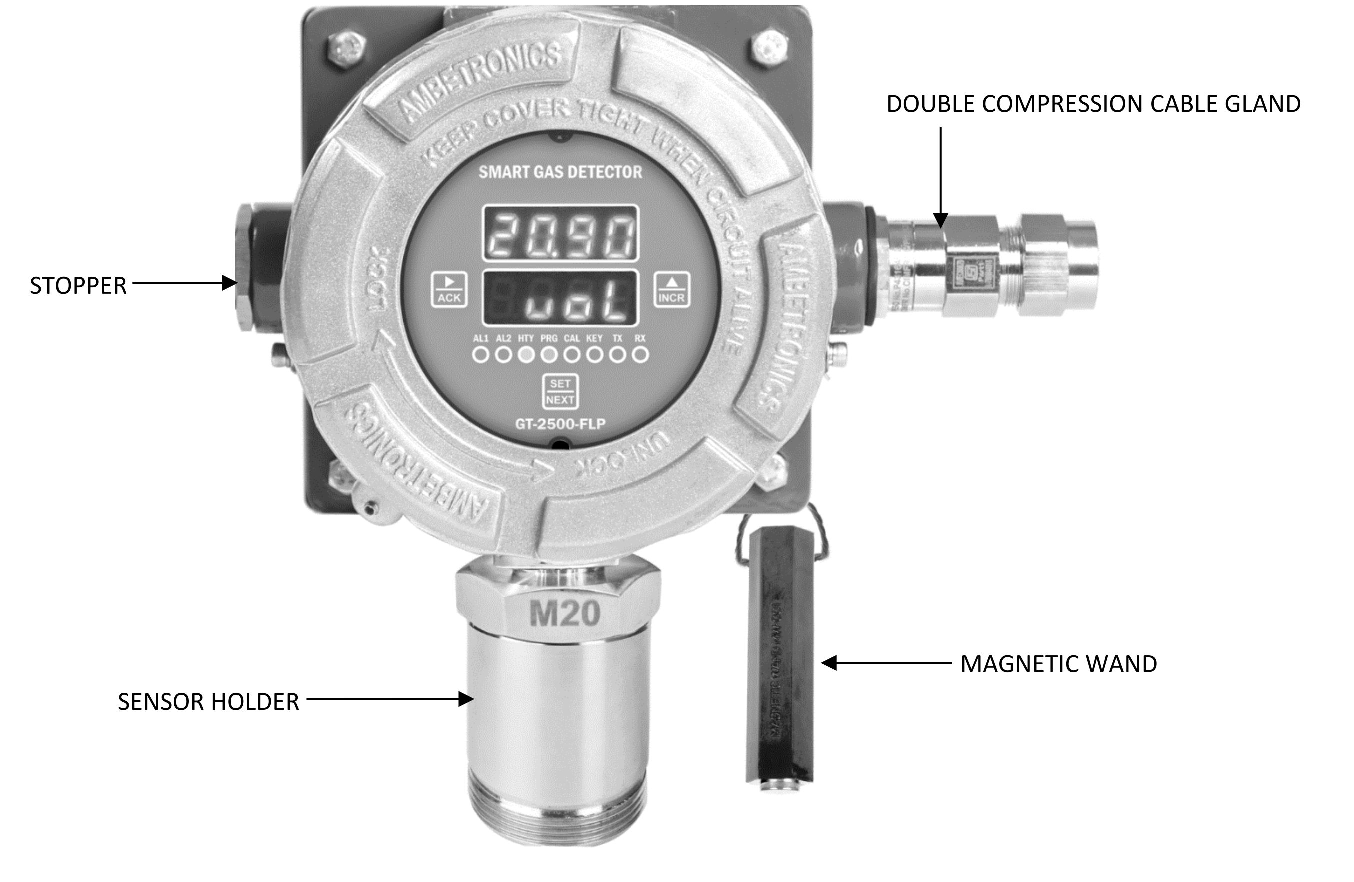


Figure 1 **GT-2500-FLP (ENC: JB-FLP-IIC)**

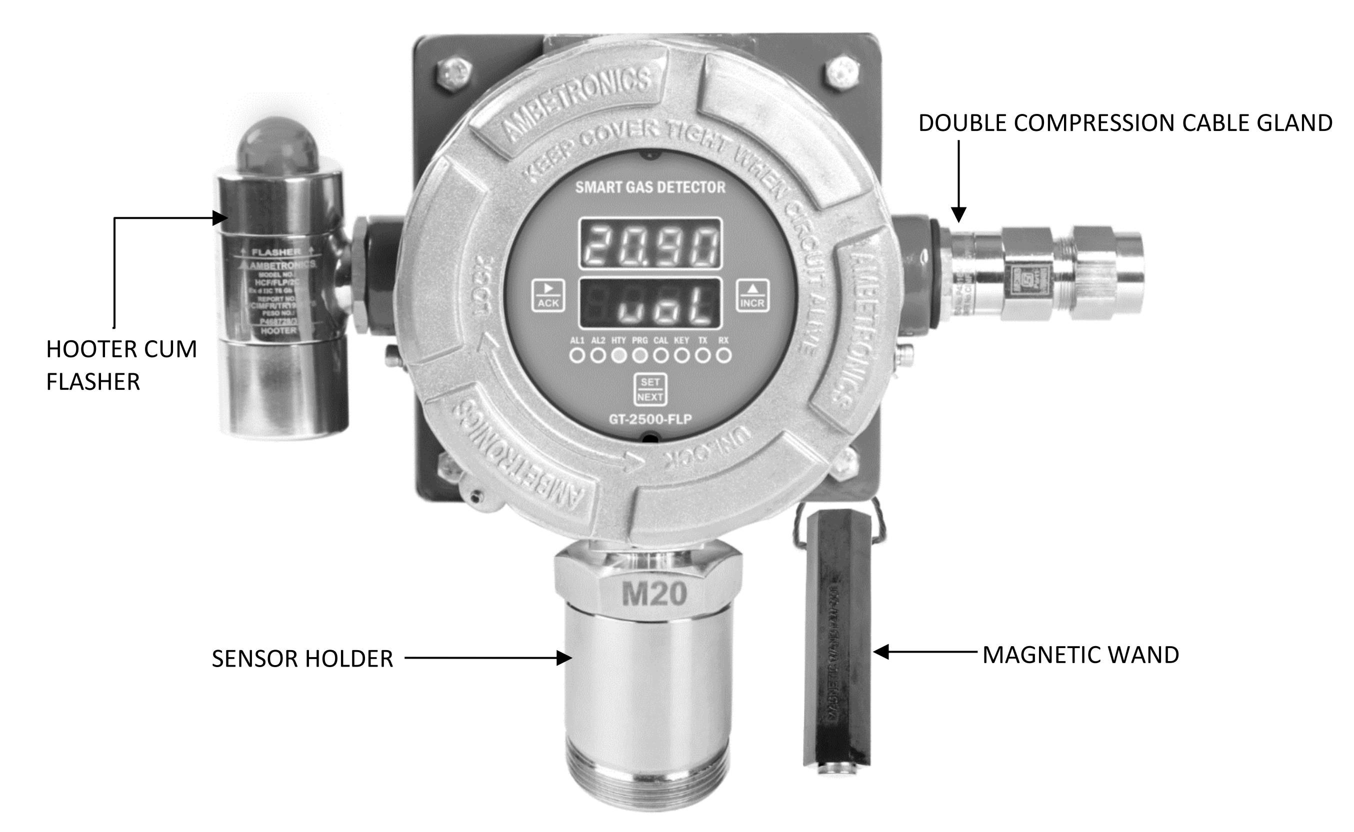


Figure 2 **GT-2500-FLP WITH HOOTER CUM FLASHER (ENC: JB-FLP-IIC)**

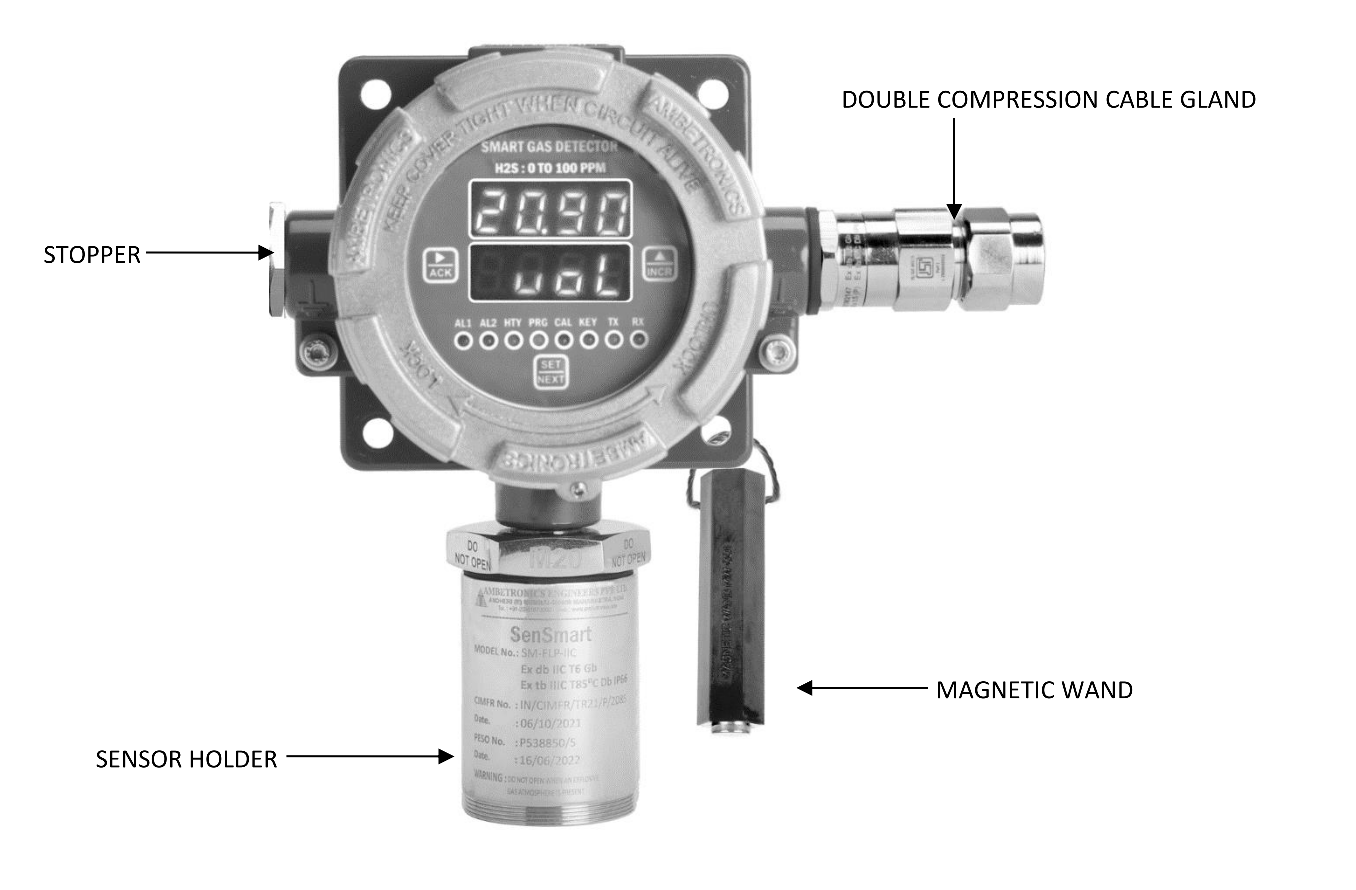


Figure 3 **GT-2500-FLP (ENC: JB90)**

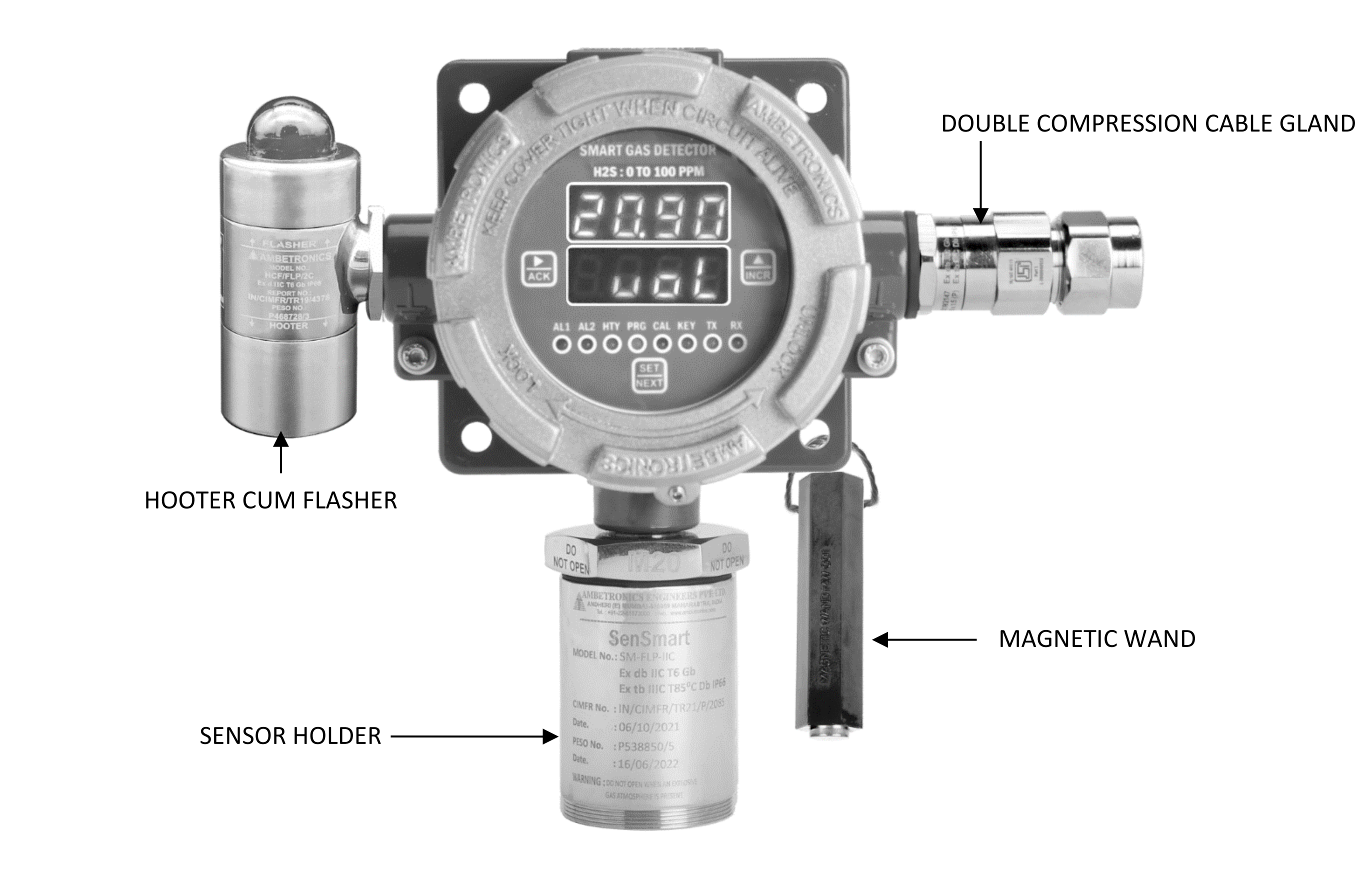


Figure 4 **GT-2500-FLP WITH HOOTER CUM FLASHER (ENC: JB90)**

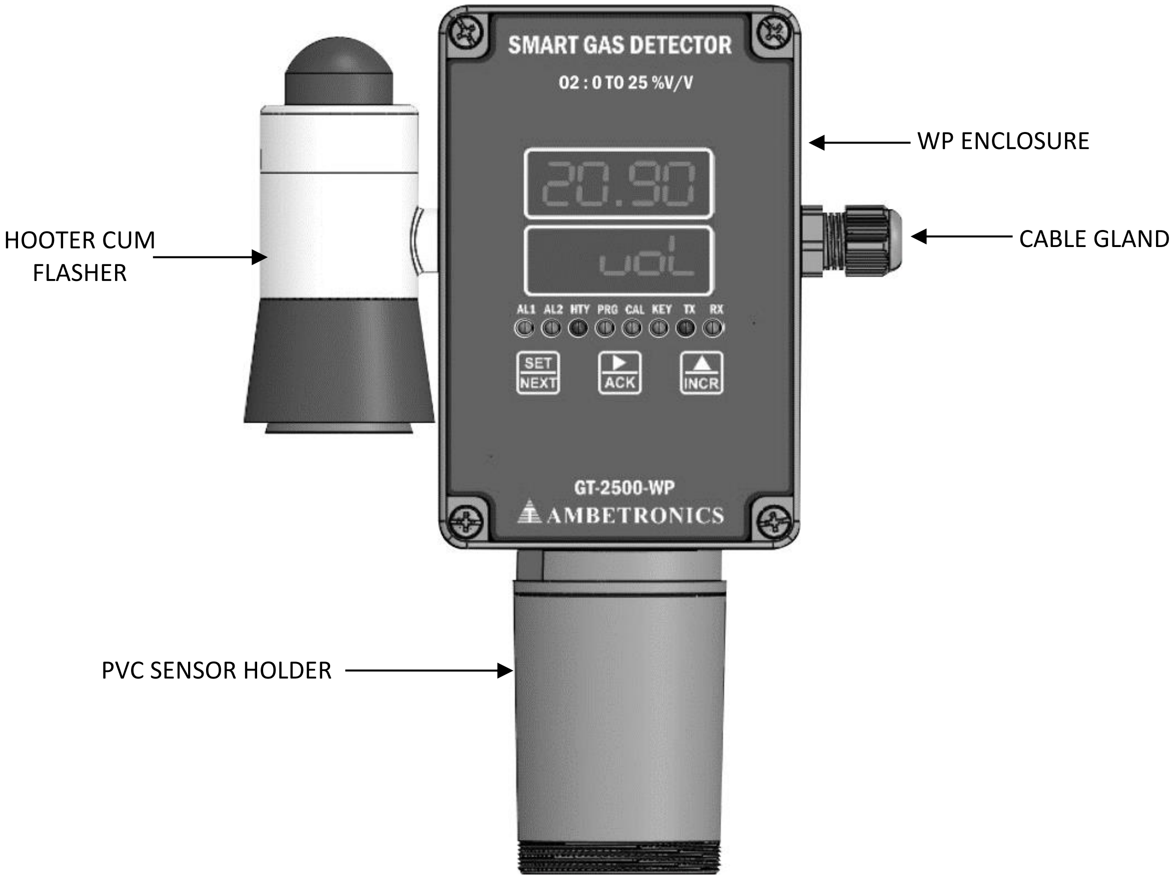


Figure 5 **GT-2500-WP WITH HOOTER CUM FLASHER (ENC: WP-P)**

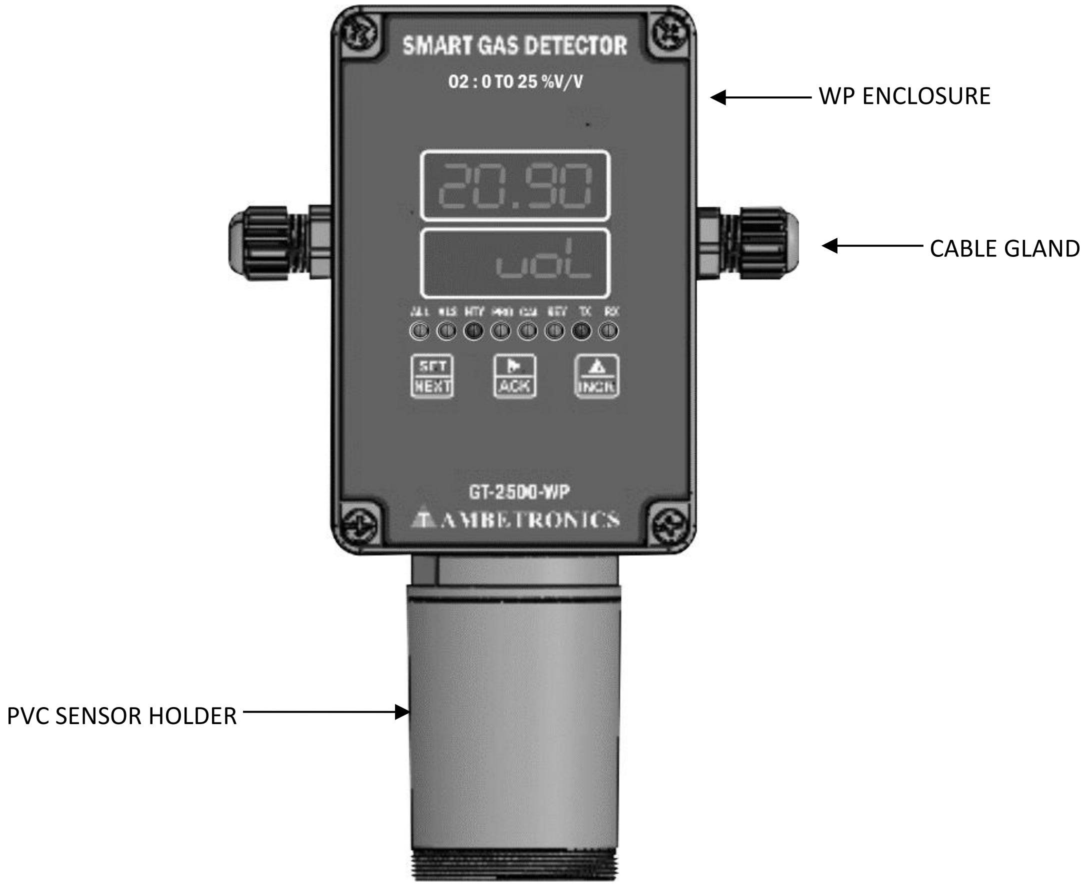


Figure 6 **GT-2500-WP WITH TWO SIDE CABLE GLAND (ENC: WP-P)**

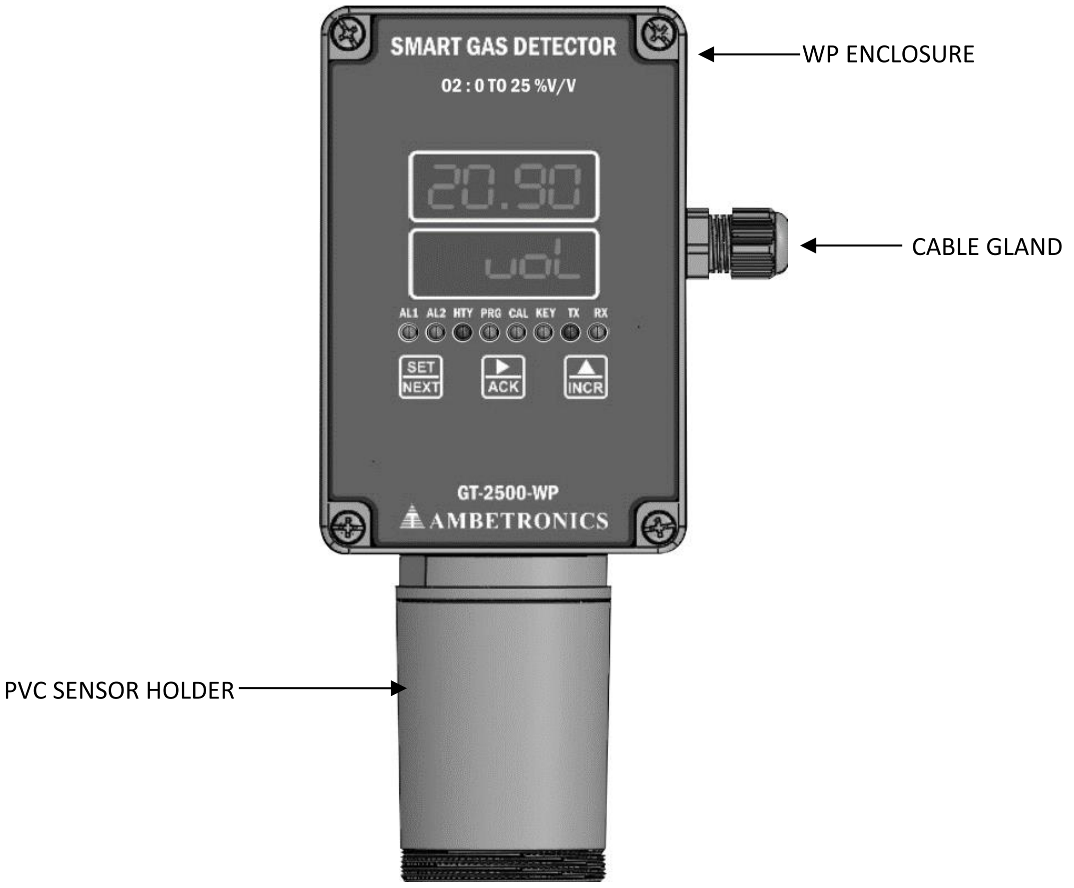


Figure 7 **GT-2500-WP WITH SINGLE CABLE GLAND (ENC: WP-P)**

Figure 8 **GT-2500-WP WITH HOOTER CUM FLASHER**

Figure 9 **GT-2500-WP WITHOUT HOOTER CUM FLASHER**

5 INSTALLATION

5.1 INSTALLATION CONSIDERATION

FLP model can be mounted in two ways Wall mounting / Stand mounting.
Regardless of the installation type (wall mounting / stand mounting), the instrument should be installed at or near the location of a possible leak or the source of emission. Installation height depends on the density of the gas being monitored along with wind speed and direction of airflow and relative position to potential leaking points should be considered.

5.2 FLAMEPROOF ENCLOSURE ASSEMBLING DIAGRAM

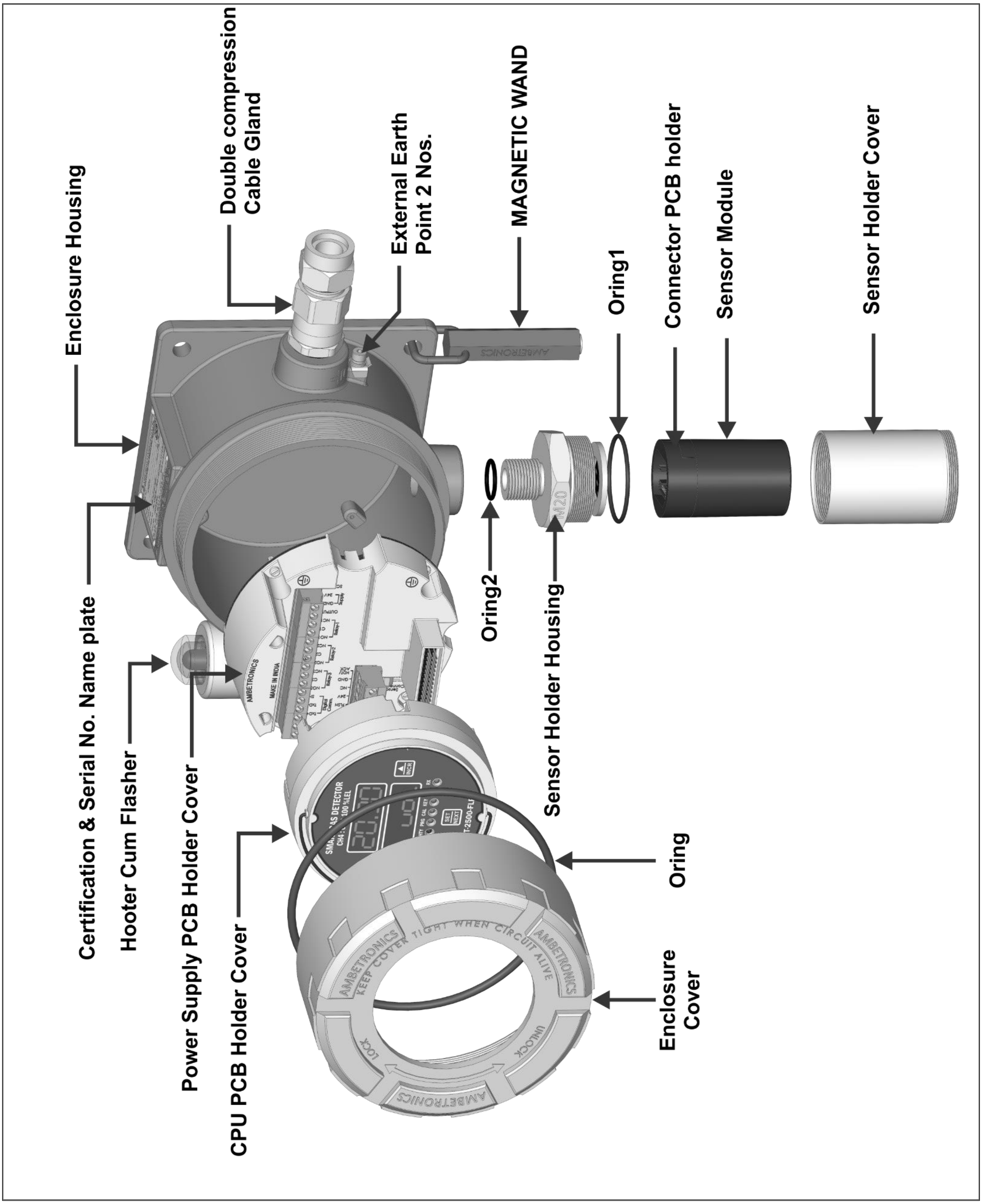


Figure 10 **GT-2500-FLP (ENC: JB-FLP-IIC)**

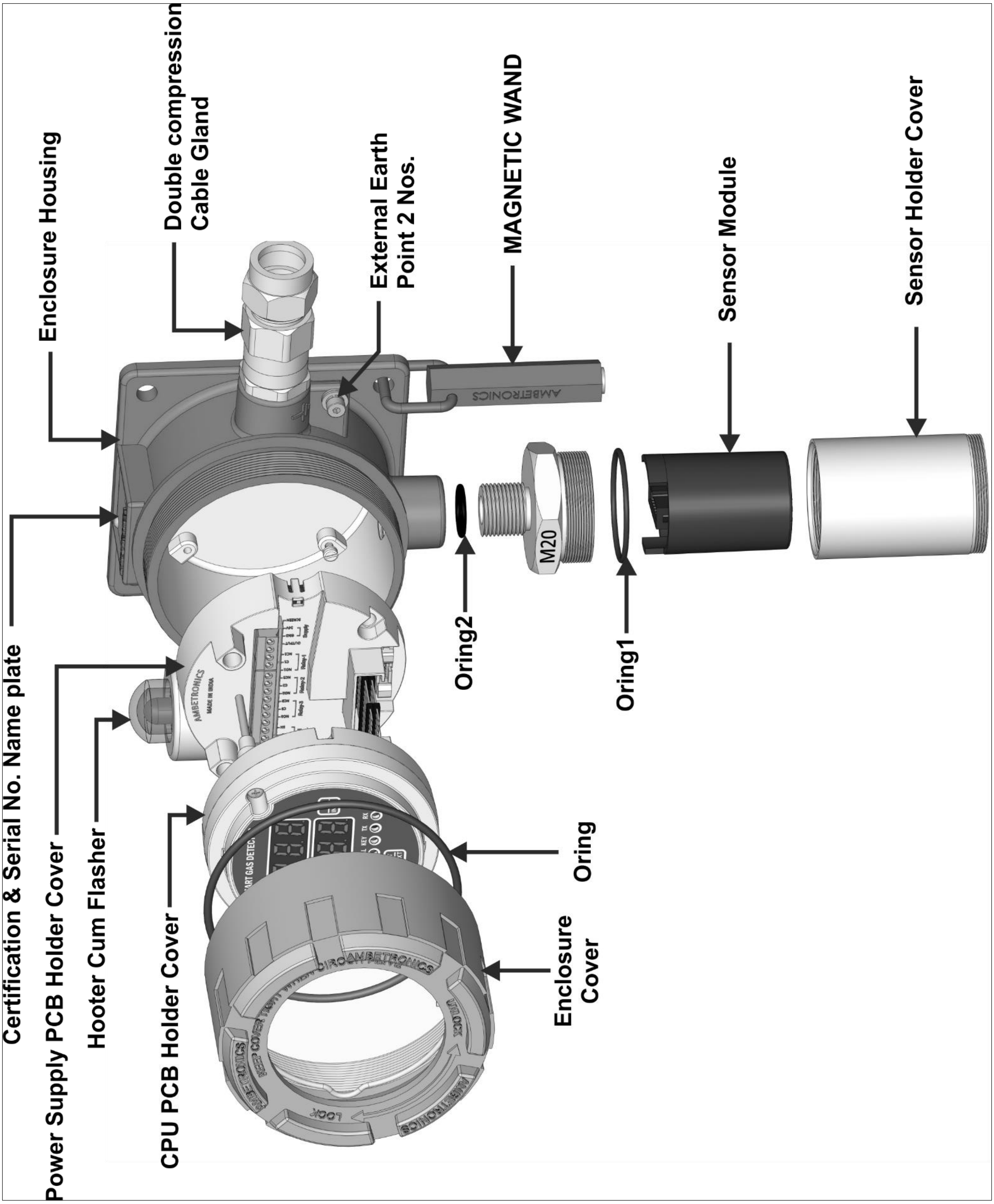


Figure 11 **GT-2500-FLP (ENC: JB90)**

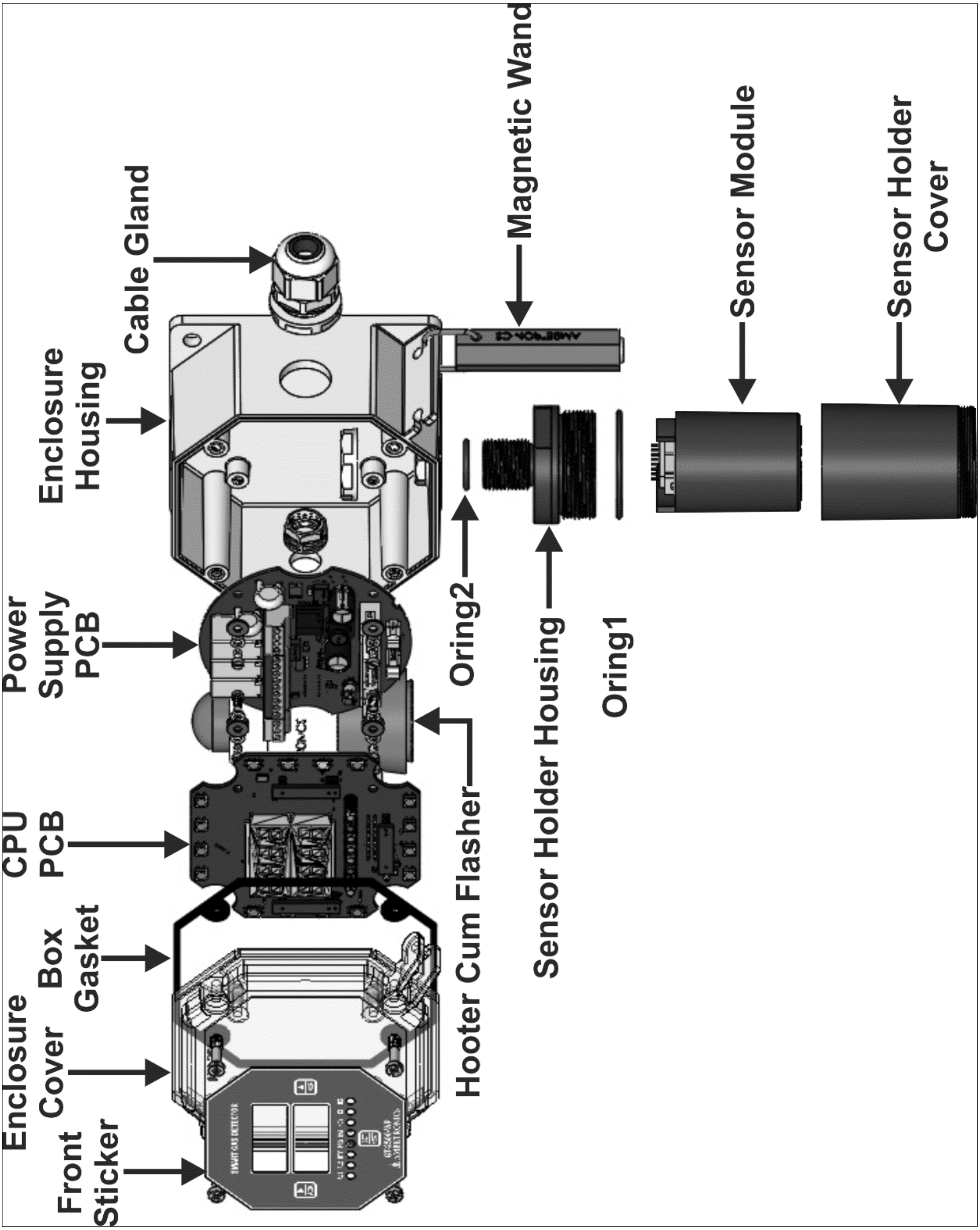


Figure 12 **GT-2500-WP**

5.3 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITHOUT HOOTER CUM FLASHER (ENC: JB-FLP-IIC)

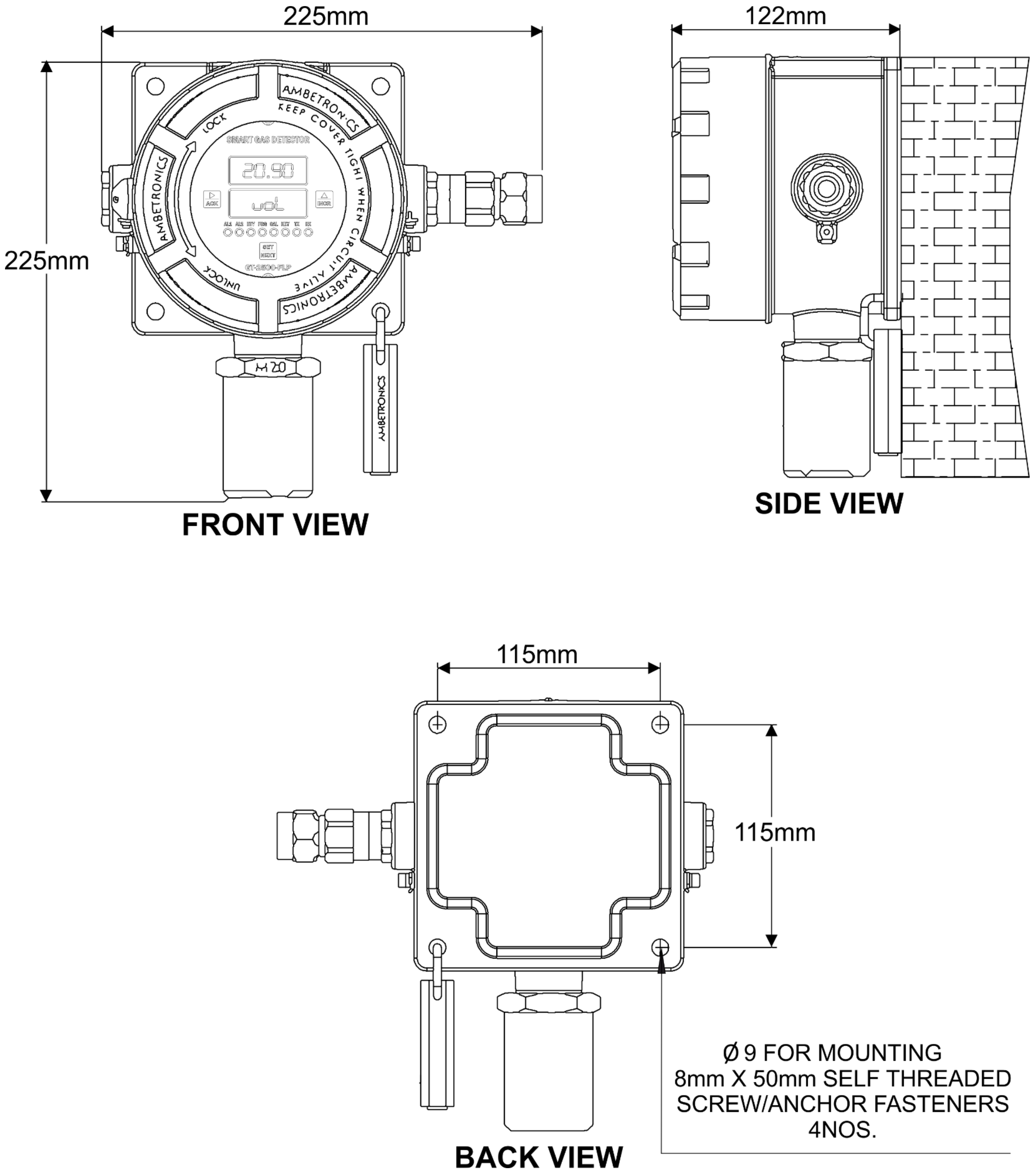


Figure 13

5.4 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITH HOOTER CUM FLASHER (ENC: JB-FLP-IIC)

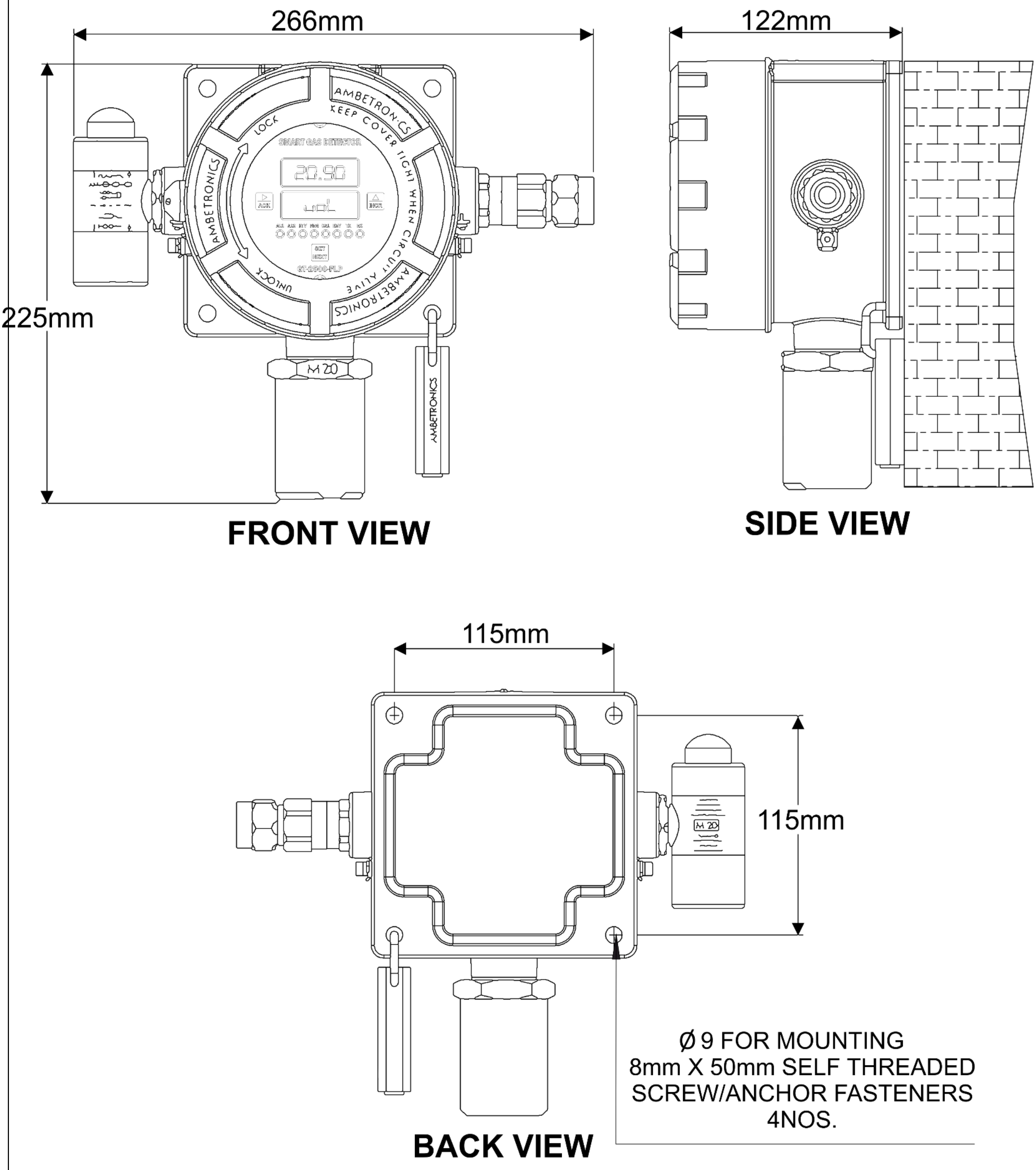


Figure 14

5.5 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITHOUT HOOTER CUM FLASHER (ENC: JB-90)

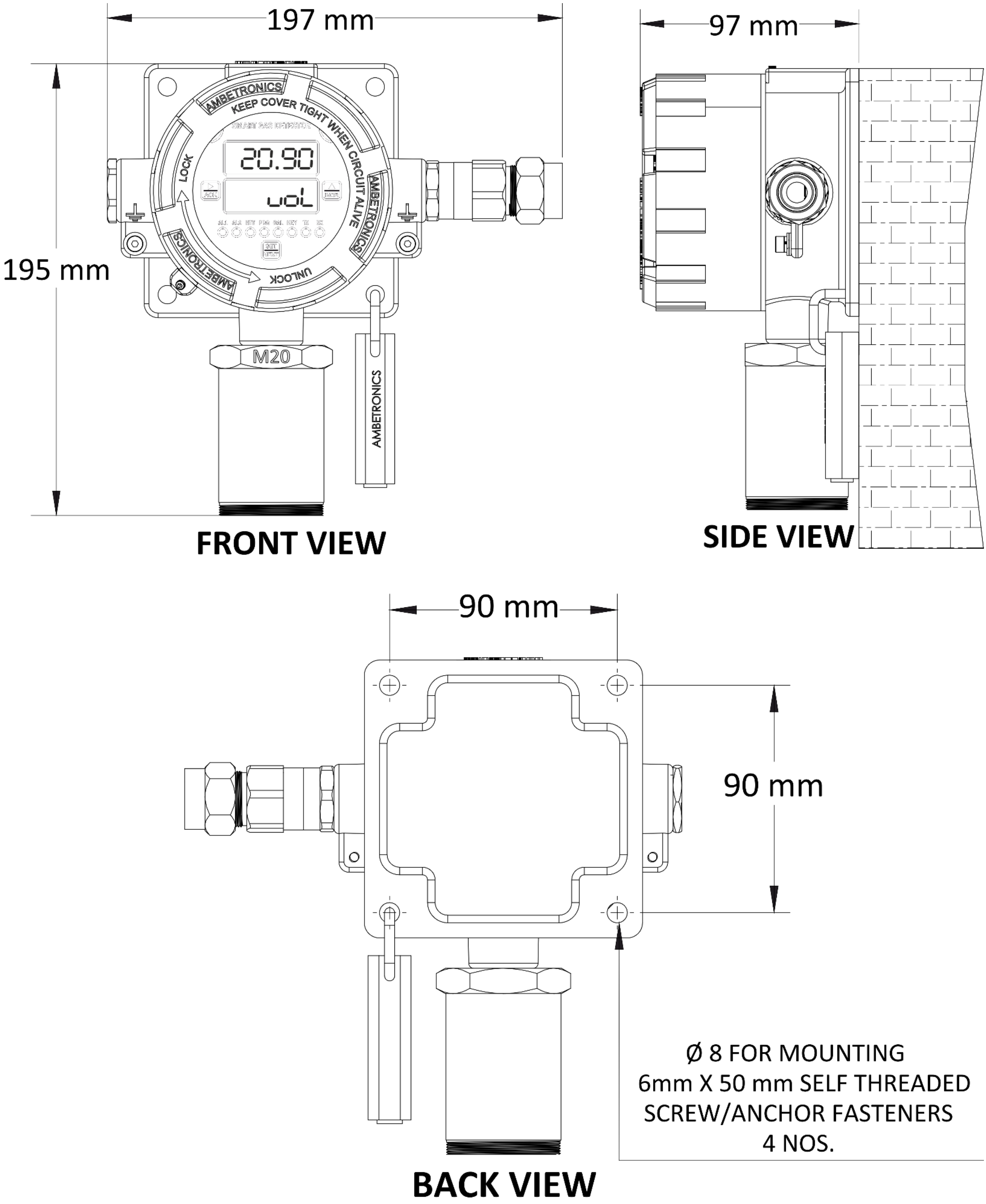


Figure 15

5.6 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITH HOOTER CUM FLASHER (ENC: JB-90)

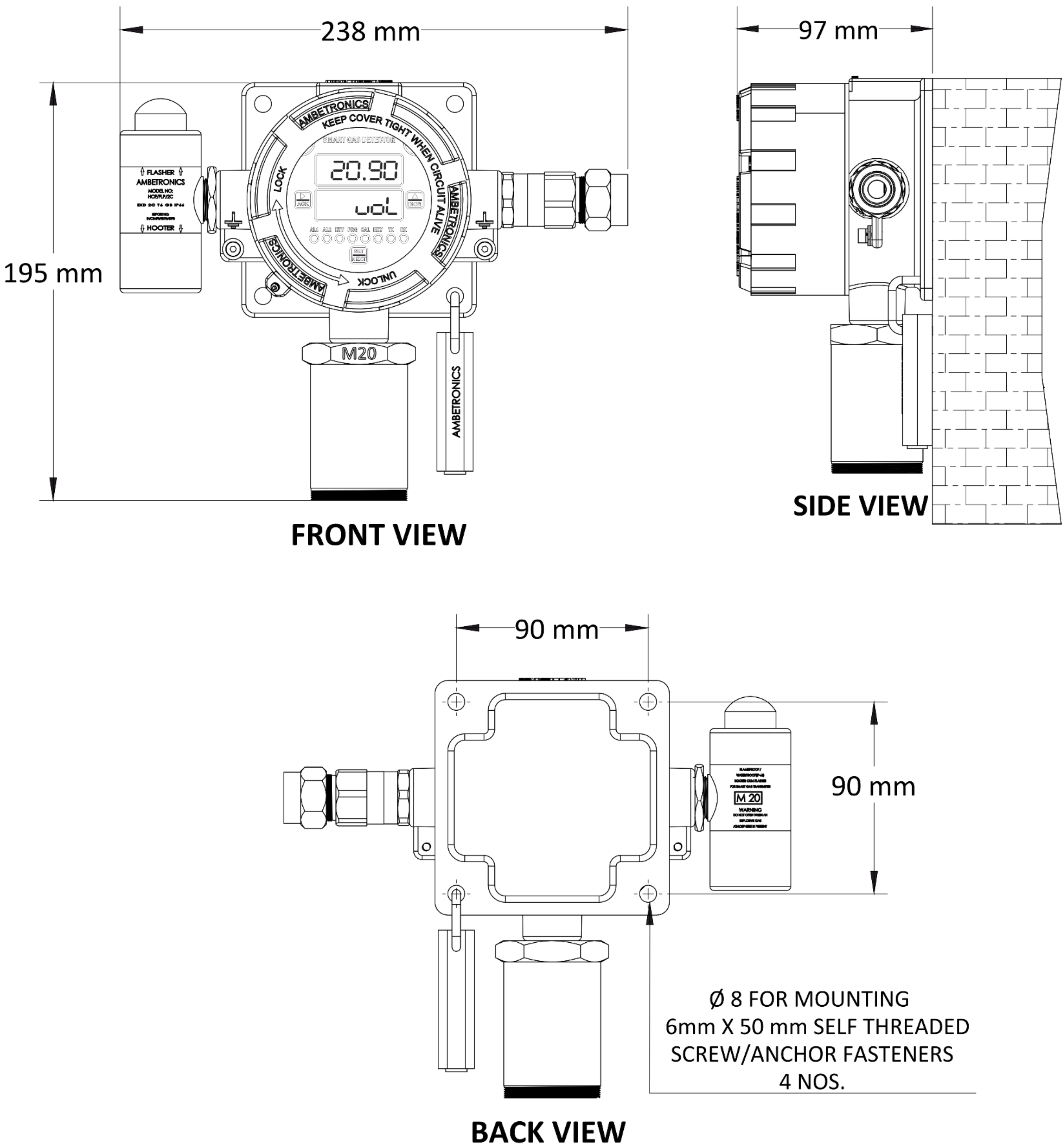


Figure 16

5.7 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITH HOOTER CUM FLASHER (ENC: WP-P)

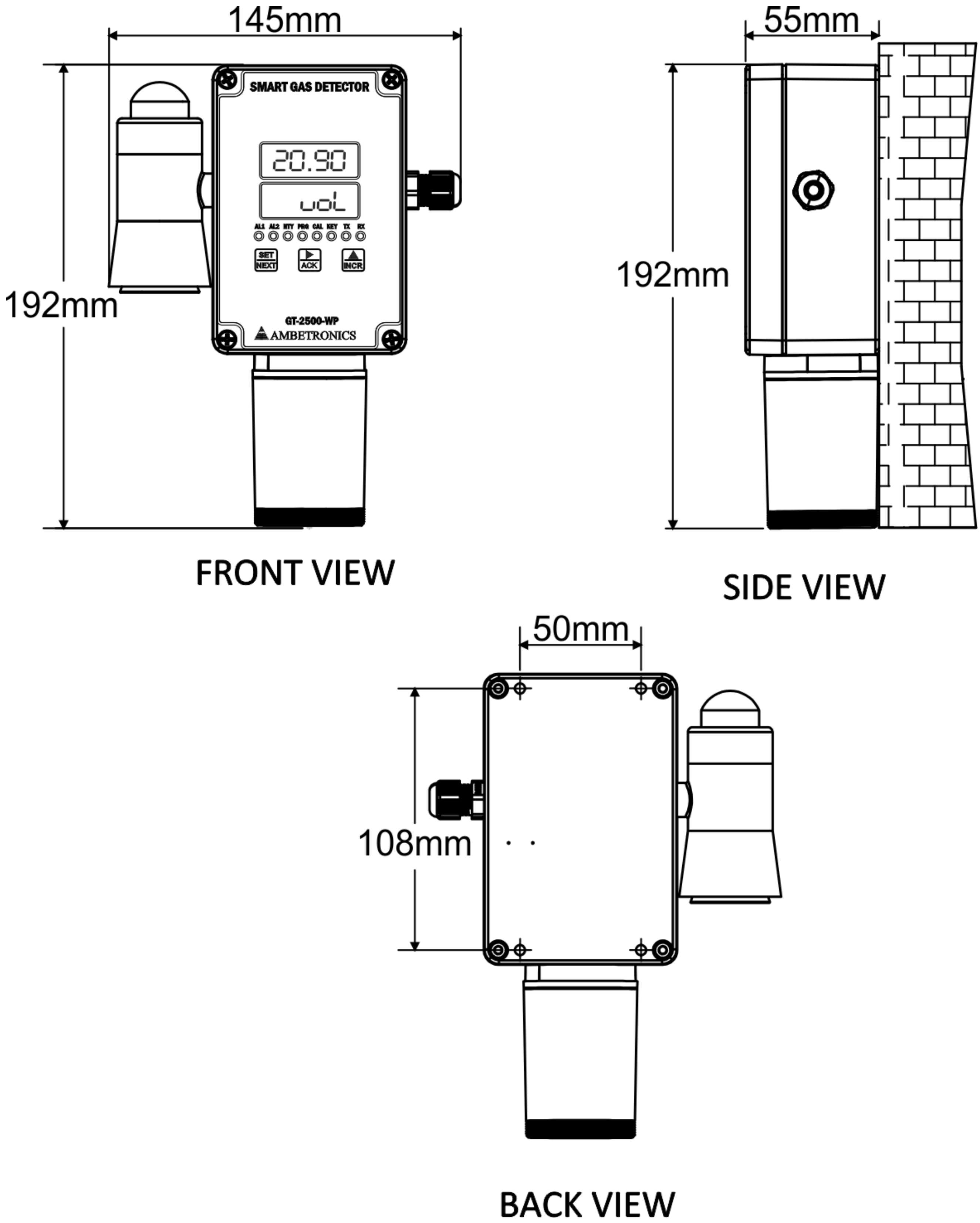


Figure 17

5.8 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS TWO CABLE GLAND (ENC: WP-P)

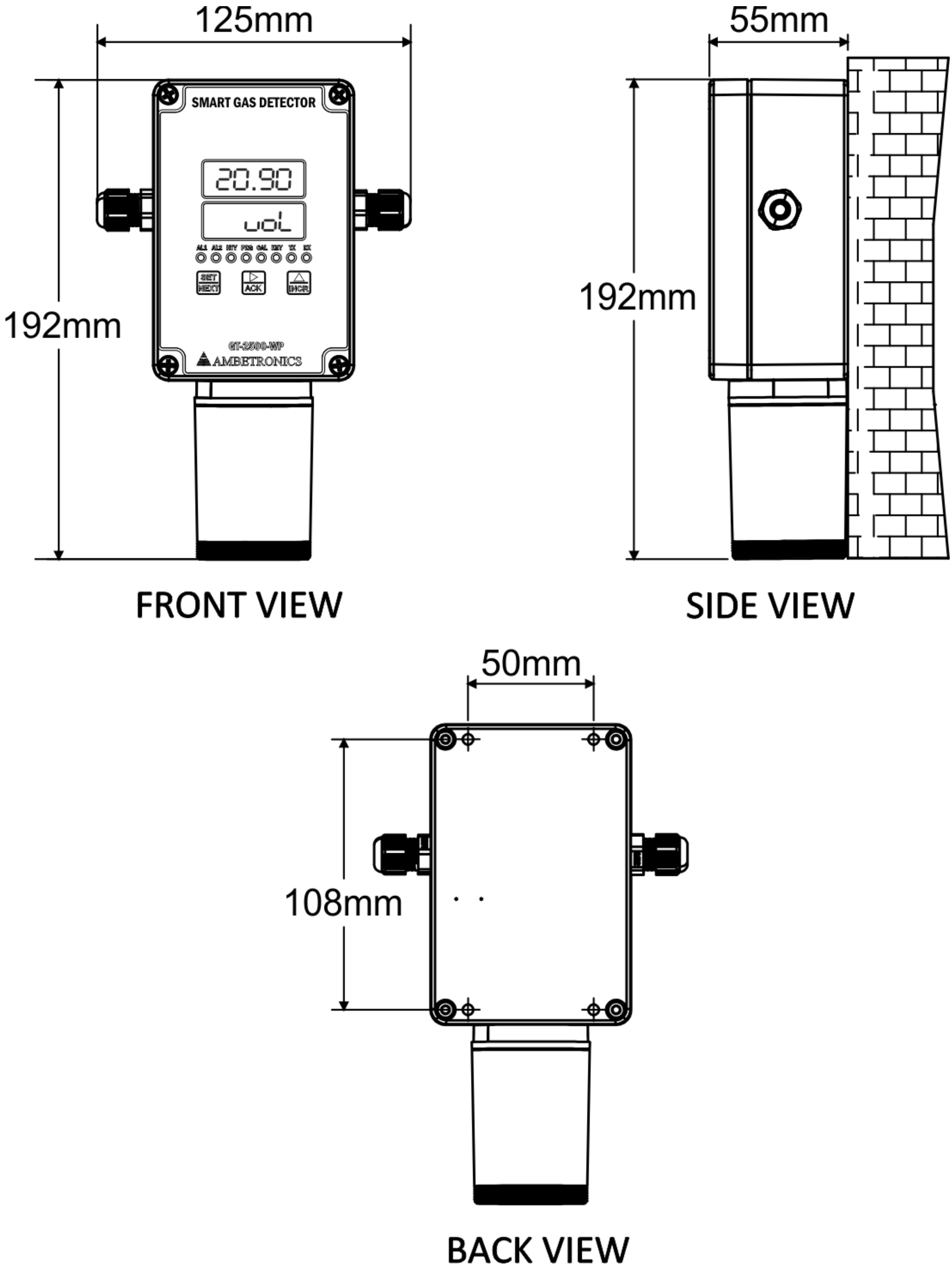


Figure 18

5.9 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITHOUT HOOTER CUM FLASHER (ENC: WP-P)

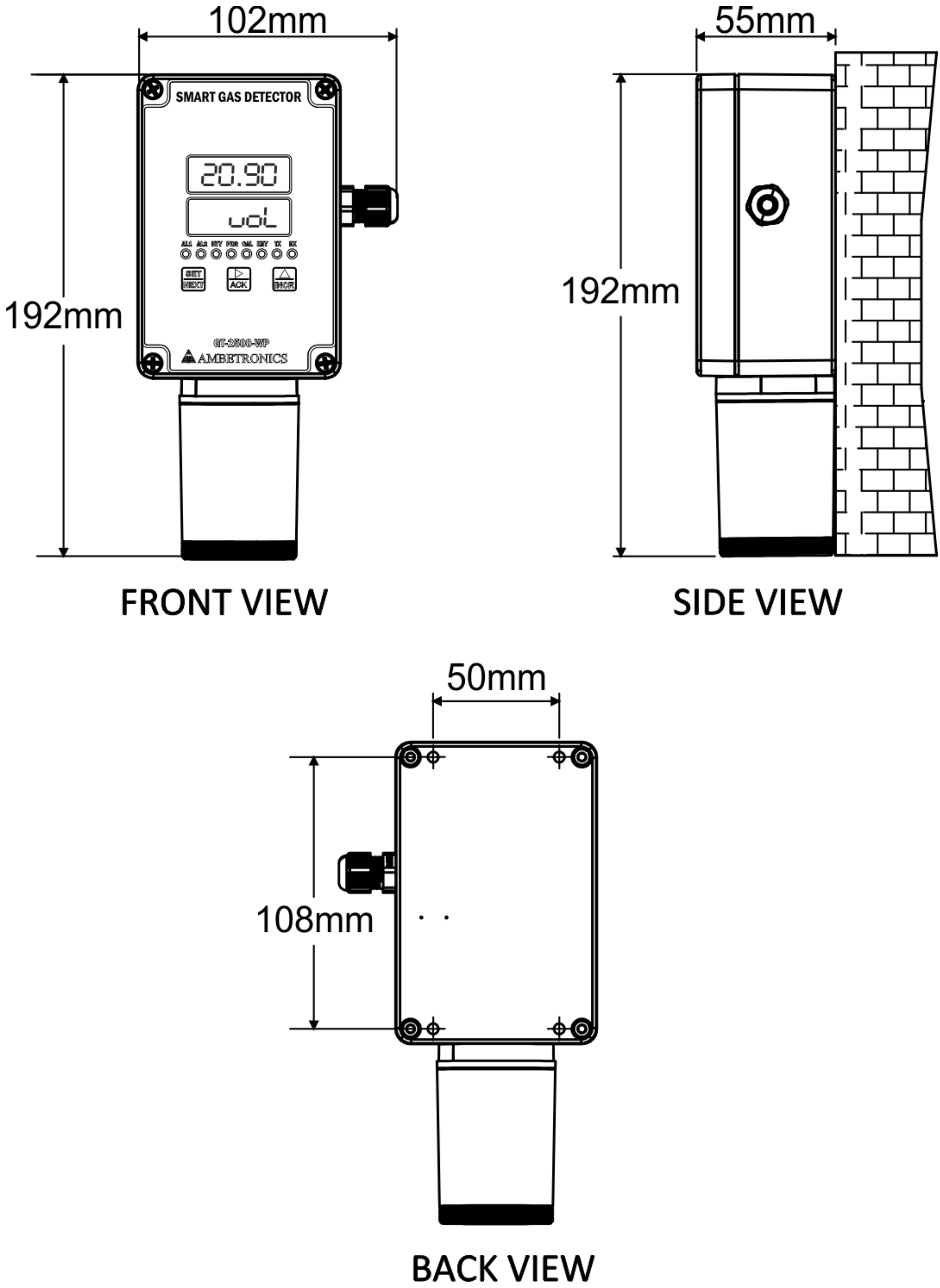


Figure 19

5.10 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITH HOOTER CUM FLASHER (ENC: WP)

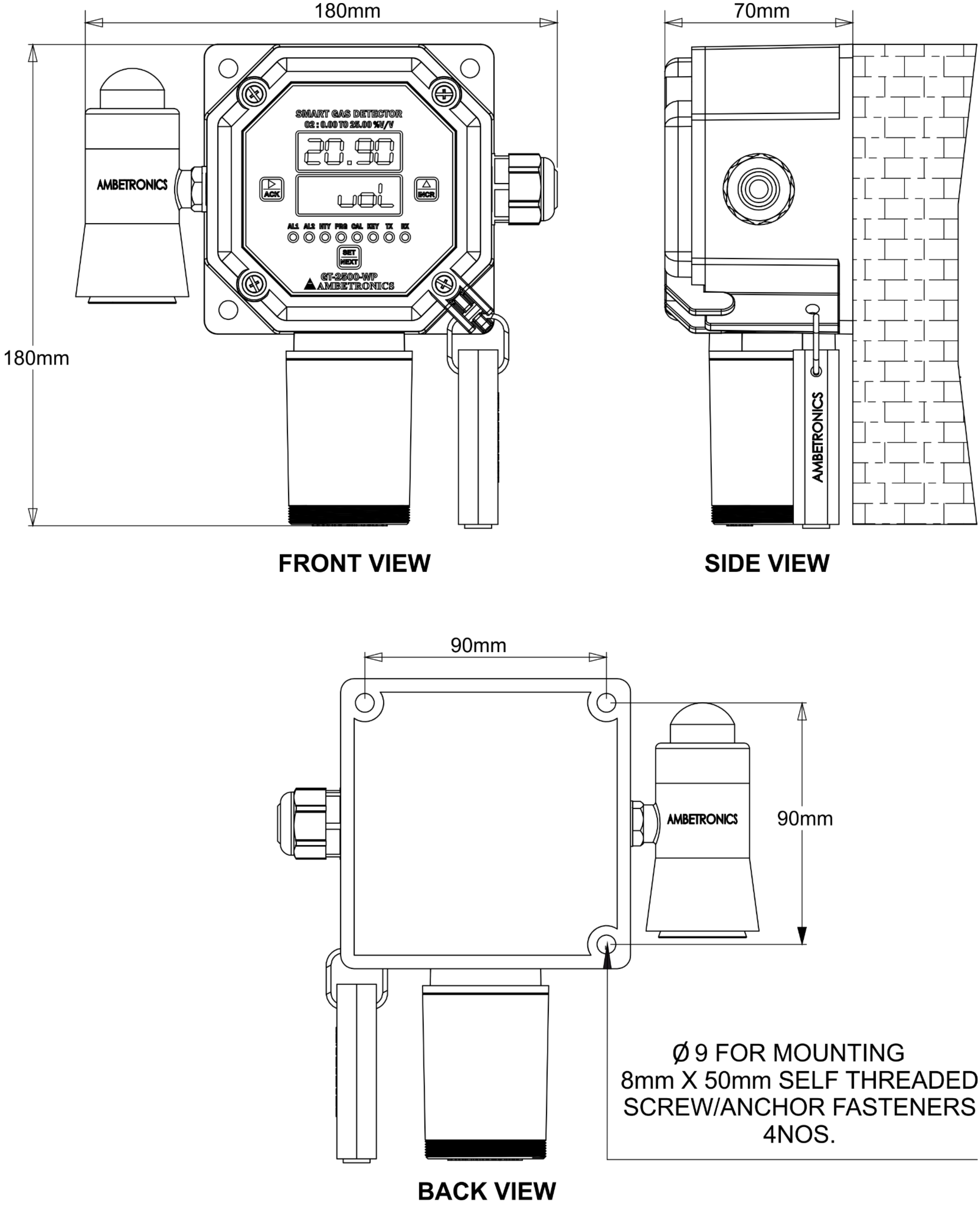


Figure 20

5.11 FLAMEPROOF ENCLOSURE DIMENSION & MOUNTING DETAILS WITHOUT HOOTER CUM FLASHER (ENC: WP)

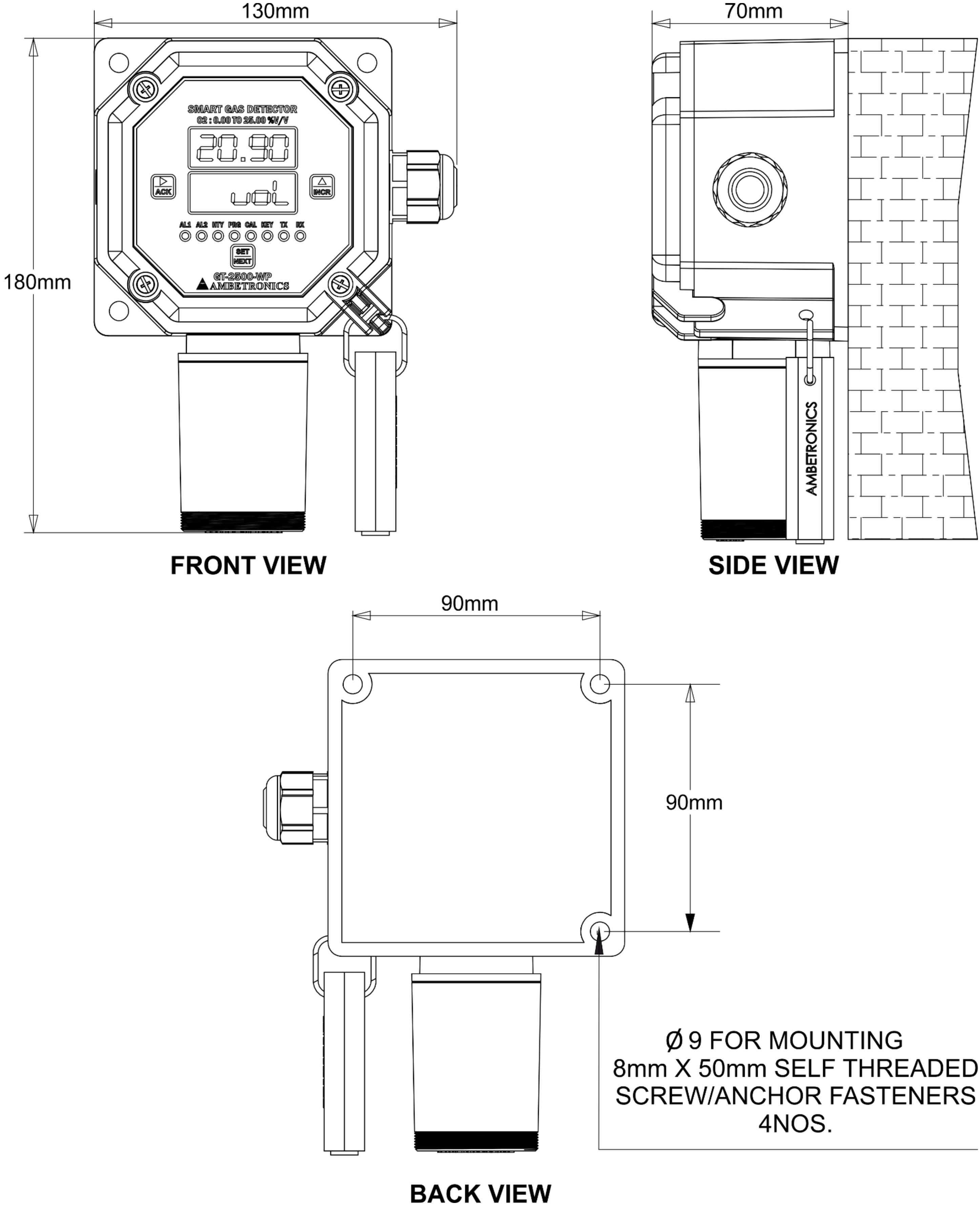


Figure 21

5.12 CANOPY WITH WALL MOUNTING (ENC: JB-FLP-IIC)

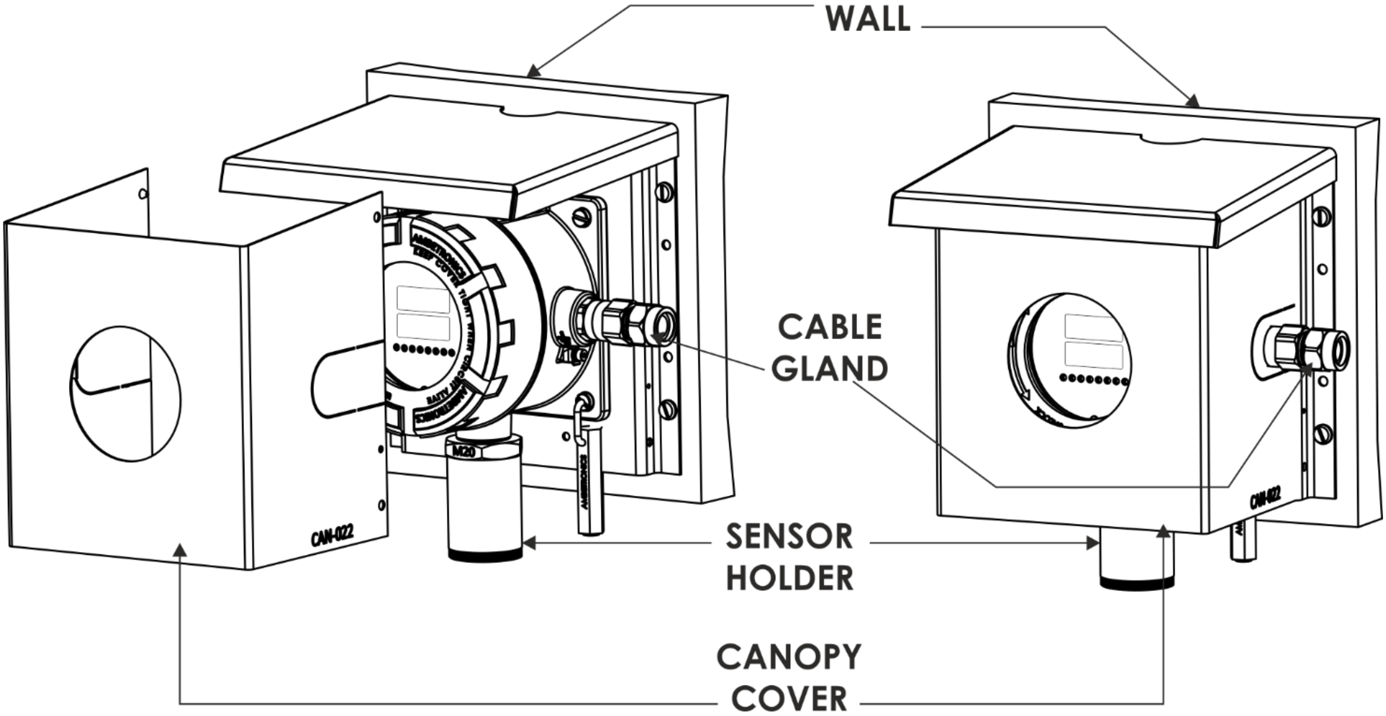


Figure 22

5.13 CANOPY WITH WALL MOUNTING (ENC: JB-90)

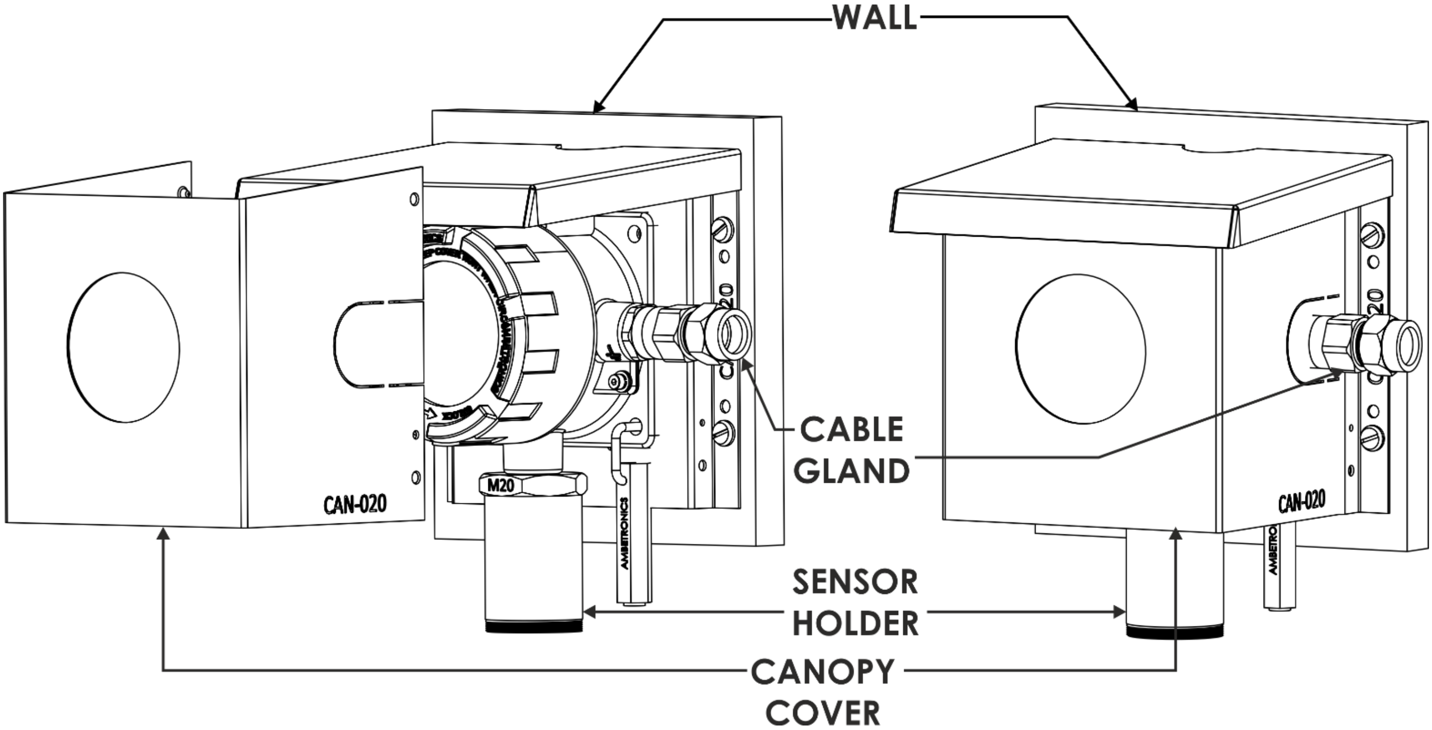


Figure 23

5.14CANOPY WITH POLE MOUNTING WITHOUT HOOTER CUM FLASHER

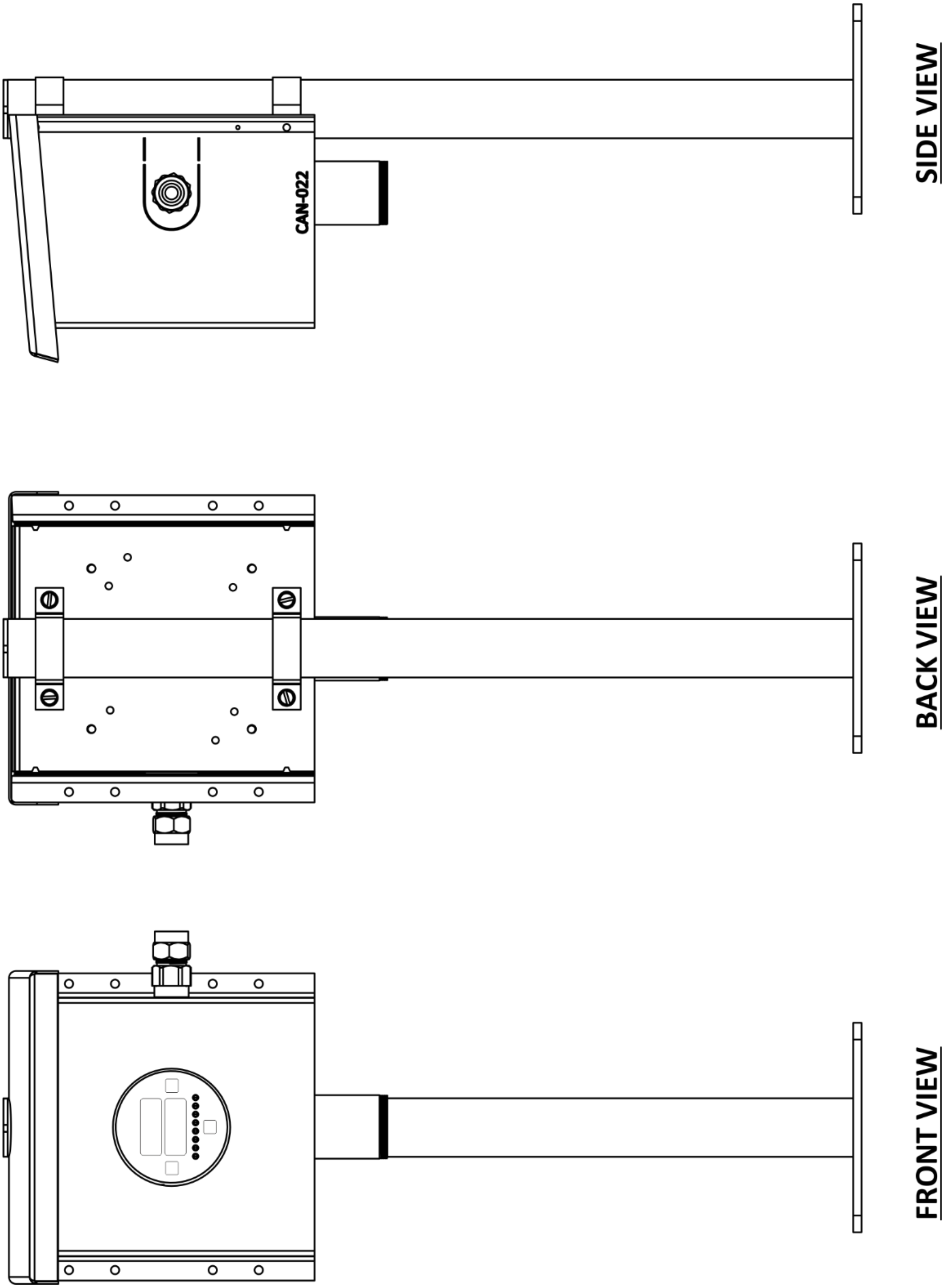


Figure 24

5.15CANOPY WITH POLE MOUNTING WITH HOOTER CUM FLASHER

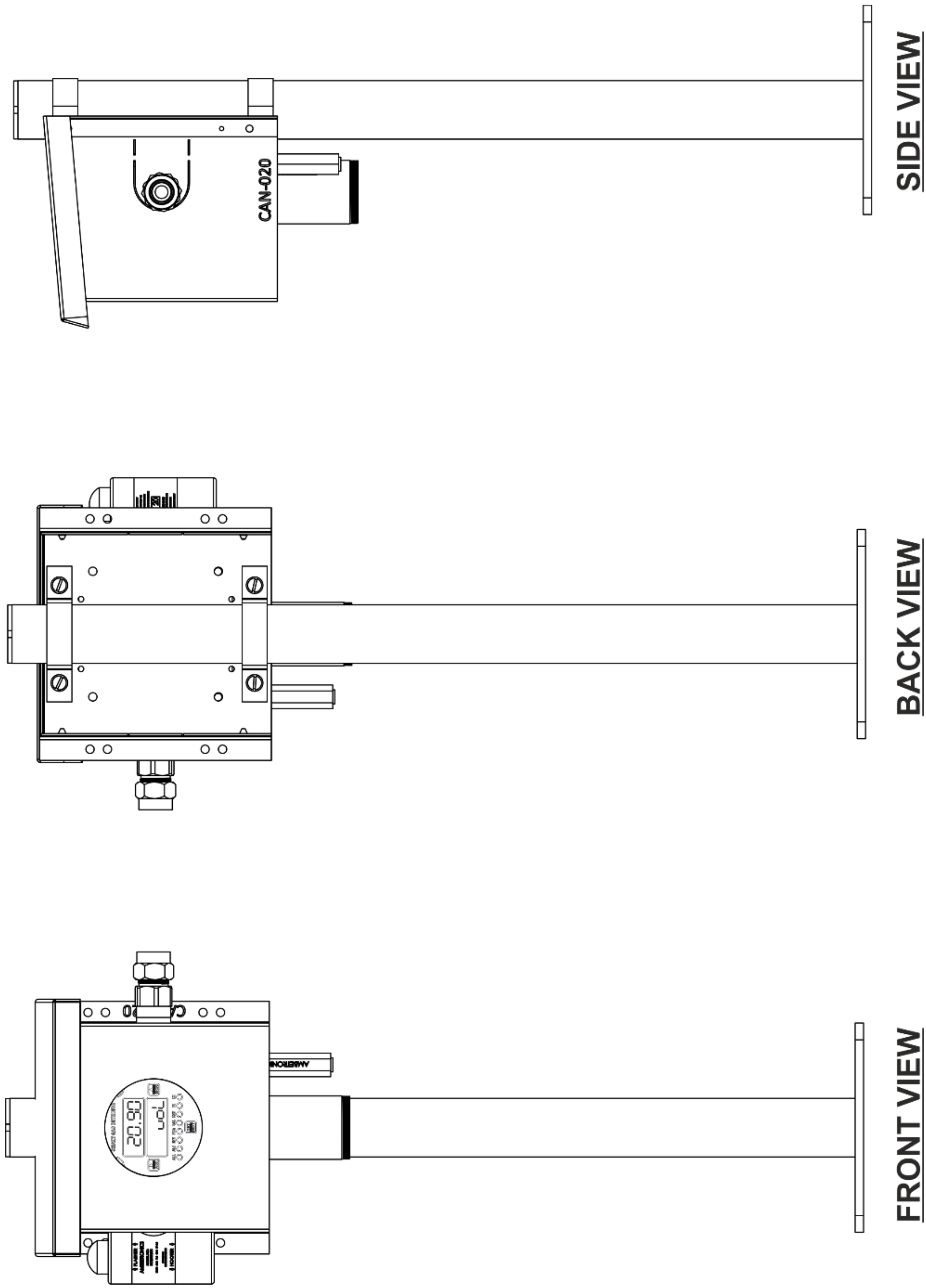


Figure 25

6 SYSTEM WIRING

6.1 WIRING INFORMATION

This topic outlines the steps in regarding for wiring the Smart Gas Detector/ Transmitter. These steps include the following.

- Wiring preparation.
- Alarm Relay wiring.
- Power & output wiring.
- Modbus interface wiring

NOTE:
1. Perform all wiring in accordance with local electrical code and local authorities wiring jurisdiction.
2. DC signal & AC power should not be run in the same conduit.
3. All filed wiring colours are arbitrary.

6.2 WIRING PREPARATION

1. Collect the appropriate type & length of wire.
   - For control wire, use insulated shielded cable with more than 80% shielding.
   - For Signal & AC power wire, use three conductor included & shielded cable with more than 80% shielding.
   - For Digital Modbus signal, use a minimum 3-conductor insulated & twisted shielded cable with more than 80% shielding.
2. Remove the top cover from the housing.
3. For wiring, Release the gland.
4. Connect control, signal & power wires in to the housing & connect shielding of cable to the GND of unit.
5. Connect Earthing cable to the Earthing Screw on the body of the instrument.
6. Do not supply power to the instrument until the connection are checked.

6.3 ALARM RELAY WIRING

Three relays are provided of rating 5A/30 VDC & 10A/250 VAC of which two relays can be used as latch/non-latch, and 3rd relay can be used as fail safe which can be kept normally ON or normally OFF, for more details refer to the connection diagram.

INFORMATION: Once wiring is complete, place the display window electronic back in the housing. Be careful, ensure any wiring should not be pinched. After that place the top cover of the enclosure & power up the instrument.

WARNING /CAUTIONS
- It is recommended that on board relay should not be used to drive loads directly. On- board relays should be used to drive a secondary higher power relay, which is connected, to the control device (e.g. strobe, siren, exhaust, fan etc.)
- For Power, Output wiring & RS-485 wiring. Refer connection diagram.

6.4 WIRING DIAGRAM

6.4.1 WIRING DIAGRAM FOR AMBETRONICS MAKE HOOTER CUM FLASHER (ENC: JB-FLP-IIC)

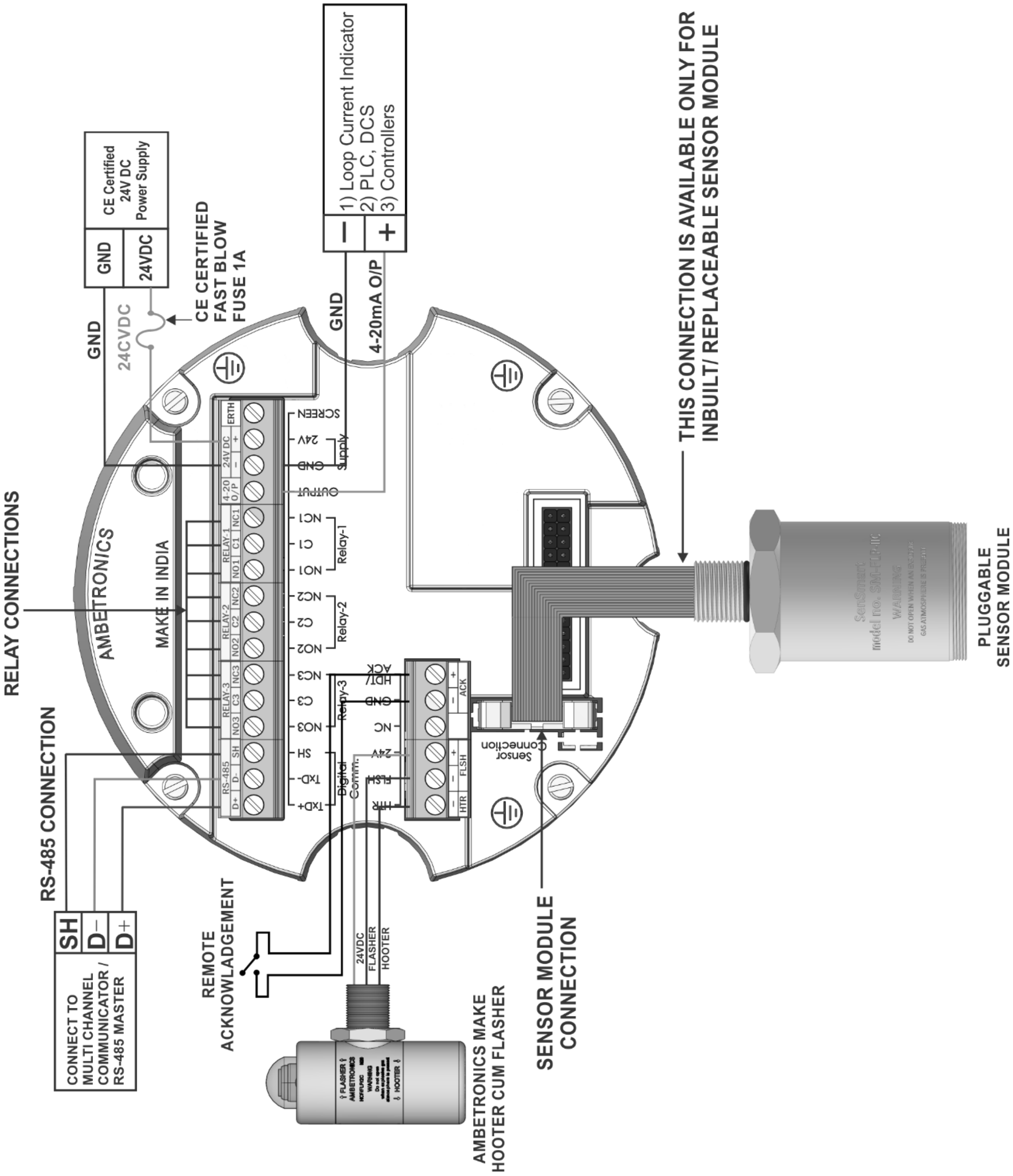


Figure 26

INFORMATION:

- **Use CE Certified 24VDC Power Supply**
- **Use CE Certified Fast Blow Fuse of Rating 1A**

6.4.2 WIRING DIAGRAM FOR AMBETRONICS MAKE HOOTER CUM FLASHER (ENC: JB-90)

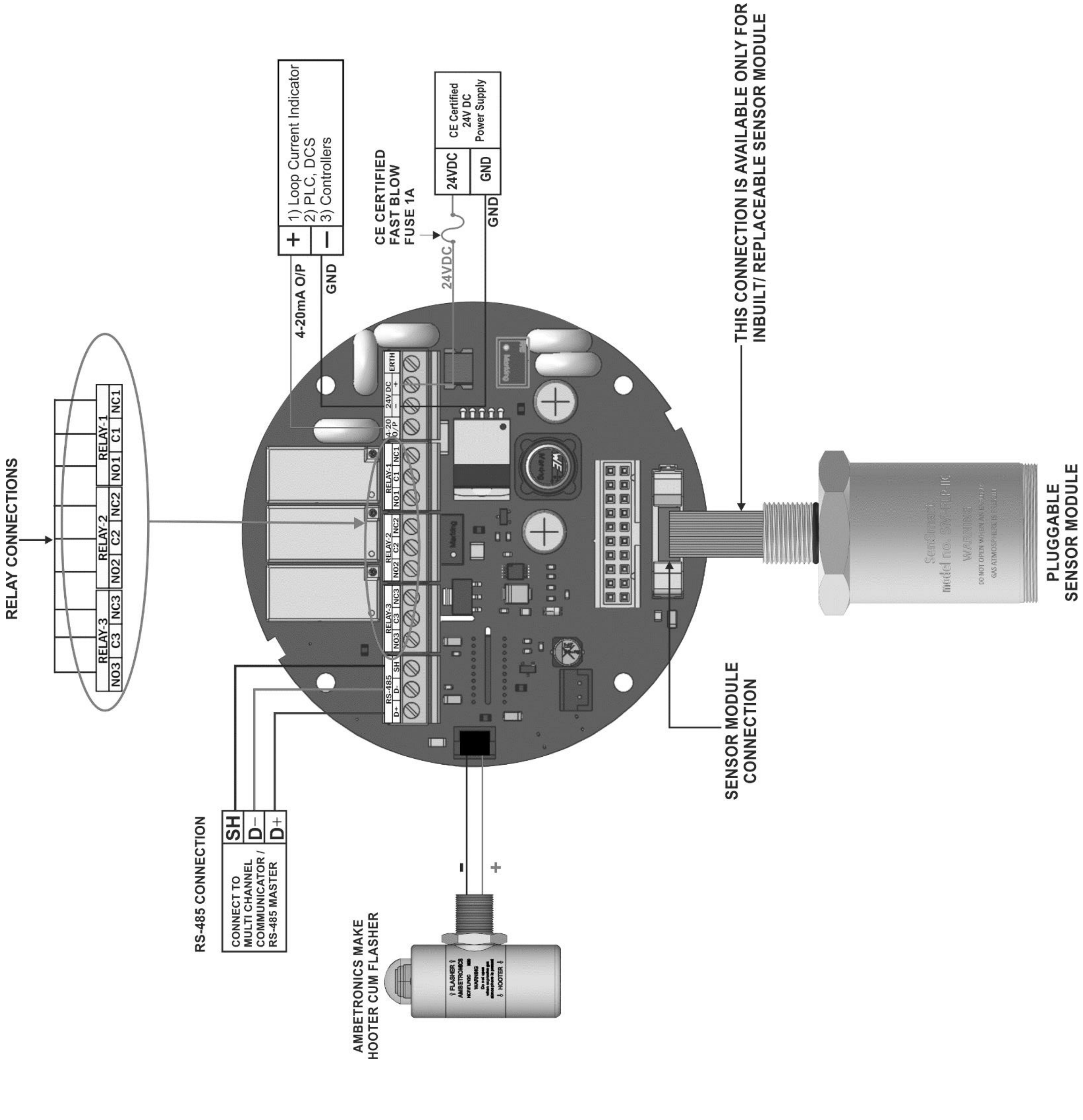


Figure 27

INFORMATION:

- **Use CE Certified 24VDC Power Supply**
- **Use CE Certified Fast Blow Fuse of Rating 1A**

6.4.3 WIRING DIAGRAM FOR AMBETRONICS MAKE HOOTER CUM FLASHER (ENC: WP-P)

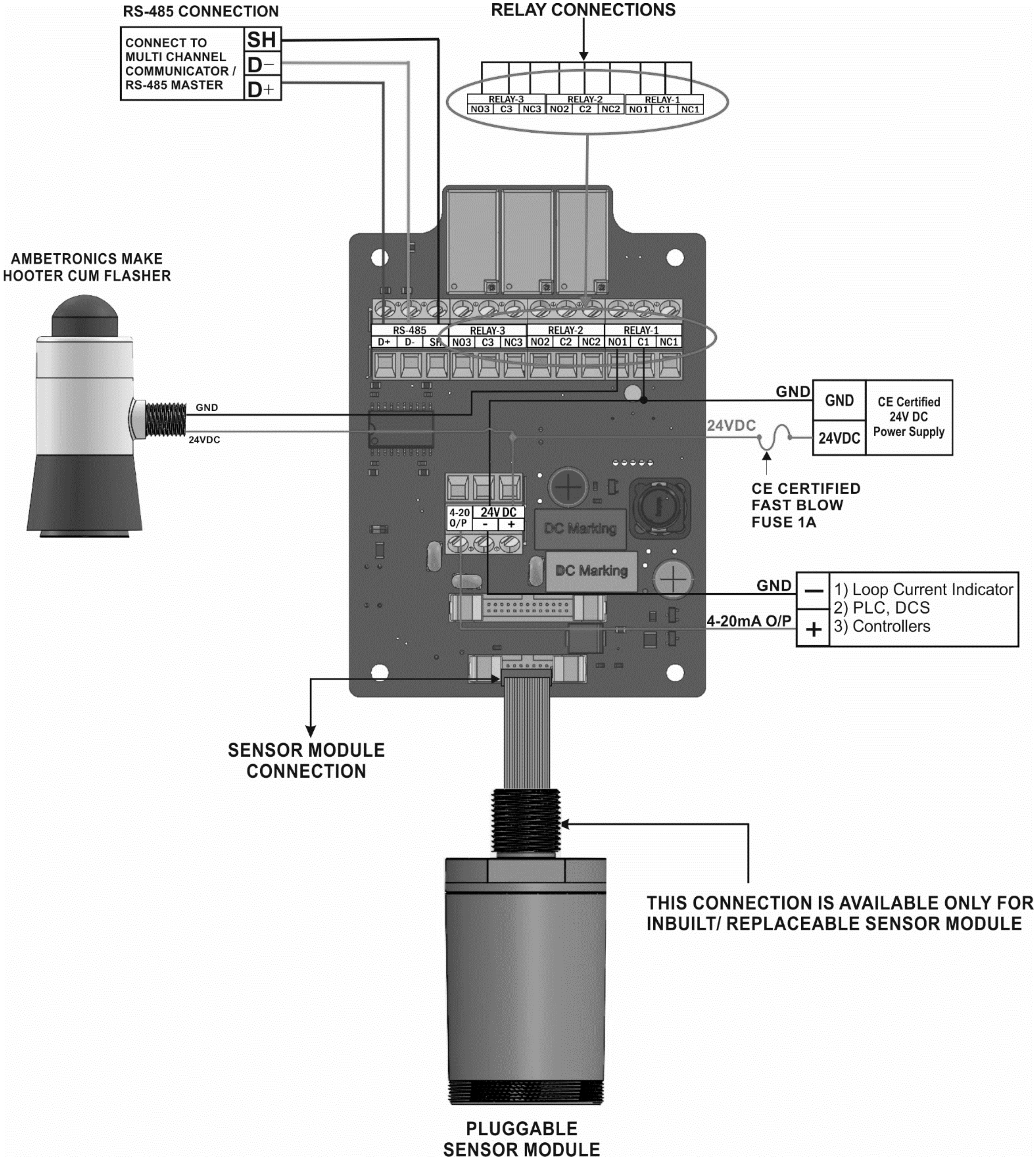


Figure 28 **GT-2500-WP**

INFORMATION:

- **Use CE Certified 24VDC Power Supply**
- **Use CE Certified Fast Blow Fuse of Rating 1A**

6.4.4 WIRING DIAGRAM FOR AMBETRONICS MAKE HOOTER CUM FLASHER (ENC: WP)

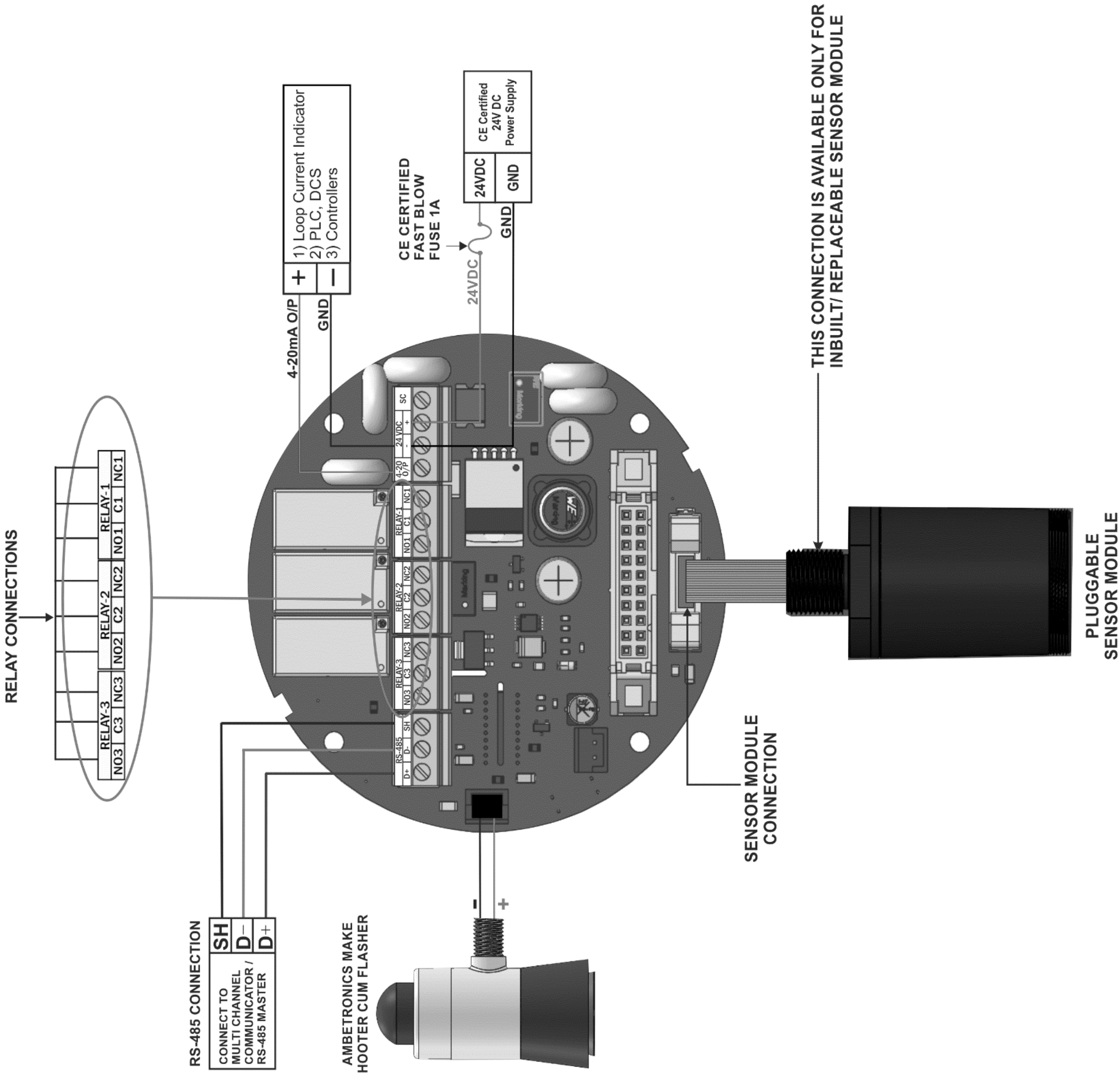


Figure 29

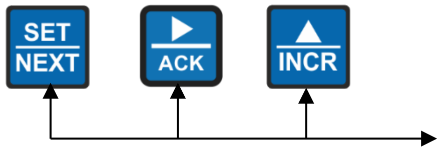
INFORMATION:

- **Use CE Certified 24VDC Power Supply**
- **Use CE Certified Fast Blow Fuse of Rating 1A**

7 KEY FUNCTIONALITY

Table 3

SR. NO.	KEY TYPE	PROGRAMMING MODE	NORMAL MODE
1.	SET/NEXT key	• Setting of parameter and selecting the parameter • Exiting the menu	• To enter the user menu when pressed for about 5 sec
2.	SHIFT/ACK key	• Shift the cursor when desired numerical value is to be edited • Selecting the parameter	• To Acknowledge alarm & Relay when pressed for 3secs. • To access status menu when pressed for 5secs. • To mute the Buzzer/Hooter when pressed for 3secs.
3.	INCREMENT KEY	• To select the desired menu	



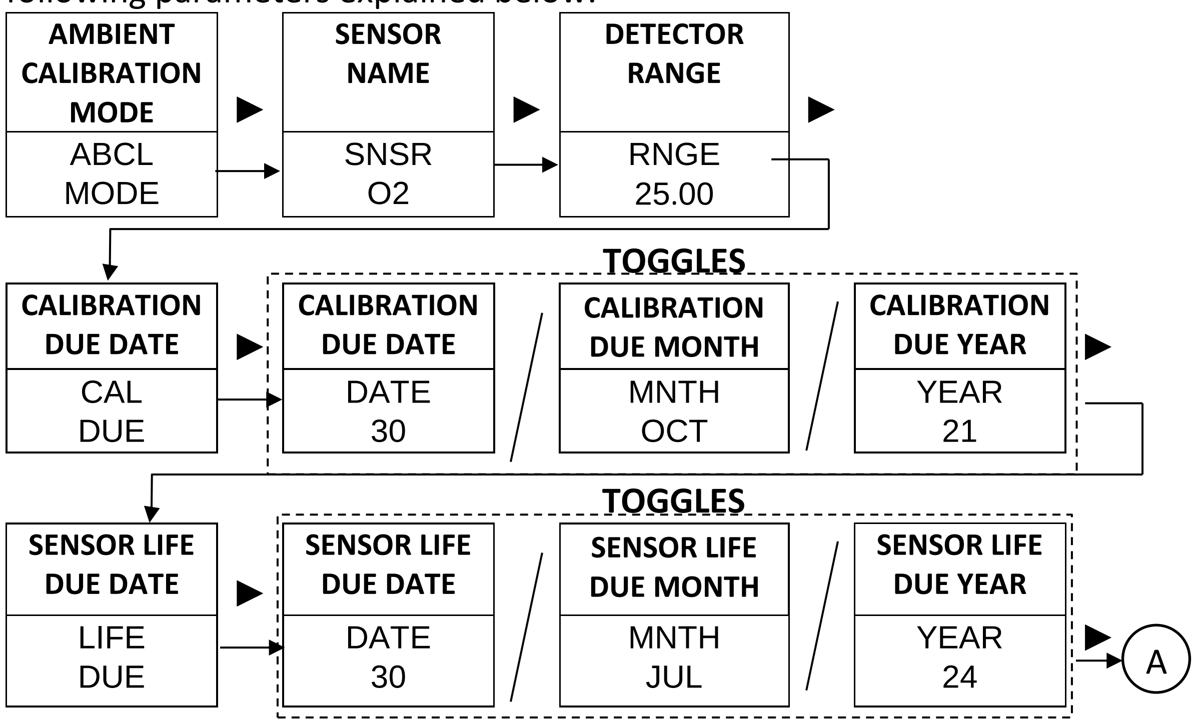
These keys can be operated using a Magnetic Wand provided with the unit & can be operated without opening the FLP & WP enclosure.

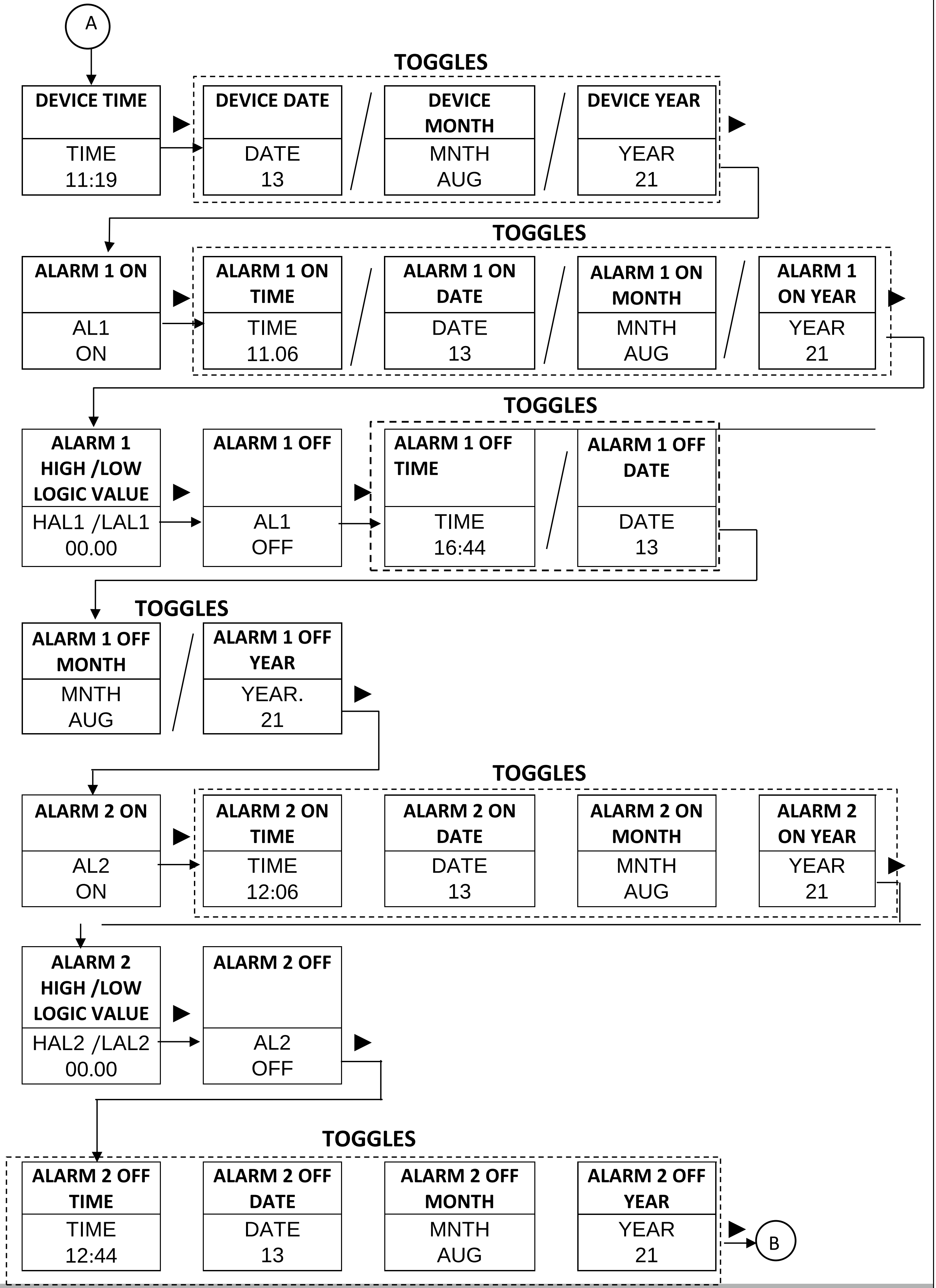
Magnetic wand

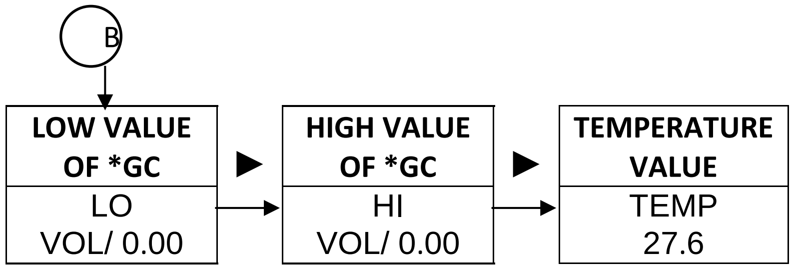
Figure 30

7.1 STATUS MENU GT-2500-FLP

When Shift/ACK key is pressed for 5secs **STATUS MENU** will open which shows following parameters explained below.




A
TOGGLES
DEVICE TIME
TIME 11:19
DEVICE DATE
DATE 13
DEVICE MONTH
MNTH AUG
DEVICE YEAR
YEAR 21
TOGGLES
ALARM 1 ON
AL1 ON
ALARM 1 ON TIME
TIME 11.06
ALARM 1 ON DATE
DATE 13
ALARM 1 ON MONTH
MNTH AUG
ALARM 1 ON YEAR
YEAR 21
TOGGLES
ALARM 1 HIGH /LOW LOGIC VALUE
HAL1 /LAL1 00.00
ALARM 1 OFF
AL1 OFF
ALARM 1 OFF TIME
TIME 16:44
ALARM 1 OFF DATE
DATE 13
TOGGLES
ALARM 1 OFF MONTH
MNTH AUG
ALARM 1 OFF YEAR
YEAR. 21
TOGGLES
ALARM 2 ON
AL2 ON
ALARM 2 ON TIME
TIME 12:06
ALARM 2 ON DATE
DATE 13
ALARM 2 ON MONTH
MNTH AUG
ALARM 2 ON YEAR
YEAR 21
ALARM 2 HIGH /LOW LOGIC VALUE
HAL2 /LAL2 00.00
ALARM 2 OFF
AL2 OFF
TOGGLES
ALARM 2 OFF TIME
TIME 12:44
ALARM 2 OFF DATE
DATE 13
ALARM 2 OFF MONTH
MNTH AUG
ALARM 2 OFF YEAR
YEAR 21
B



Event values

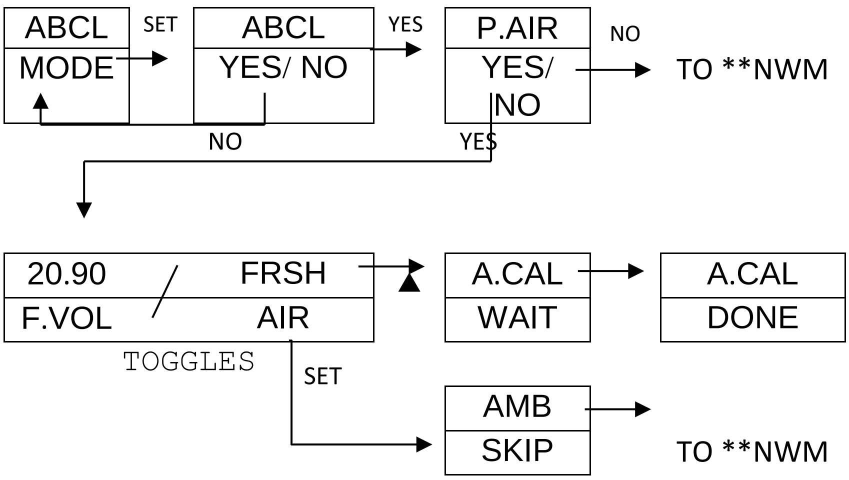
ALARM 1/ ALARM 2 ON TIME & DATE
ALARM 1/ ALARM 2 LOW/ HIGH LOGIC VALUE
ALARM 1/ ALARM 2 OFF TIME & DATE

- This values get updated when Alarm1/ Alarm2 are triggered. We can reset these values in Alarm menu. Refer Page No.35.
- Alarm 1/2 Low/High logic value will depend on logic setting in Alarm Menu. i.e. High or Low

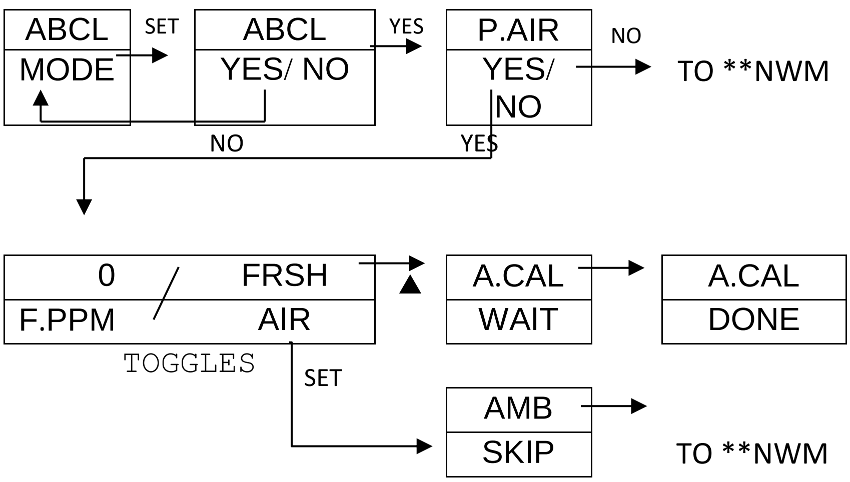
AMBIENT CALIBRATION MODE

Press key to go inside Ambient Calibration Mode

For Oxygen



For other Gas types



Description

ABCL MODE	**AMBIENT CALIBRATION MODE**	F.PPM	**FRESH PPM**
P.AIR	**PURGE AIR**	FRSH AIR	**FRESH AIR**
F.VOL	**FRESH VOLUME**	A.CAL	**AMBIENT CALIBRATION**

NOTE: This menu is used to perform ambient calibration of Electrochemical/ Catalytic/ PID/ NDIR gas sensors.

8 LED INDICATION

1. **ALARM 1 LED:** This LED will start blinking whenever the gas concentration crosses the ALARM1 set point. In Latch condition, when gas concentration crosses the set point value then after pressing 

key for 3 sec then it will glow steadily and whenever the gas concentration comes within the set point it will turn off. During normal operation it will be in OFF condition.
2. **ALARM 2 LED:** This LED will start blinking whenever the gas concentration crosses the ALARM2 set point. In Latch condition, when gas concentration crosses the set point value then after pressing 

key for 3sec then it will glow steadily and whenever the gas concentration comes within the set point it will turn off. During normal operation it will be in OFF condition.
3. **HEALTHY LED:** This LED will start blinking whenever there is an occurrence of warning i.e. Sensor Open / Over Range. During normal operation this LED will be in ‘STEADY ON’ condition which shows that unit is working fine
4. **PROGRAMMING MODE LED:** This LED will start blinking whenever user enter into ‘OPERATOR SETTING MODE’/ TEST MODE’ after entering the correct password. During normal operation it will be in OFF condition.
5. **CALIBRATION MODE LED:** This LED will start blinking whenever user enter into ‘CALIBRATION MODE’.
6. **KEY LED:** This LED glows when a key is pressed or magnetic wand placed on keys. If no key is pressed it remains off.
7. **TX LED:** This is ‘Transmit LED’ & it blinks when device transmits the data through RS-485.
8. **RX LED:** This is ‘Receive LED’ & it blinks when device receives the data through RS-485.

9 CAPTION MEANING

A. MAIN MENU AND GENERAL FUNCTIONS

MENU ESC	Esc To Main Menu	MENU CODE	Menu Code
MENU ALRM	Menu Alarm	MENU OFST	Menu Offset
MENU CAL	Menu Calibration	MENU LOHI	Menu LOW/HIGH
MENU COMM	Menu Communication	MENU OUT	Menu Output
MENU TEST	Menu Test	MENU BUMP	Menu Bumptest

MENU RLOK	Menu Range Lock	MENU RTC	Menu RTC
MENU INFO	Menu Information		

B. CODE MENU

ENTR PSWD	Enter Password to Set New Password	PSWD 0000	Enter Password
NPSW 0000	Enter new password	CNFM YES/NO	Confirm password Change Yes/No
CHNG SUCC	Password Change Successful		

C. ALARM MENU

ALRM PSWD	Enter Password to edit Alarm settings		
PARA BACK	Back to main menu	PARA AL-1	Parameter Alarm 1
PARA AL-2	Parameter Alarm 2	PARA BUZZ	Parameter Buzzer
PARA HTR	Parameter Hooter	PARA FLSR	Parameter Flasher
PARA KBUZ	Parameter Key Buzz	PARA SNOZ	Parameter Snooze Time
PARA F-RL	Parameter Fail Safe	PARA F-DL	Parameter Fail Delay
PARA EVNT	Parameter Event	PARA LDEN	Parameter LED Enable
ALRM BACK	Back to Alarm		
AL-1/2 ENBL	Enable Alarm 1 or 2	ENBL YES/NO	Alarm Enable (Yes or No)
AL-1/2 ST-P	Alarm 1 or 2 Set Point	AL-1/2 HYST	Alarm 1 or 2 Hysteresis
AL-1/2 LACH	Alarm 1 or 2 Latch	LACH ENBL /DISB	Latch Enable / Disable
AL-1/2 LOGC	Alarm 1 or 2 Logic	LOGC HIGH / LOW	Alarm Logic High / Low
AL-1/2 DELY 0000	Alarm 1 or 2 Delay	AL-1/2 ALED RED/ BLUE/YELO/CYAN/VIOL	Alarm 1 or 2 Alarm LED Red/ Blue/ Yellow/ Cyan/ Violet
BUZZ DISB/ENBL	Buzzer (Enable or Disable)	HTR DISB/ENBL	Hooter (Enable or Disable)
FLSR DISB/ENBL	Flasher (Enable or Disable)	KBUZ ON / OFF	Key Buzz On/Off
SNOZ 0000	Snooze 0000	F-RL ON / OFF	Fail Relay ON/OFF
F-DL 0000	Set Delay For Fail Safe Relay	EVNT CLR	Event Clear
CLR YES/ NO	Clear YES/ NO	LDEN OFF/ ON/ BLIK	LED Enable OFF/ON/ BLINK

Note: ‘LED EN’ & ‘LED’ (AL1 & AL2) this feature is only present in **WP-JB90** Model.

D. OFFSET MENU

OFST PSWD	Enter Password to edit Offset settings
OFST +000	View or edit offset Parameter

E. CALIBRATION MENU

CAL PSWD	Enter Password to edit Calibration settings		
CALB	Calibration	CFEN	Calibration Fail Enable
CDEN	Calibration Due Enable	SREN	Sensor Replace Enable
PIDS	PID Status Enable		

For Oxygen/ Nitrogen Detector / Transmitter

SPAN LOW	Low span	SPAN HIGH	High Span
LOW 00.00	Set Low Span Cal* value	HIGH 20.90	Set High Span Cal* value
CAL HIGH	Calibration High	CAL LOW	Calibration Low
HIGH SUCC	High CAL* success	LOW SUCC	Low CAL* success
HIGH FAIL	High CAL* fail	LOW FAIL	Low CAL* fail
HIGH SKIP	High CAL* skip	LOW SKIP	Low CAL* skip
CAL ESC	Escape Calibration menu		
CFEN ENBL/DISB	CAL Fail Enable / Disable	CDEN ENBL/DISB	Calibration Due Enable / Disable
SREN ENBL/DISB	Sensor Replace Enable / Disable		

For Toxic / Combustible (Catalytic Pallister), PID Detector / Transmitter

SET SPAN	Set Span	CAL ESC	Escape Calibration menu
CAL SPAN	Apply Span gas	CAL ZERO	Apply zero gas
ZERO SUCC	Zero CAL* success	ZERO SUCC	Span CAL* success
ZERO FAIL	Zero CAL* fail	SPAN FAIL	Span CAL* fail
ZERO SKIP	Zero CAL* skip	SPAN SKIP	Span CAL* skip
CFEN ENBL/DISB	CAL FAIL Enable / Disable	CDEN ENAB/DISB	Calibration Due ENABLE / DISABLE
SREN ENBL/DISB	Sensor Replace Enable / Disable	PIDS ENAB/DISB	PID Status Enable / Disable
NOTE: PIDs option is shown only for PID sensor module.			

For NDIR (CH_4/CO_2/C_3H_8) Detector / Transmitter

SET SPAN	Set Span	CAL ESC	Escape Calibration menu
CAL SPAN	Apply Span gas	CAL ZERO	Apply zero gas
ZERO SUCC	Zero CAL* success	SPAN SUCC	Span CAL* success
ZERO FAIL	Zero CAL* fail	SPAN FAIL	Span CAL* fail
ZERO SKIP	Zero CAL* skip	SPAN SKIP	Span CAL* skip
ZGAS WAIT	ZERO GAS WAIT…	SGAS WAIT	SPAN GAS WAIT…

GAS INPT / OUT OFRG	GAS INPUT OUT OF RANGE	SPAN CNF	SPAN CONFIRM
CFEN ENBL/DISB	CAL FAIL ENABLE / DISABLE	CDEN ENBL/DISB	Calibration Due ENABLE / DISABLE
SREN ENBL/DISB	Sensor Replace ENABLE / DISABLE		

F. MIN-MAX (LOW/ HIGH) MENU

LOHI PSWD	Enter Password to edit LOW/HIGH settings
HI / LO CLR YES/NO	low-high Clear value of the Gas Yes/no

G. COMMUNICATION MENU

COMM PSWD	Enter Password to edit Communication settings		
COMM ID	Communication ID	ID 0001	ID change or view
COMM BAUD	Communication Baud rate	BAUD 9.6 /19.2/ 38.4/ 57.6	Baud rate 9600 /19200/ 38400/ 57600
COMM PRTY	Communication Parity bit	PRTY NONE	Parity none
PRTY ODD	Parity odd	PRTY EVEN	Parity Even
COMM STOP	Communication Stop bit	STOP ONE/TWO	Stop bit (one or two)
COMM TEST	Communication Test	COMM ESC YES/NO	Communication Esc from menu

H. OUTPUT MENU

For Current Output

OUT PSWD	Enter Password to edit Output settings		
OUT BACK	Back To Main Menu	OUT HIGH	Current Output for High Range Value
OUT LOW	Current Output for Low Range Value	1MA	1mA
3.7MA	3.8 mA	4MA	4mA
8MA	8mA	12MA	12mA
16MA	16mA	20MA	20mA
22MA	22mA	OUT SCAL DOWN/UP	Output Scale Down/Up
OUT INHB	Inhibit Mode for output	OUT LOOP	Output Loop

For 0-10V Voltage Output

OUT PSWD	Enter Password to edit Output settings		
OUT BACK	Back To Main Menu		
OUT HIGH	Voltage Output for Low Range Value	OUT LOW	Voltage Output for High Range Value

0V	0VOLT	2V	2VOLT
4V	4VOLT	6V	6VOLT
8V	8VOLT	10V	10VOLT
OUT SCAL DOWN/UP	Output Scale Down/Up		
OUTINHB	Inhibit Mode for output		

For 0-5V Voltage Output

OUT PSWD	Enter Password to edit Output settings		
OUT BACK	Back To Main Menu		
OUT HIGH	Voltage Output for Low Range Value	OUT LOW	Voltage Output for High Range Value
0V	0VOLT	1V	1VOLT
2V	2VOLT	3V	3VOLT
4V	4VOLT	5V	5VOLT
OUT SCAL DOWN/UP	Output Scale Down/Up		
OUT INHB	Inhibit Mode for output		

For 0-1V Voltage Output

OUT PSWD	Enter Password to edit Output settings		
OUT BACK	Back To Main Menu		
OUT HIGH	Voltage Output for Low Range Value	OUT LOW	Voltage Output for High Range Value
0MV	0mV	0.2V	200mV
0.4V	400mV	0.6V	600mV
0.8V	800mV	1V	1000mV
OUT SCAL DOWN/UP	Output Scale Down/Up		
OUT INHB	Inhibit Mode for output		

I. TEST MENU

TEST PSWD	Enter Password to edit Test settings
MIN 0000	Set min increment value for virtual gas concentration
0 TEST / H2S	Gas Concentration Value

J. BUMP TEST MENU

BUMP PSWD	Enter Password to operate Bump test settings
APLY GAS	Apply Gas

K. RANG-LOCK MENU

RLOK PSWD	Enter Password to edit Range-Lock settings
RLOK NO/YES	Range Lock NO/YES

L. RTC MENU

RTC PSWD	Enter Password to operate RTC MENU settings

RTC TIME	RTC TIME	TIME 00.00	Time
CHNG YES/NO	Time Change Yes / No	HOUR 00	Hour
MIN 00	Minute	SEC 00	Seconds
SAVE YES/NO	Save Yes /No	RTC DATE	Set date
DATE 00	Date	MNTH XXXX	Set month
YEAR 00	Set year		

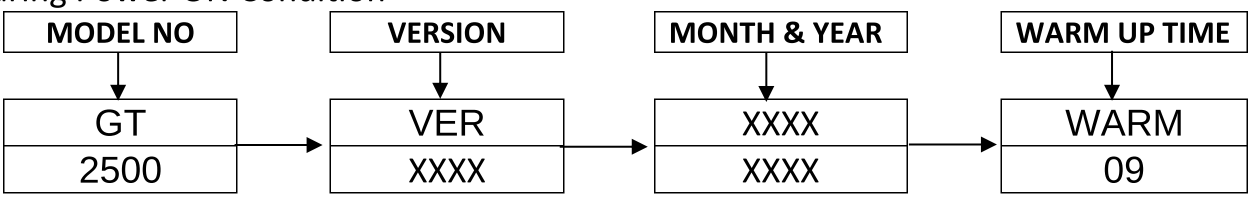
M. INFO MENU

INFO PSWD	Enter Password to operate ABOUT MENU settings		
INFO DVSR	Device serial No	DEV SRNO 12345678	Device serial No 12345678
INFO SNSR	Sensor serial No	SNSR 87654321	Sensor serial No 87654321
INFO VER	Software Version	SOFT VER VER XX.XX	Software Version VERSION XX.XX
INFO ESC	Back to main menu		

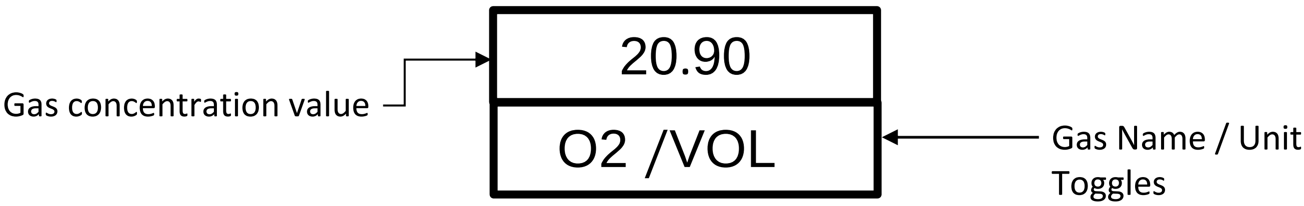
10 DISPLAY DETAILS

10.1 POWER ON INDICATION ON THE DISPLAY

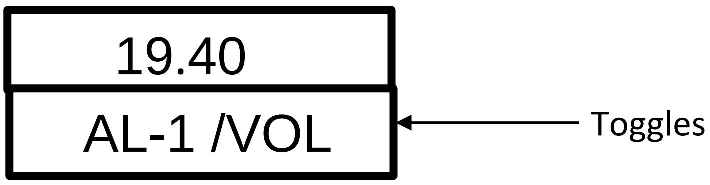
During Power ON Condition



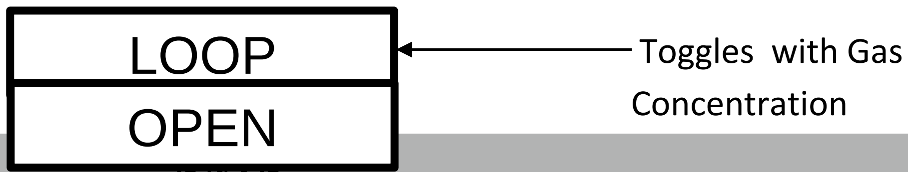
After warm up, the device switches to NWM* displaying the GC* and its units as shown in the below.



1) For Alarm condition



2) If Output Current connection is in open condition then display shows as below



NOTE: The values shown above are for representation purpose only.
As per Gas Unit, Gas Name, Gas Concentration value will be different.

10.2 SOME IMPORTANT INDICATIONS

Table 4

SNSR OPEN	Indicates sensor is disconnected
OVER RNGE	Indicates the GC* has exceeded its range of instrument
LOOP OPEN	Indicates that 4-20mA loop connection is open.

11 MENU OPERATION

To enter the programming mode, press set key in the play mode for about 5 sec. Once we enter the programming mode, display shows **MENU**. Below that we see **ESC MENU**. Here on pressing **SET KEY**, we exit the menu.

MENU
ESC

To view menu headers, press **INCR KEY,** we see the headers in following order.

MENU ESC → MENU CODE → MENU ALRM → MENU OFST → MENU CAL →
MENU LOHI → MENU COMM → MENU OUT → MENU TEST → MENU BUMP →
MENU RLOK → MENU RTC → MENU INFO

Below now we explain the options available in the different menu functions. To view the steps, refer to flowchart in the next section.

Table 5

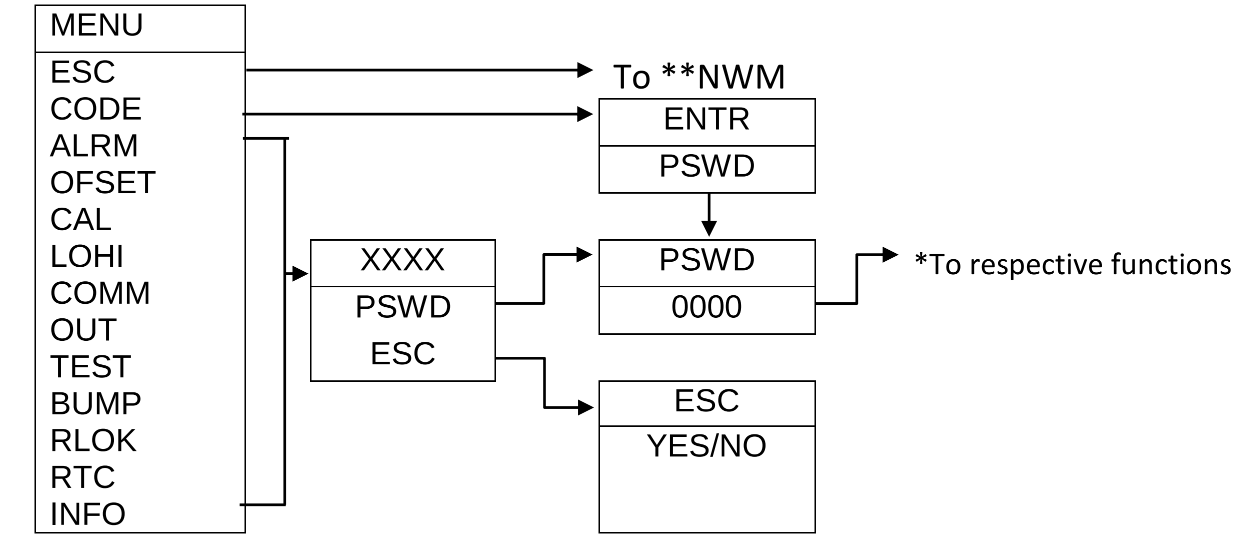
CODE MENU (Refer Flowchart)	This mode is used to change the user password used for making changes in the menus.
ALARM MENU (Refer Flowchart)	This mode is used to make the changes in the Alarm set points. There are two Alarms Alarm 1 and Alarm 2. The Alarm condition can be set to high or low in the alarm logic submenu and alarm value can be set in Alarm set point submenu as per the requirement. When the GC* exceeds the set point limits, the buzzer alerts the operator if enabled. AL1/AL2 LED is used to indicate the Alarm. Increment key is used to silence the buzzer. ALARM LED shall remain ON till the GC* comes back in set point limits for gas. Alarm menu also contains **STEL** and **TWA** settings for Toxic & VOC gases. Can change colour of RGB LED for different alarm condition
OFFSET MENU	This menu is used to adjust any errors due to drift/calibration by setting the offset in the GC* of up to +/-25% of the full scale. The menu won't allow user to set more than its limit. Enter and exit Offset menu shall be the similar Alarm menu.

CALIBRATION MENU	This menu is used to perform the calibration of the detector. This calibration must be performed by qualified personnel only. Ambetronics shall not be responsible for any changes done due to invalid procedure followed for calibration.
LOHI MENU	This menu is used to view the Low / High (min/max) value of GC*
COMMUNICATION MENU	This menu is used to set the device ID, baud rate, parity, stop bit and also test the RS-485 communication.
OUTPUT MENU	This menu is used to set/adjust (4-20mA) Current / (0-10V/0-5V/0-1V) Voltage output so that if any linearity issues present are aligned.
TEST MENU	This menu is used to check the electronics of the system by simulating the value of GC * for testing purpose.
BUMP TEST MENU	A Gas response check is periodically required & carry out, is often called as 'Bump Test'. This test is performed by using calibration gas required to apply to the sensor via Calibration Cap. Bump Test helps user to take decision about recalibration of detector.
RANGE LOCK MENU	This menu is used to lock the Gas Concentration at higher & lower range. 1) If RANGE LOCK is selected 'Yes' Whenever display value > Detector High range value, Display shows higher range value & Display value < Detector Low range value, Display shows lower range value. 2) If RANGE LOCK selected 'No' Whenever display value > Detector High range value, Display shows "OVER RNGE" & Display value < Detector Low range value, Display shows "SNSR OPEN".
RTC MENU	This menu is used to set date & time of the device
INFO (ABOUT) MENU	This menu displays information about device such as Device Serial No, Sensor Module Serial No and Device Software Version.

11.1 FLOWCHART

In the below flowchart below menu are available in the device. To scroll down in the list press '▲' key and to select the menu and save the setting, press SET key.

In all menus if incorrect password is entered we can enter the menu but we can't set any values. Exceptions are Password, Calibration and Bump test menu. In these menus if wrong password is entered then we can see the WRONG PASSWORD on the display and the unit will return to PASSWORD screen. If you want to ESC then press ▲ key to return to NWM*



* Note: Respective menu name will be seen in all menus except password menu

* Refer Appendix

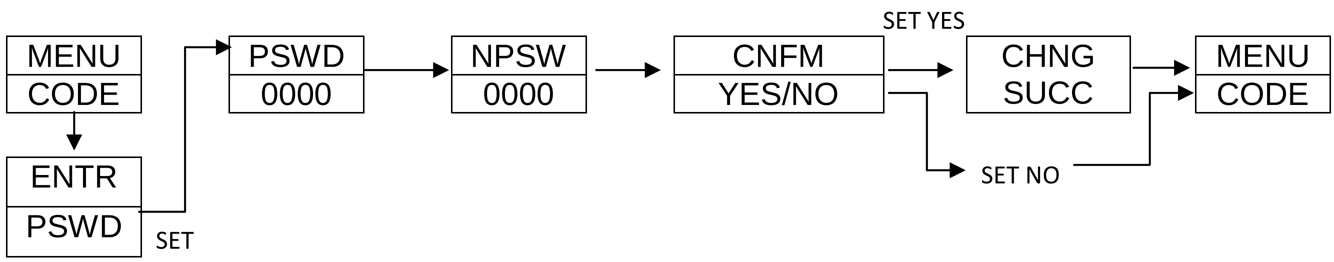
Time out Delay:

If no key is pressed for 2 minutes in any menu (except Calibration & Bump Test menu) the display will return to normal working mode.

However, in calibration menu time out is 10 minutes & in Bump test menu time out is 20 minutes.

11.2 CODE MENU / PASSWORD MENU

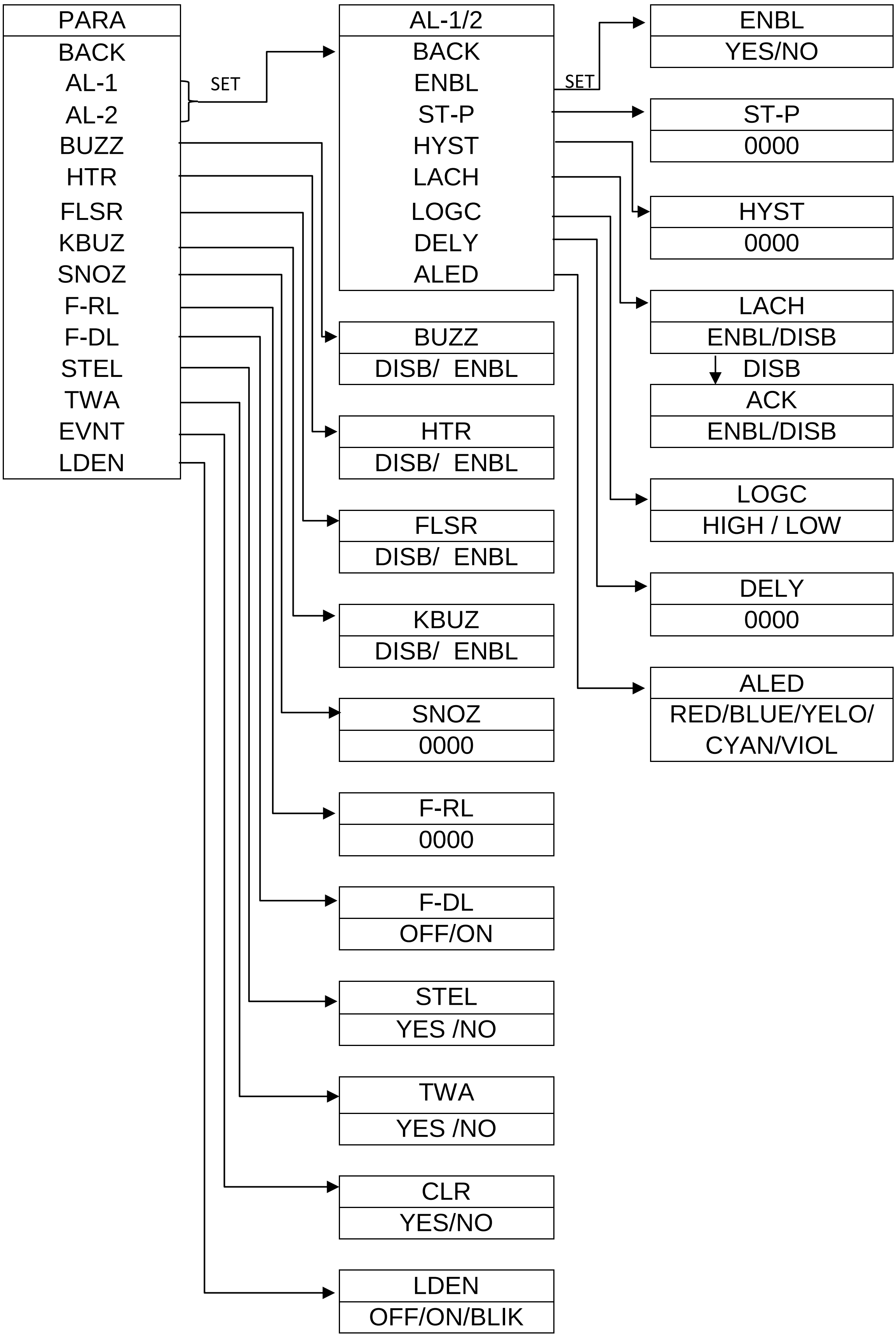
This menu is used to change the user password used for making changes in the menus. In main menu select code menu and enter correct password.Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ "Back"to go out of the setting parameter / menu. After entering the CODE MENU, the following will be displayed **Note: Default password is 0001**



11.3 ALARM MENU

This menu is used to set Alarm set points.

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu. After entering the ALARM MENU, the following will be displayed.



Note: 'LED EN' & 'LED' (AL1 & AL2) this feature is only present in **WP-JB90** Model.

Table 6

ALARM1 AND ALARM2	AL1/AL2's set point for GC * can be set. Upon violation of AL1/AL2 set points "AL1/AL2" indication on LCD display is seen as shown in display details and automatically disappear when GC * value comes with in set point limits. Settable Range: 0 to High Range of device. For Healthy condition it will show green color & for sensor open it will show blue color of RGB & During Alarm condition it operates as per user requirement.
BUZZER	Buzzer generates sound to alert the user for alarm violation.
HOOTER	Hooter generates sound to alert the user for alarm violation.
FLASHER	Flasher generates RED light to alert the user for alarm violation.
KEY BUZZ	Buzzer generates sound while pressing individual key
SNOOZE	Turns ON the alarm again after set seconds if the alarm condition still holds true. Settable range*:* 0 to 999 seconds. Snooze time starts after ACK key is pressed & Relay are acknowledged.
FAIL SAFE RELAY	• 'NORMALY-OFF': Fail safe relay will be 'OFF' in normal conditions and will be 'ON' if any fault occurs. Fault maybe OPEN /OVER. • 'NORMALY -ON': Failsafe relay will be 'ON' in normal conditions and will be 'OFF' if any fault occurs. Fault maybe OPEN /OVER.
FAIL-DELAY (Fail safe delay)	Failsafe delay setting is for failsafe relay. When fault such as Open/Over occurs, failsafe relay will be operated after set fail safe delay. Fail safe delay can be set from 0 to 999 sec.
HYSTERESIS	Hysteresis of up to 10% of the full scale range can be set. The Gas level may sometimes fluctuate during an Alarm condition, which causes repeated Alarm indications. To avoid repeated Alarms, Hysteresis is used. Hysteresis works with the High & Low Alarm SP* When Logic is High, Alarm is on when GC * > SP* & Alarm is turned off when GC * < (SP* - Hysteresis). When Logic is Low, Alarm is ON when GC * < SP* & Alarm is OFF when GC * is > (SP*+Hysteresis).

LATCH

Latch Disable: ACK Disable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Disable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
BEFORE ACK	BLINK	BLINK	ON	ON	ON	ON
AFTER ACK	BLINK	BLINK	ON	ON	OFF	ON

Latch Disable: ACK Disable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Disable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
NO ACK	OFF	OFF	OFF	OFF	OFF	OFF

Latch Disable: ACK Enable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Enable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
BEFORE ACK	BLINK	BLINK	ON	ON	ON	ON
AFTER ACK	BLINK	BLINK	OFF	OFF	OFF	ON

Latch Disable: ACK Enable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Enable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
NO ACK	OFF	OFF	OFF	OFF	OFF	OFF

Latch: Enable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch: Enable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
BEFORE ACK	BLINK	BLINK	ON	ON	ON	ON
AFTER ACK	STABLE	STABLE	OFF	OFF	OFF	ON

Latch: Enable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch: Enable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY	HOOTER	FLASHER
BEFORE ACK	BLINK	BLINK	ON	ON	OFF	ON
AFTER ACK	OFF	OFF	OFF	OFF	OFF	OFF

NOTE: Acknowledgement is compulsory in Latch: Enable condition

LATCH (FOR WP & JB90-FLP MODEL)

Latch Disable: ACK Disable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Disable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
BEFORE ACK	BLINK	BLINK	ON	ON
AFTER ACK	BLINK	BLINK	ON	ON

Latch Disable: ACK Disable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Disable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
NO ACK	OFF	OFF	OFF	OFF

Latch Disable: ACK Enable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Enable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
BEFORE ACK	BLINK	BLINK	ON	ON
AFTER ACK	BLINK	BLINK	OFF	OFF

Latch Disable: ACK Enable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch Disable: ACK Enable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
NO ACK	OFF	OFF	OFF	OFF

Latch: Enable **LOGIC: HIGH** **GC > AL1 or AL2 or Both SET POINTS**	**OR**	**Latch: Enable** **LOGIC: LOW** **GC < AL1 or AL2 or Both SET POINTS**

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
BEFORE ACK	BLINK	BLINK	ON	ON
AFTER ACK	STABLE	STABLE	OFF	OFF

Latch: Enable **LOGIC: HIGH** **GC < AL1 or AL2 or Both SET POINTS**	**OR**	**Latch: Enable** **LOGIC: LOW** **GC > AL1 or AL2 or Both SET POINTS**

NOTE: Gas value crossed set points & reduced below set point. No acknowledgement was done during this condition.

CONDITION	AL1 LED	AL2 LED	AL1 RELAY	AL2 RELAY
BEFORE ACK	BLINK	BLINK	ON	ON
AFTER ACK	OFF	OFF	OFF	OFF

NOTE: Acknowledgement is compulsory in Latch: Enable condition

LOGIC	If Logic is High for set point value AL1/AL2 then particular alarm indication appears on display if GC * value > set point value. If Logic is Low for set point value AL1/AL2, then particular alarm indication appears on display if GC * value < set point value.
STEL & TWA	Short Term Exposure Limits (STEL) and Time Weighted Average (TWA) These settings are only for Toxic & VOC gases to warn that user is exposed for more than 15 minutes or 8 hours respectively.
EVENT CLEAR	This setting will clean the event values such as Alarm 1/ Alarm 2 ON Time & Date, OFF Time & date, Gas value recorded in Alarm 1/ Alarm 2 Low/High Logic value. Refer status menu on Page No. 30
LED ENABLE ON/OFF/BLINK	With help of this setting user can change setting of RGB LED to ON/OFF/BLINK.

11.4 OFFSET MENU

To adjust any errors due to drift/calibration can be done by setting an Offset.
An offset of maximum ±25% of full scale value / range can be set.
Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu.

After entering the OFFSET MENU, the following will be displayed.

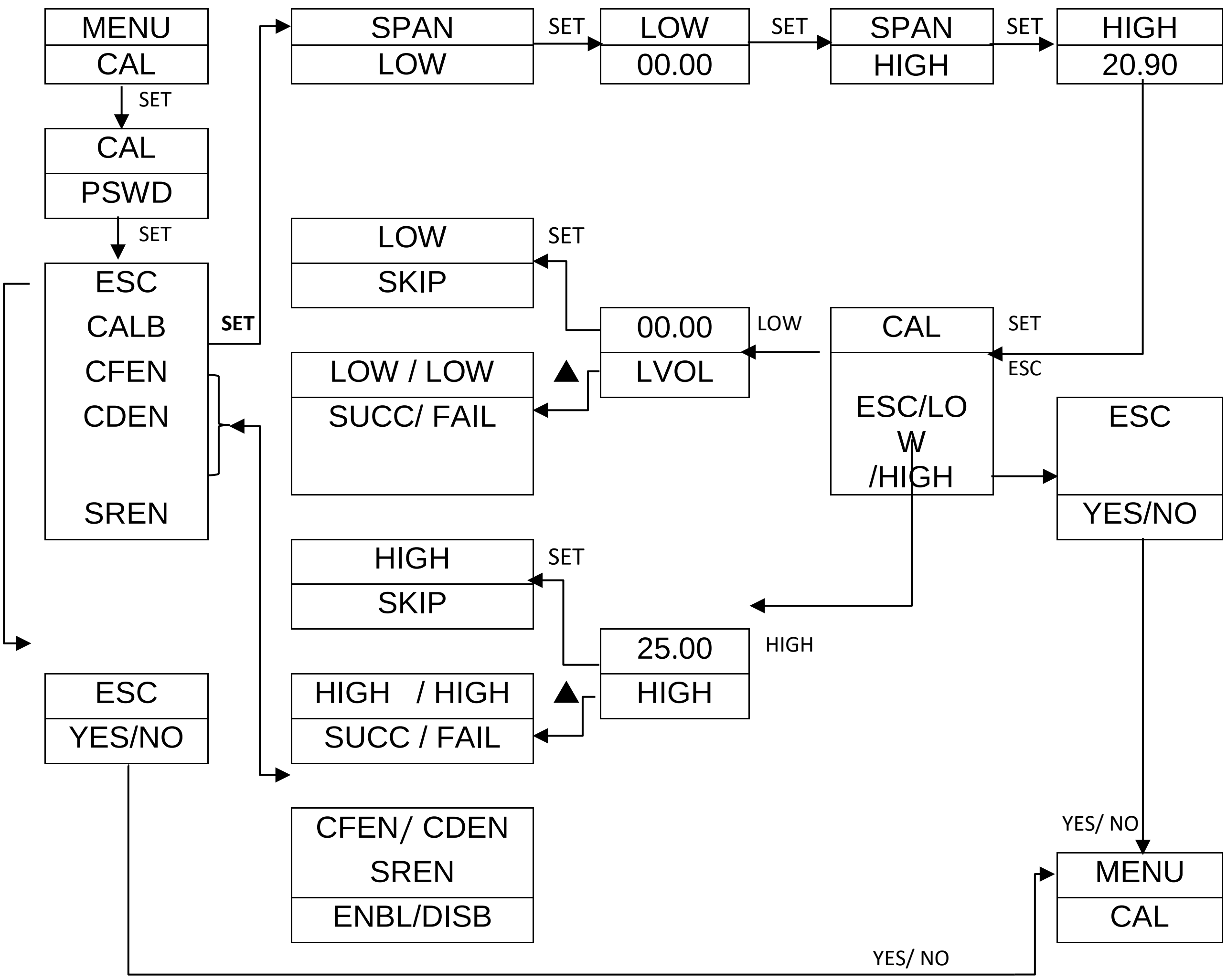
OFST
±1000

11.5 CALIBRATION MENU

Before initial calibration, allow the detector to stabilize as per the warm up time after applying power. To calibrate the detector, use an appropriate span calibration gas cylinder, constant Gas flow regulator & Ambetronics calibration cap & user manual for calibration procedure.

Press set key to enter the menu and set/save the parameter. Use '▲' key to select parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ "BACK" to go out of the setting parameter / menu. After entering the password, the following menu will be displayed.

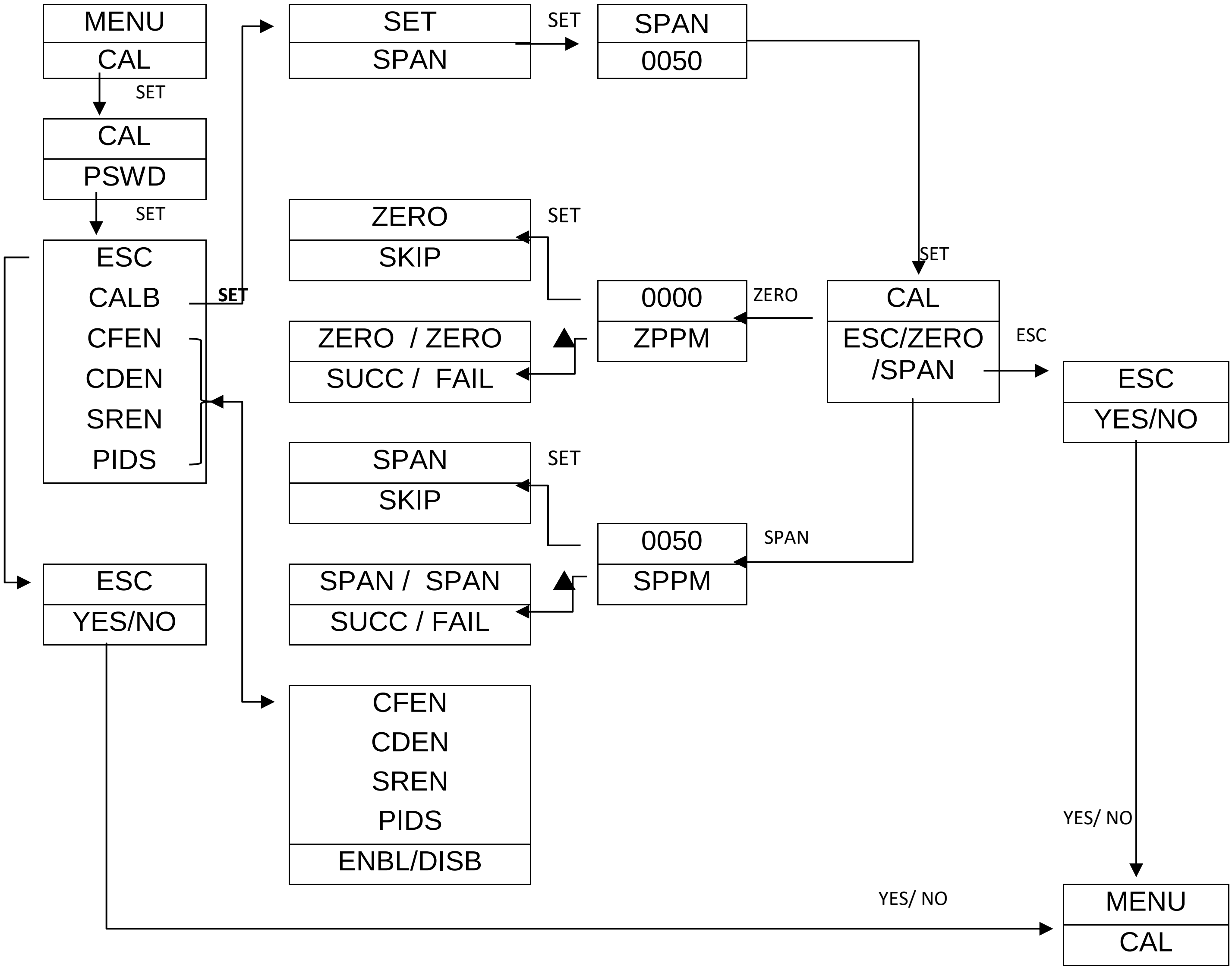
For Oxygen / Nitrogen Detector/ Transmitter:



After cal success/fail, Display return to CAL PSWD

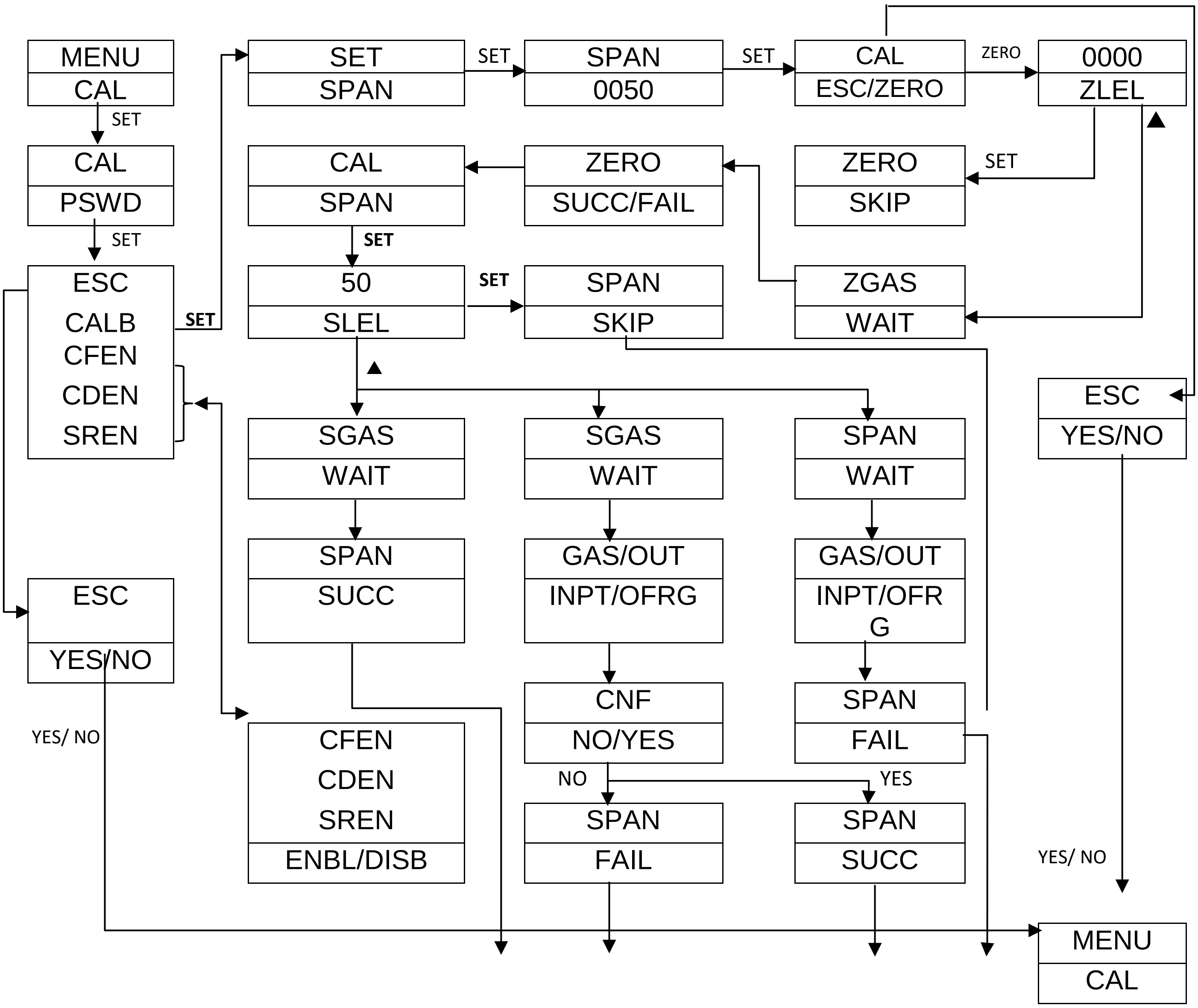
As per Gas Unit, Gas Name, Gas Concentration value will be different.

For Toxic, Combustible & PID Detector/ Transmitter:



After cal success/fail, Display return to CAL MENU
As per Gas Unit, Gas Name, Gas Concentration value will be different.

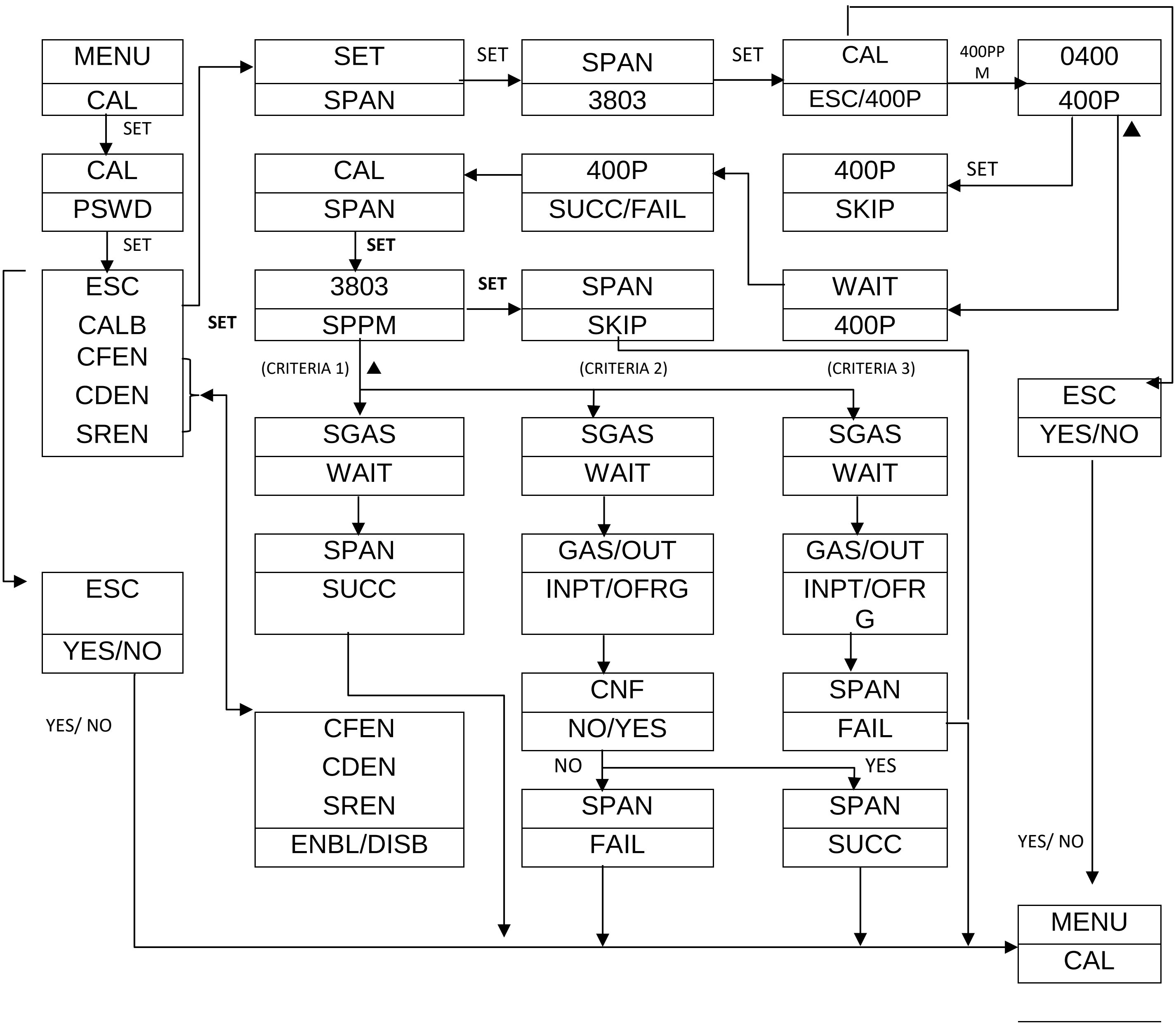
For NDIR (CH_4/C_3H_8) Detector/ Transmitter:



After cal success/fail, Display return to CAL MENU

As per Gas Unit, Gas Name, Gas Concentration value will be different.

For NDIR (CO_2) Detector/ Transmitter:



After cal success/fail, Display return to CAL MENU
As per Gas Unit, Gas Name, Gas Concentration value will be different.

Table 7

LOW/ZERO SKIP	:	Skips the Low/Zero calibration, when set key is pressed while calibration.
HIGH /SPAN SKIP	:	Skips the High/Span calibration, when set key is pressed while calibration.
LOW /ZERO SUCCESS or LOW/ ZERO FAIL	:	} This message is displayed to inform the status of the calibration whether the particular calibration is done successfully or has failed.
HIGH /SPAN SUCCESS or	:	

HIGH / SPAN FAIL		
CALIBRATION FAIL ENABLE (CFEN)	:	Calibration Fail message shown on main display can be enable or disabled through here.
CALIBRATION DUE ENABLE (CDEN)	:	Calibration Due message shown on main display can be enable or disabled through here.
SENSOR REPLACE ENABLE (SREN)	:	Sensor Replace message shown on main display can be enable or disabled through here.
PID STATUS ENABLE (PIDS)	:	PID status message shown on main display can be enable or disabled through here. **Note**: PIDS option is shown only for PID sensor module.

Table 8

Calibration STEPS & Policy:			
HIGH/SPAN	**LOW/ZERO**	**Status**	**REMARKS**
Success	Success	CAL* success	Unit will work as per new CAL* data
Success	fail	CAL* fail	Unit will work as per previous CAL* data
Fail	X	CAL* fail	Unit will work as per previous CAL* data
Success	X	LOW/ZERO CAL* not done	Unit will work with new GAS SPAN & old LOW SPAN data
X	Success	HIGH/SPAN CAL* not done	Unit will work with old GAS SPAN & new LOW SPAN data
X	Fail	CAL* fail	Unit will work as per previous CAL* data
X	X	CAL* not done	Unit will work as per previous CAL* data

NOTE:

1) After **'Span / High Cal Success / Fail'** Display return to 'Calibration Password'
2) After **'Zero / Low Cal Fail'** Display return to 'Calibration Password'
3) For PID sensor (Range: 4000 PPM), use Gas in 3500 to 4000 PPM Range for doing SPAN calibration.

11.5.1 CALIBRATION INSTRUCTION FOR OXYGEN / NITROGEN ANALYZER / DETECTOR/ TRANSMITTER

For Low calibration use: Set 'Low Cal' between 0% V/V to 5 % V/V

- For 0 % V/V use Pure Nitrogen gas **(99.999%V/V [5.0 Grade], Moisture & Oxygen Level <2 PPM & CO+ CO_2 level < 0.5PPM & T.H.C. < 0.2 PPM & other components should be nil.)**

- 1 % V/V to 5 % V/V use Oxygen gas Balance Nitrogen
  (**Use O2 Gas for accurate linearity**)

For High Calibration use: **Set 'High Cal' between 18% V/V to 23 % V/V**

- Normally set 20.9 % V/V & for 20.9 % V/V
  Use **Ambient fresh air** or **Compressed Air Cylinder**
  (20.9 % V/V, O_2 Balance Nitrogen) as calibration gas for **'High Cal'.**
- OR use 18 % V/V to 23 % V/V Oxygen gas Balance Nitrogen as calibration gas for **'High Cal'.**
- Regulator flow Rate = 0.5 LPM for Low & High Calibration.
- Low / High calibration can be skipped.
- For Oxygen, Nitrogen Detector warm up time is 2 hours.

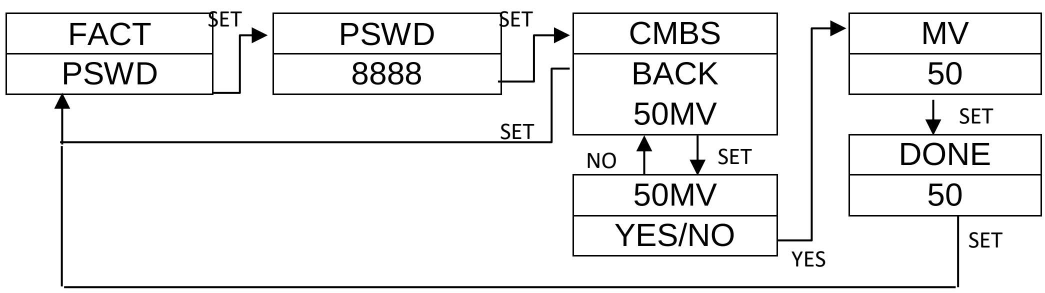
11.5.2 CALIBRATION INSTRUCTION FOR TOXIC, PID, COMBUSTIBLE CATALYTIC OR PELLISTOR, NDIR- CH_4 , NDIR- C_3H_8 DETECTOR / TRANSMITTER.

Zero millivolt adjustment menu for combustible sensor

❖ When increment key is pressed for 15 seconds, device will go into factory menu. Enter 8888 as password to open zero millivolt adjustment menu for combustible sensor

❖ For combustible gas type setting to adjust zero offset millivolts (mV) of sensor
For NE-C2 sensor, mV will auto adjust between 50-100mV

For KC-C1 sensor, mV will auto adjust between 50-150mV



ZERO CALIBRATION:

Compressed Air Cylinder (20.9 % V/V, O_2 Balance Nitrogen) should be used to perform the Zero calibration if the surrounding area contains any residual amount of Target Gas.

If no residual gas is present then atmospheric background Ambient fresh air can be used to perform the Zero Calibration.

SPAN CALIBRATION:

Use Target gas concentration with balance air ¼ th or ½ of Target gas Detector range.

- Regulator flow Rate = 0.5 LPM for Zero & Span Calibration.
- For Toxic / PID / Combustible Detectors Zero / Span calibration can be skipped.
- For NDIR-CH4 & C3H8, Zero calibration is recommended & cannot be skipped.

- For NDIR-CH4 & C3H8, warm up time is 5 minutes.

11.5.3 IMPORTANT NOTE FOR TOXIC GAS ANALYZER / DETECTOR/ TRANSMITTER

- Use Surrogate gas for specified Toxic gas Detector as recommended by manufacturer or refer calibration & Test report for factor.
- For ETO/NO Detector warm up time is 1 day & for other detectors warm up time is 2 hours.

11.5.4 IMPORTANT NOTE FOR COMBUSTIBLE GAS ANALYZER / DETECTOR/ TRANSMITTER

For Combustible Catalytic / Pellistor gas detector other than Methane / LPG / Hydrogen, Other Combustible Gas detector are calibrated with methane & factors for those gases are mentioned in the Calibration & Test report.
For Combustible Gas Detector warm up time is 1 hour.

11.5.5 IMPORTANT NOTE FOR PID DETECTOR / TRANSMITTER

- All VOCs are available in PID detection principle in PPM ranges.
- PID detector will be provided by calibration with Isobutylene gas.
- In PID detector, VOC other than Isobutylene is calibrated with Isobutylene gas by Setting VOC correction factor.
- In Calibration Report, VOC factor with respect to Isobutylene gas will be Mentioned.
- Detection value of VOC = Isobutylene gas concentration value x factor.
- For PID Detector warm up time is 1 hour.
- While Calibration of PID Detector, Ensure environment should be free from VOC or other Gases.

11.5.6 CALIBRATION INSTRUCTION FOR NDIR- CO_2 DETECTOR / TRANSMITTER

ZERO CALIBRATION:

Use Pure Nitrogen gas **(99.999%V/V Moisture & Oxygen Level <2 PPM & CO+ CO_2 level < 0.5PPM & T.H.C. < 0.2 PPM & other components should be nil.)**

SPAN CALIBRATION:

Use CO_2 gas concentration with balance Nitrogen ¼ th or ½ of CO2 gas range.
Regulator flow Rate = 0.5 LPM for Zero & Span Calibration
For NDIR- CO_2, Zero calibration is recommended & cannot be skipped.
For NDIR-CO_2, warm up time is 5 minutes

11.5.7 STANDARD CALIBRATION SET UP OF FLAMEPROOF MODEL

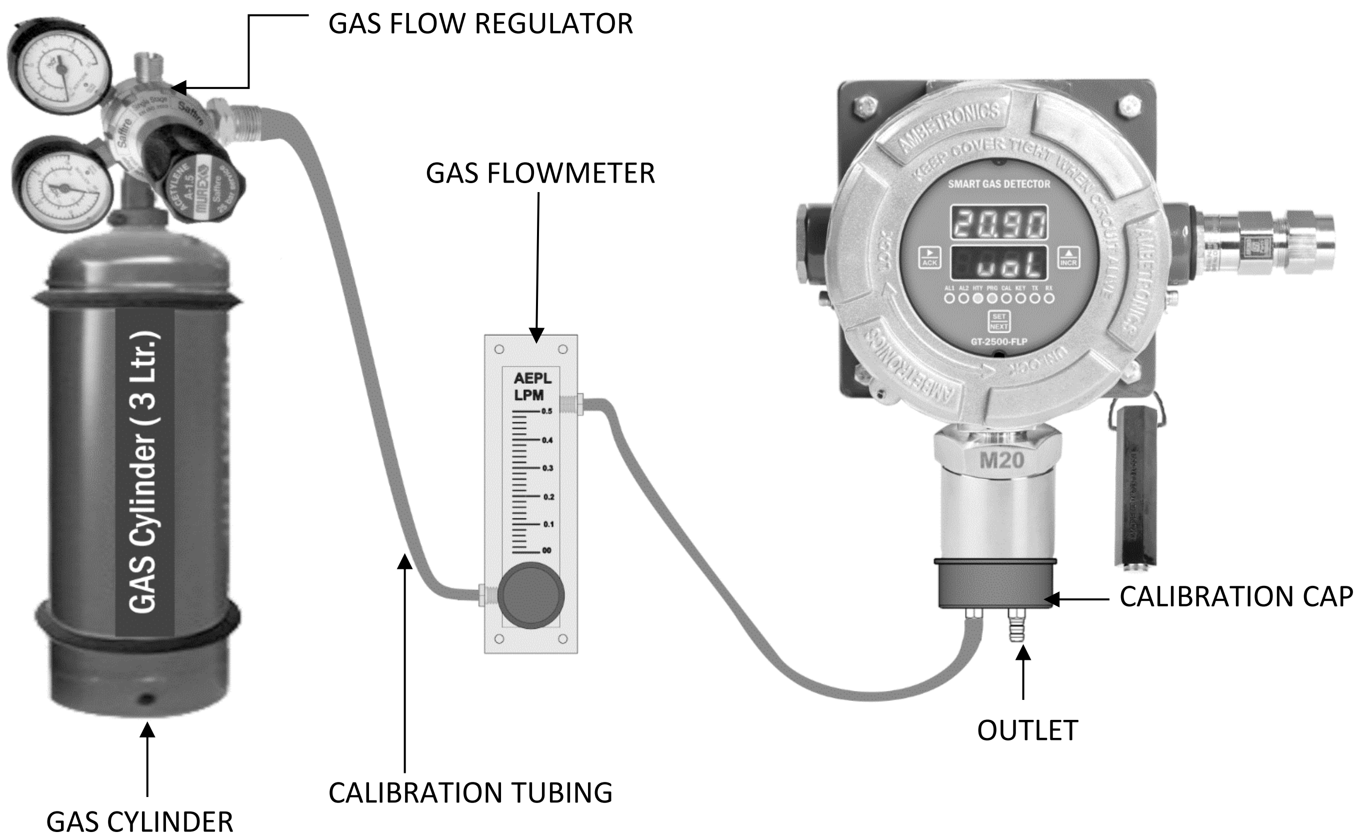


Figure 31

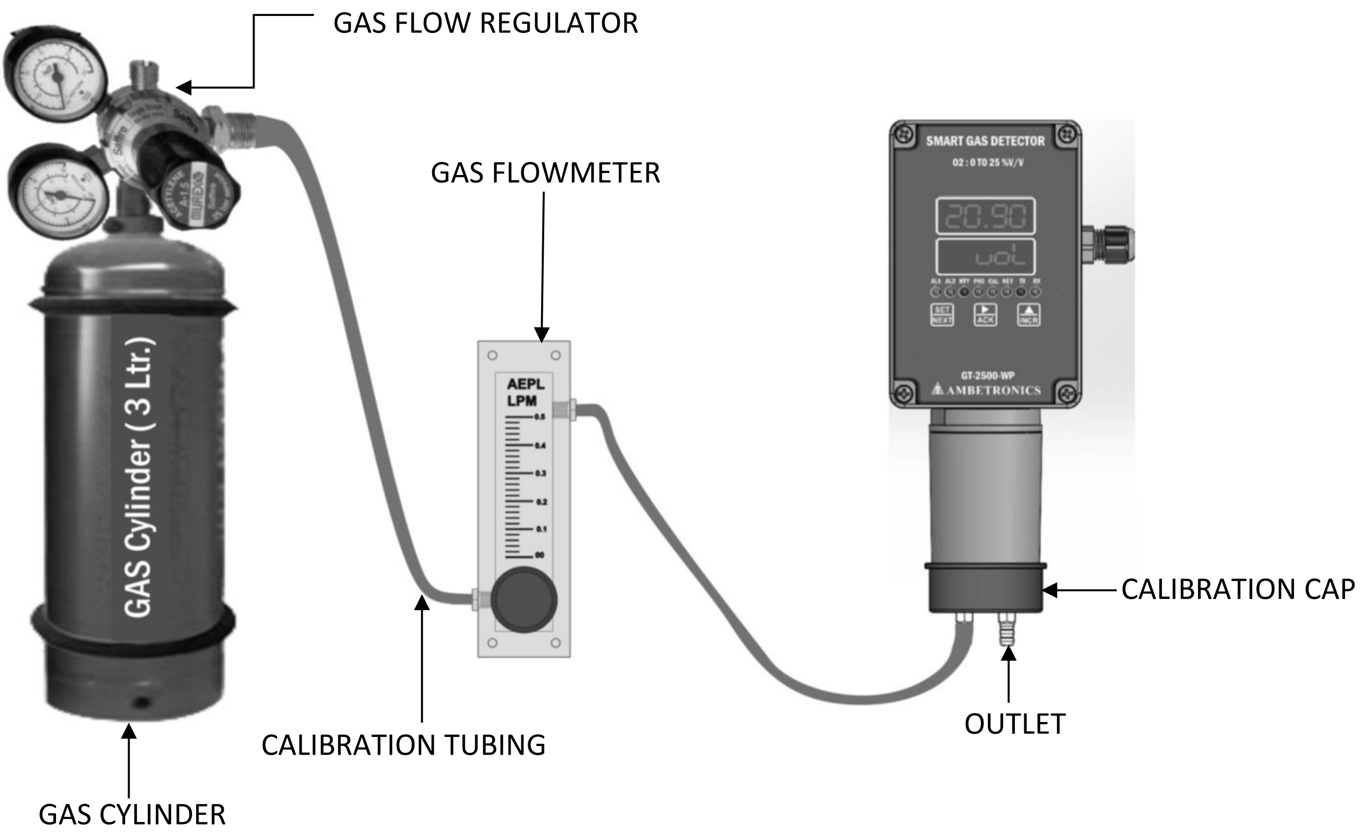


Figure 32

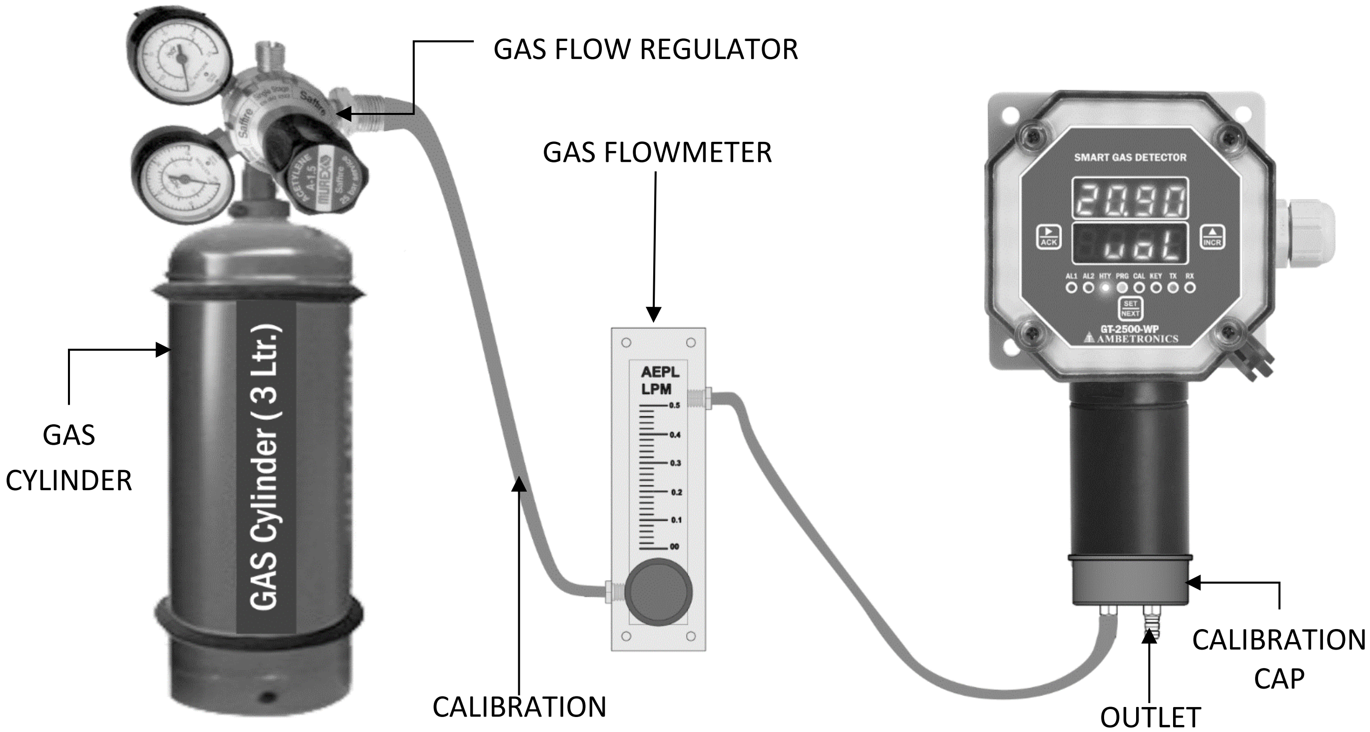


Figure 33

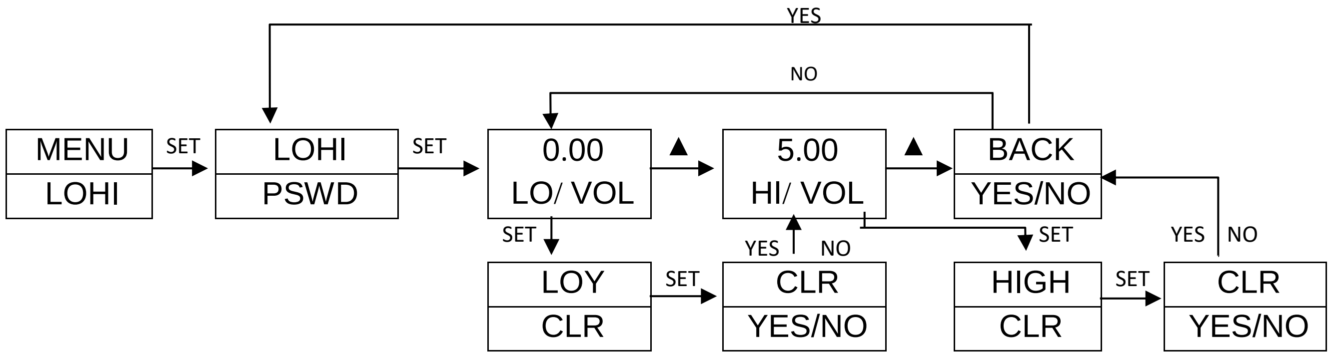
STEPS FOR PREPARATION OF CALIBRATION SET UP:

1) Arrange all require component like Calibration Gas Cylinder with Gas flow Regulator, Calibration Cap, Calibration Tubing, and Detector to be calibrated & connect as shown in Calibration set up.
2) Keep Calibration tubing length as short as possible.
3) While connecting tubing use short piece of rubber tube.
4) Before starting calibration, ensure Calibration Cap, Calibration Tubing, are connected properly to avoid leakage.
5) Use soap water to observe leakage.
   Pour soap water over joints. If leakage is there, bubbles will be seen & if leakage is not there, bubble will not be seen.
   Use Teflon tape between joints to avoid leakage.
6) After ensuring leakage is not found , open Calibration Gas Cylinder & set flow rate as recommended & connect Calibration tubing to Detector to be calibrated.
7) For Toxic corrosive gas such as CL2, HCl, H2S, SO2, VOC, NH3, HF, NO2 etc. Use Teflon tubing or recommended by manufacturer.
8) For Combustible gas /NDIR-CH4 /NDIR-CO2 use normal Tygon tubing or recommended by manufacturer.
9) Follow the calibration procedure mentioned in Operator / User manual.
10) For Zero / Low calibration, Apply Gas maximum 2 min or up to Stabilisation of reading & save Zero / Low calibration as per Calibration procedure.
11) For Span/ High calibration, Apply Gas maximum 2 min or up to Stabilisation of reading & save Span/ High calibration as per Calibration procedure.
12) For any assistance contact factory.

11.6 LOHI MENU (MIN-MAX MENU)

This menu is used to view Low/High (Min/Max) of the Gas Concentration. This value can be cleared to by selecting YES.

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu. After entering the password, the following menu will be displayed



Min-Max value for oxygen as an example is given. As per Gas Unit, Gas Name, Min-Max value will be different.

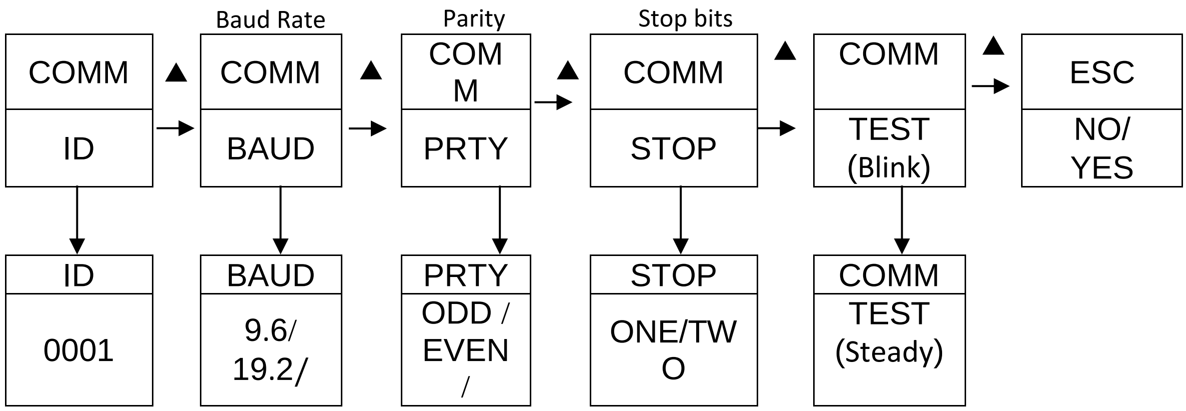
NOTE:

1. **This menu is used to see Low peak & High peak of Gas Concentration during operation. After Power On Min & Max value of Gas Concentration are updated after 1 minute only once. After that you can clear the Min & Max value of Gas Concentration as per your requirement & will be updated in online operation as per Low Peak & High peak of Gas concentration.**
2. **Minimum Value is only for Oxygen Gases and for Toxic Gases.**
3. **Maximum value is for Oxygen Gases and for Toxic Gases & Combustible Gases.**

11.7 COMMUNICATION MENU

It is used to set Serial Communication parameters to communicate with remote terminal/ PC.

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu. After entering the COMMNUNICATION MENU, the following will be displayed



	38.4/ 57.6	NONE /		

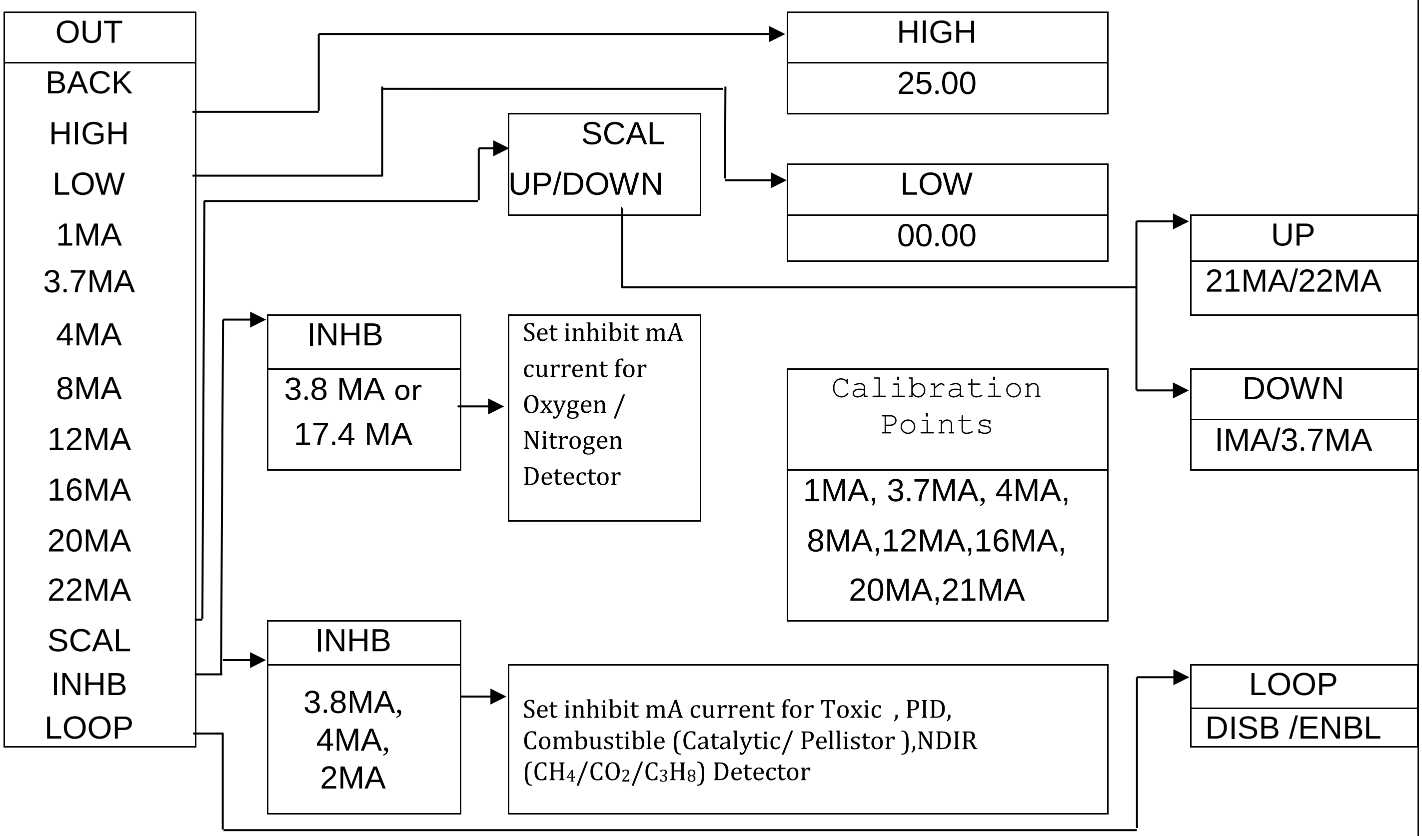
Table 9

DEVICE ID	:	ID can be set from 1 to 250 as per the requirement
BAUD RATE	:	This value can be set at 9.6 / 19.2/38.4/57.6kbps.
PARITY BIT	:	This is the parity bit ODD, EVEN & NONE can be set. Same setting is to be done in computer software also.
STOP BITS	:	The stop bits indicate the end of data string; selection can be done as 1/ 2 bits. It is usually set ONE
DATA BITS	:	Data bits are not shown but should be consider as 8.
TEST	:	When 'Test' is selected 'Test" on display get steady & "**Ambetronics Engineers Pvt Ltd"** on Hyper-Terminal will be displayed.

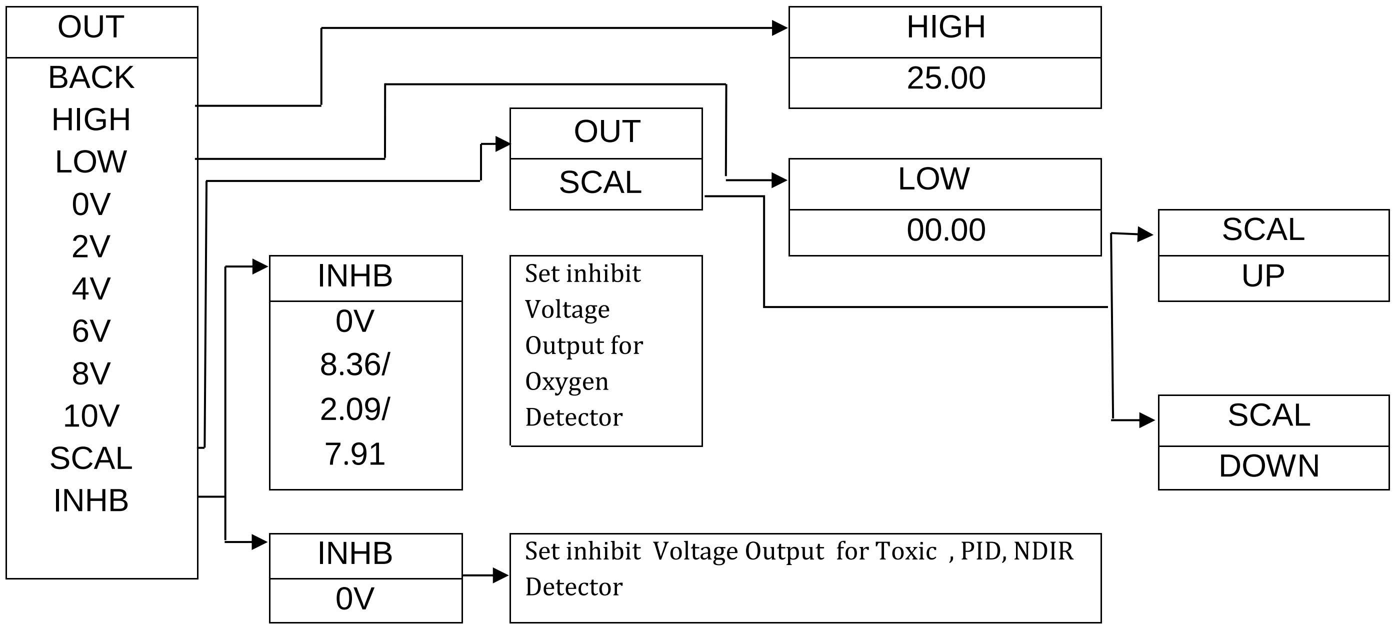
11.8 OUTPUT MENU

This menu is used to set output range & current output of Parameter. The current output must be checked with the Multimeter.

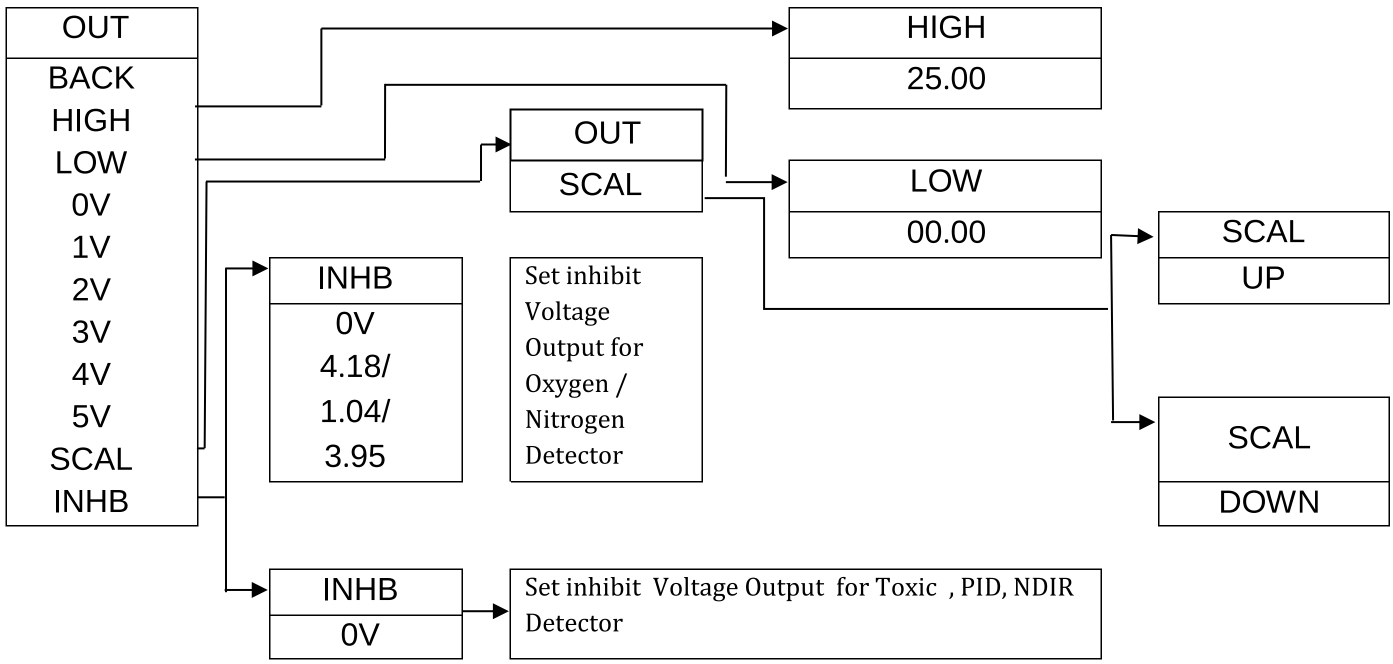
Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu. After entering the OUTPUT MENU, the following will be displayed.



For 0-10 Voltage Output



For 0-5 Voltage Output



For 0-1 Voltage Output

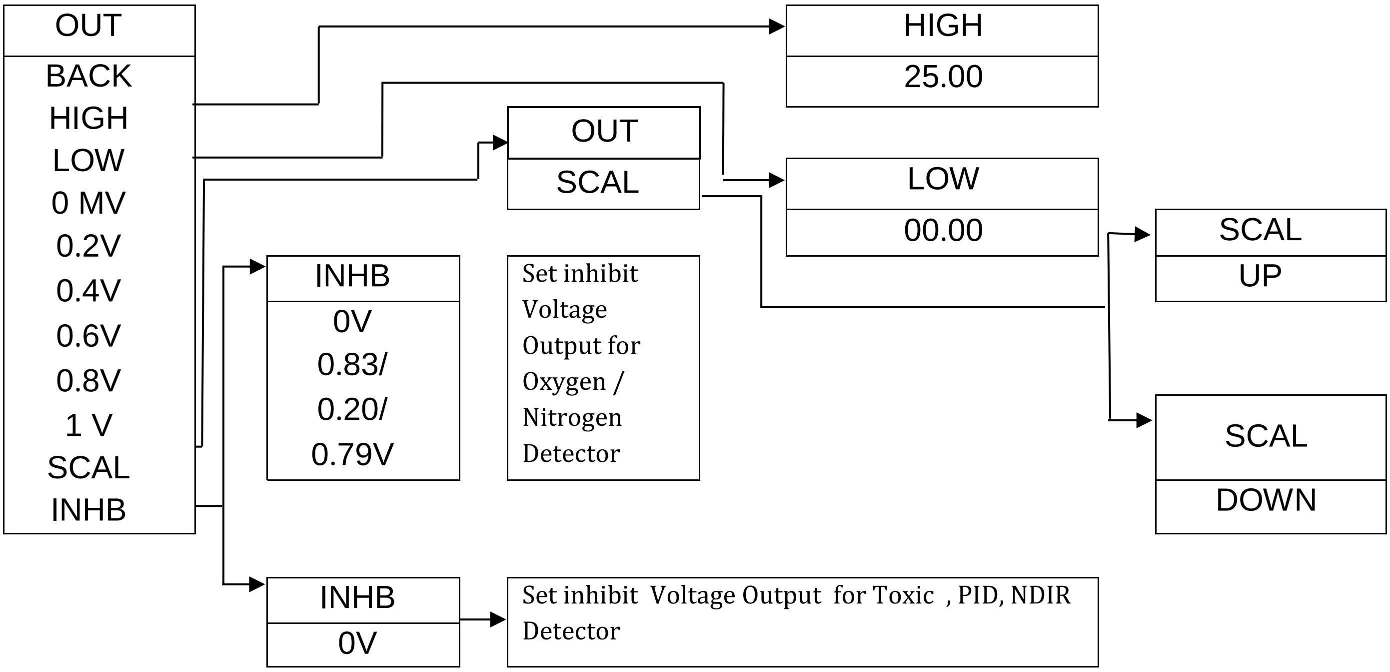


Table 10

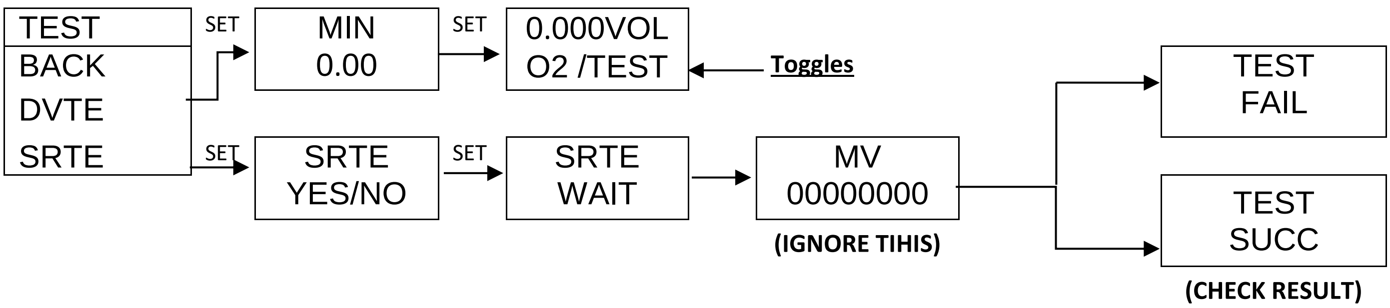
HIGH (High)	:	Set High range of Detector for which 20mA & 10V/5V/1V output is required.
LOW (Low)	:	Set Low range of Detector for which 4mA & 0V output is required.
SCALE	:	This menu is used to set scale of the output Current / Voltage in fault condition, which may be “Sensor Open/Over Range”. Upscale Current/ Voltage = 21 mA/22mA & 10V/5V/1V Down scale Current/ Voltage = 1mA/3.7 mA & 0V
Current/ Voltge calibration point	:	This calibration point is used to set Current / Voltage output. For this Digital multimeter is required to connect to Detector/ Transmitter
INHB	:	This mode is required for servicing & programming other parameters in detector setting to avoid false alarm & to give information about servicing & setting of parameters are going on. • Inhibit mode current is adjustable & user selective Oxygen for 25% V/V range : 3.8mA / 17.4mA Oxygen for 100% V/V range : 3.8mA / 7.34mA Nitrogen for 100% V/V range : 3.8mA/ 16.656mA Toxic / combustible / PID /NDIR : 2mA / 3.8mA / 4mA • Inhibit mode for 0-10V voltage output is adjustable & user selective Oxygen for 25% V/V range : 0V / 8.36V Oxygen/N2 for 100% V/V range : 0V/ 2.09V Nitrogen for 100% V/V range : 0V/7.91V Toxic / combustible / PID /NDIR : 0V

		• Inhibit mode for 0-5V voltage output is adjustable & user selective Oxygen for 25% V/V range : 0V / 4.18V Oxygen/N2 for 100% V/V range : 0V/ 1.045V Nitrogen for 100% V/V range : 0V/3.955V Toxic / combustible / PID /NDIR : 0V • Inhibit mode for 0-1V voltage output is adjustable & user selective Oxygen for 25% V/V range : 0V / 836mV Oxygen/N2 for 100% V/V range : 0V/ 209mV Nitrogen for 100% V/V range : 0V/791mV Toxic / combustible / PID /NDIR : 0mV *** Note: Refer last point of important notes for more details**
LOOP	:	While this option is enabled if 4-20mA Loop connection is broken or gets disconnected. mA Loop Open message will display on screen.

11.9 TEST MENU

This menu is used to check the electronics of the system by simulating the value of the virtual Gas Concentration for testing purpose.

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ "Back" to go out of the setting parameter / menu. After entering the TEST MENU, the following will be displayed.



DESCRIPTION

TEST	**TEST MODE**
DVTE	**DEVICE TEST MODE**
SRTE	**SENSOR TEST MODE**
MIN	**MINIMUM GAS INCREMENT VALUE**
TEST FAIL	**SENSOR TEST FAIL**
TEST SUCC	**SENSOR TEST SUCCESS**

DEVICE TEST MODE

Set Minimum Value of the Virtual Gas concentration manually to check working of Alarm settings, Indication for Alarms & current output with respect to virtual gas concentration.

Test mode operation: In Test mode if increment in virtual GC* is done above High range 'Over range' will be displayed & corresponding current output will be as per upscale/downscale.

In Test mode if Decrement in virtual GC* is done below Low range 'Sensor Open' will be displayed & corresponding current output will be as per upscale/downscale

SENSOR TEST MODE

1. This mode is used to check the condition of sensor Toxic sensor.
2. Oxygen, Nitric oxide (NO) & ethylene oxide (ETO) cannot be checked with this mode.
3. If sensor test succeeds then sensor is ok.
4. If sensor test fails then sensor needs to be replaced.

⚠ **Warning:**

Use 'Test mode' only to check the device is working properly or not. That is 4-20mA output, Relay, LED Indication are working properly without connecting the sensor to device just by incrementing and decrementing the virtual GC* with the help of '▲' &'▶' keys. Do not use 'Test mode' when the detector is completely installed on the site. For the operation take permission from your senior authority; otherwise it may be very dangerous.

11.10 BUMP TEST MENU

NOTE: A Gas response check is periodically required & carried out is often called as 'Bump Test'. This test is performed by using Calibration Gas required to apply to the sensor via calibration cap. Bump test helps user to take decision about calibration required for detector.

Press SET key to enter the menu and set the parameter. Use '▲' key to select parameters. Use '▲' & '▶' keys to edit the parameter value. Use "Back" to go to the previous menu or setting and use "ESC"/ "BACK" to go out of the setting parameter / menu.

After entering the password, the following menu will be displayed.

APLY	→	0000		
GAS	→	VOL / BUMP	←	TOGGLES

When 'Apply Gas' will be displayed, apply particular calibration gas for one minute & Corresponding Gas Concentration will be displayed.

During Bump Test current output will be as per Set current in inhibit mode.

⚠ **Warning:**

After Gas Concentration observation, do not exit directly from 'Bump Test' menu.

Ensure that Applied Gas Concentration reaches within safe limit to avoid false alarm while exit.

11.11 RANGE LOCK MENU

This menu is used to lock the Gas Concentration at Higher & Lower range.

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu.

After entering RANGE LOCK MENU, below setting will be displayed.

RLOK
YES/NO

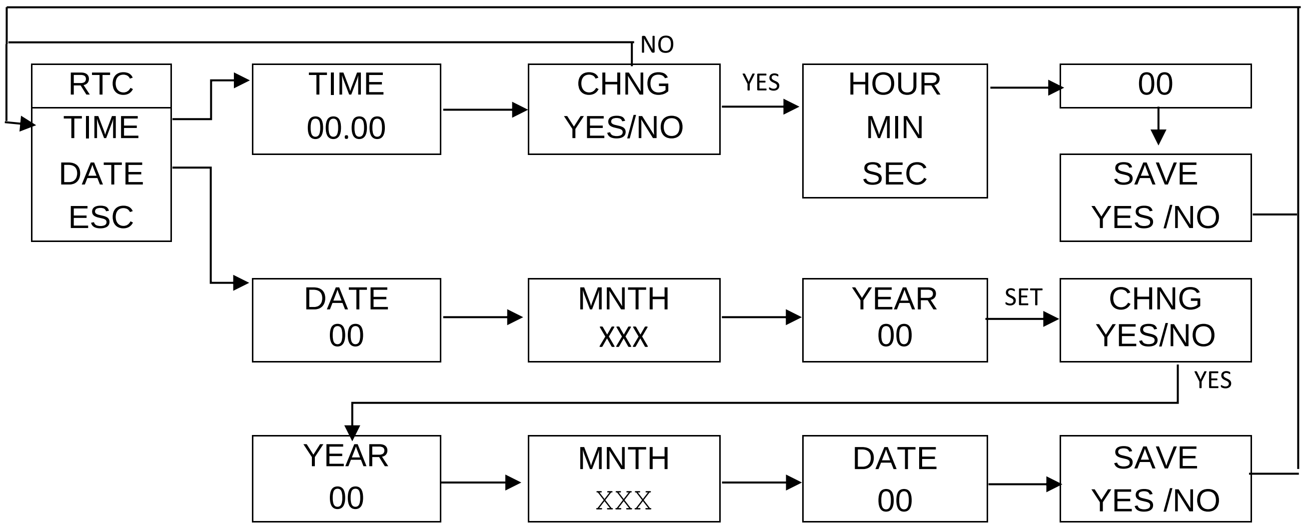
This menu is used to lock the Gas Concentration at higher & lower range.

1) If RANGE LOCK is selected 'Yes'
   Whenever display value > Detector High range value, Display shows higher range value & Display value < Detector Low range value, Display shows lower range value.
2) If RANGE LOCK selected 'No'
   Whenever display value > Detector High range value, Display shows "OVER RNGE" & Display value < Detector Low range value, Display shows "SNSR OPEN".

11.12 RTC MENU

This menu is used to set RTC (Real Time Clock).

Press SET key to enter the menu and set/save the parameter. Use '▲' key to select parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ " Back "to go out of the setting parameter / menu. After entering the password, the following menu will be displayed.



11.13 INFO (ABOUT) MENU

This menu displays information about device.

Press **SET** key to enter the menu and set/save the parameter. Use '▲' key to select menu parameters. Use '▲' & '▶' keys to edit the parameter value. Use "BACK" to go to the previous menu or setting and use "ESC"/ "Back" to go out of the setting parameter / menu. After entering the ABOUT MENU, the following will be displayed.

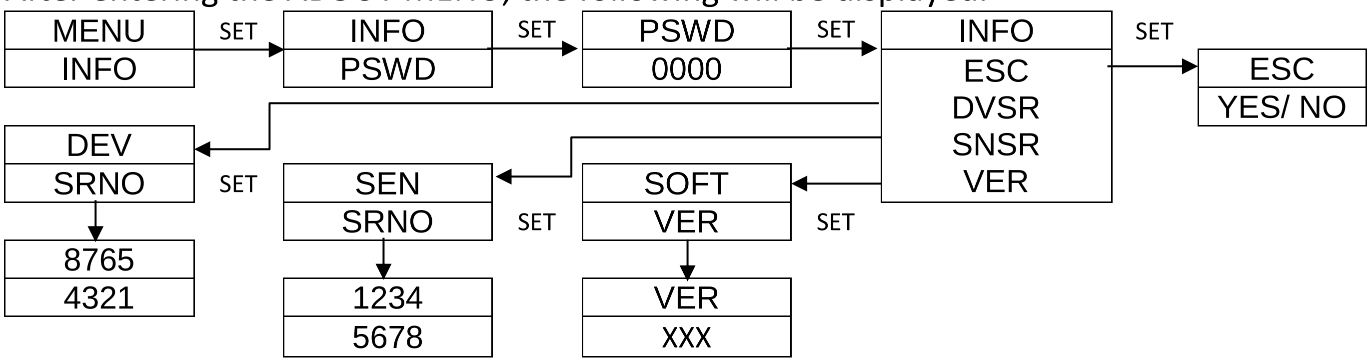


Table 11

DEVICE SERIAL NO	This menu shows Device Serial No stored in device memory. User cannot edit this number.
SENSOR SERIAL NO	This menu shows Sensor Serial No stored in sensor module memory. User cannot edit this number.
SOFTWARE VERSION	This menu shows Software Version number of device.

12 APPENDIX

12.1 ACRONYMS USED IN THIS MANUAL

NWM* – Normal Working Mode
GC* – Gas Concentration
SP* – Set point
PV*– Process Value
PV* & GC* are of same meaning
Operator Setting menu & User menu are of same meaning.
Device/ Detector/ Transmitter/ Instrument/ Unit are of same meaning.
Operator/ User are of same meaning

13 MODBUS ADDRESS DESCRIPTIONS

13.1 READ/ WRITE REGISTER DETAILS

Table 12

READ/ WRITE REGISTER DETAILS	
0X01	READ COIL REGISTER
0X03	READ HOLDING REGISTER
0X04	READ MULTIPLE INPUT REGISTER
0X05	WRITE COIL REGISTER
0X06	WRITE SINGLE REGISTER
0X16	WRITE MULTIPLE REGISTER

13.2 MODBUS ADDRESS

Table 13

HOLDING REGISTERS			
SR.NO	**PARAMETERS**	**ADDRESS**	**DESCRIPTION**
1	Gas Concentration value	40000	Holds the Gas Concentration value.
2	Use AL1	40001	Holds alarm setting condition. 01=Alarm 1 Enabled 00=Alarm 1 Disabled
3	AL1 Set Point	40002	Holds the set point value for Alarm 1. Can be set from 0 to range of instrument
4	AL1 Hysteresis	40003	Holds the hysteresis value for Alarm 1. Can be set up to 10% of range selected.

5	AL1 Logic	40004	Holds the logic value for alarm 1. 01=Alarm 1 Logic High. 00=Alarm 1 Logic Low.
6	AL1 Time Delay	40005	Holds the delay time for alarm 1. can be set from 0 to 999 sec
7	Use AL2	40006	Holds alarm enable condition. 01=Alarm 2 Enabled 00=Alarm 2 Disabled
8	AL2 Set Point	40007	Holds the set point value for Alarm 2. Can be set from 0 to range of instrument.
9	AL2 Hysteresis	40008	Holds the hysteresis value for Alarm 2. Can be set up to 10% of range selected.
10	AL2 Logic	40009	Holds the logic value for alarm 2. 01=Alarm 1 logic High. 00=Alarm logic Low.
11	AL2 Time Delay	40010	Holds the delay time for Alarm 2. Can be set from 0 to 999 sec
12	STEL ENABLE / DISABLE	40011	Holds the STEL setting condition. 01=STEL enabled. 00=STEL Disabled
13	STEL Set Point	40012	Holds the set point value for STEL. Can be set from 1 to range of the instrument
14	TWA ENABLE / DISABLE	40013	Holds the TWA setting condition. 01=TWA enabled. 00=TWA Disabled
15	TWA Set Point	40014	Holds the set point value for TWA. Can be set from 1 to range of the instrument.
16	BUZZER ENABLE / DISABLE	40015	Holds the Buzzer condition for Alarms 01= Buzzer ON 00= Buzzer OFF
17	Buzzer On Key	40016	Holds the Buzzer on key condition for Alarms 01= Key Buzz ON 00= Key Buzz OFF
18	Hooter On/Off	40017	Holds the Hooter status 0 = Hooter OFF 1= Hooter ON
19	Flasher On / Off	40018	Holds the Flasher status 0 = Flasher OFF 1= Flasher ON
20	Snooze	40019	Holds the value for snooze. Can be set Min: 000 Max : 999

21	-	40020	-
22	Relay1 Latch ENB / DIS	40021	Holds the Relay1 Latch setting condition. 01= Relay1 enabled. 00= Relay1 Disabled
23	-	40022	-
24	Relay2 Latch ENB / DIS	40023	Holds the Relay2 Latch setting condition. 01= Relay2 enabled. 00= Relay2 Disabled
25	FAIL safe Relay On/ Off	40024	Holds the Relay3 status 0 = Normally OFF 1= Normally ON
26	-	40025	-
27	-	40026	-
28	-	40027	-
29	Device ID	40028	Here Device ID can be set from 1 to 250
30	Baud rate	40029	Holds the Baud Rate setting 0 = 9.6 kbps 1 = 19.2 kbps 2 = 38.4 kbps 3 = 57.6 kbps
31	Parity	40030	Holds the Parity Bit setting 0 = None 1 = Odd 2 = Even
32	Stop bit	40031	Holds the stop bit setting 0=one 1= two
For Current Output			
33	Output High Range	40032	Holds the Higher set value for 4-20 mA output.
34	Output Low Range	40033	Holds the Lower set value for 4-20 mA output.
35	Counts Adj ENB / DIS	40034	This setting allows you to adjust the counts for current calibration points. 0=Disable 1=Enable
36	1mA counts Set	40035	Holds the value for 1mA. Default value is 0 adjust the counts if needed.
37	3.7mA counts Set	40036	Holds the value for 3.7mA. Default value is 0 adjust the counts if needed.
38	4mA counts Set	40037	Holds the value for 4mA. Default value is 0 adjust the counts if needed.
39	8mA counts Set	40038	Holds the value for 8mA. Default value is 0 adjust the counts if needed.

40	12mA counts Set	40039	Holds the value for 12mA. Default value is 0 adjust the counts if needed.
41	16mA counts Set	40040	Holds the value for 16mA. Default value is 0 adjust the counts if needed.
42	20mA counts Set	40041	Holds the value for 20mA. Default value is 0 adjust the counts if needed.
43	22mA counts Set	40042	Holds the value for 22mA. Default value is 0 adjust the counts if needed.
44	Scale select	40043	Holds the value for Scale Current 0=down 1=up
45	Inhibit select *** Note: Refer last point of important notes for more details**	40044	Holds the value for Inhibit Current a. For O2/N2 gas type 0=3.8mA 1=17.4mA /7.34mA/ 16.656mA b. For other gas types 0=2mA 1=3.8mA 2=4mA
For Voltage Output (RANGE : 0 TO 10V / 0 TO 5V / 0 TO 1V)			
46	0 VOLT/ 0 mV Set	40035	Holds the value for 0V/0mV. Default value is 0 **Note:** Counts can be adjusted by adding or subtracting from existing value
47	2 VOLT/ 1 VOLT/ 200 mV Set	40036	Holds the value for 2V/1V/200mV. Default value is 0
48	4 VOLT/ 2 VOLT/ 400 mV Set	40037	Holds the value for 4V/2V/400mV. Default value is 0
49	6 VOLT/ 3 VOLT/ 600 mV Set	40038	Holds the value for 6V/3V/600mV. Default value is 0
50	8 VOLT/ 4 VOLT/ 800 mV Set	40039	Holds the value for 8V/4V/800mV. Default value is 0
51	10 VOLT/ 5 VOLT/ 1000 mV Set	40040	Holds the value for 10V/5V/1000mV. Default value is 0
52	-	40041	-
53	-	40042	-
54	Scale select	40043	Holds the value for Scale Voltage 0=down 1=up
55	Inhibit select *** Note: Refer last point of important notes for more details**	40044	Holds the value for Inhibit Voltage a. For O2/N2 gas type 0= 0V 1= 8.36V / 2.09V / 7.91V & 1.045V/ 3.955V & 209mV/ 791mV b. For other gas types 0= 0V

56	-	40045	-
57	-	40046	-
58	-	40047	-
59	-	40048	-
60	Backlight on / off	40049	Holds the Backlight status 0 = Backlight OFF 1= Backlight ON
61	Flashing on / off	40050	Holds the flashing status 0 = FLASHING OFF 1= FLASHING ON
62	Range lock ENB/DIS	40051	Holds the value for range lock 0= NO 1= YES
63	Password	40052	Holds the value for password Eg. 0000 For setting a different password just enter new value Eg. 1234
64	OFFSET	40053	Holds the Offset value for Alarm 1 & 2. Can be set up to 25% of range selected.
CALIBRATION POINTS			
1	Calibration Mode Enter	40080	1= Enables Calibration Mode 0= Disable Calibration Mode
2	Low Span Enter	40081	1= Enters Low Span Setting.
3	Zero/Low Span Calibration Start	40082	1= Starts Zero/Low Span Calibration
4	Zero/Low Span Calibration Success	40083	1= Succeeds Zero/Low Span Calibration
5	Span/High Span Calibration Enter	40084	1= Enter Span/High Calibration Setting.
6	High Span/Span Range Enter	40085	Enter High Span/Span Range Value as required
7	High Span/Span Calibration Start	40086	1= Starts High Span/Span Calibration
8	High Span/Span Calibration Success	40087	1= Succeeds High Span/Span Calibration
9	Temperature CAL Value	40088	Shows Temperature of the sensor in real time. Applicable only for O_2 & toxic sensors if assembled.
10	Gas Concentration Value	40089	Shows Gas Concentration value in real time.

INPUT REGISTERS			
1	Gas Concentration value	30000	Holds the Gas Concentration value.
2	Channel Status	30001	Refer to Channel Status table 17 & 18.
3	Sensor Status	30002	Refer to Sensor Status table 19.
4	Decimal point	30003	Holds the decimal point. 0 = NO RESOLUTION 1 = 0.1 RESOLUTION (1DP) 2 = 0.01 RESOLUTION (2DP)
5	MIN Value	30004	Holds the Minimum value of gas concentration
6	MAX Value	30005	Holds the Maximum value of gas concentration
7	Sensor Name 1st Part	30006	Holds the 1st half of sensor name
8	Sensor Name 2nd Part	30007	Holds the 2nd half of sensor name
9	Calibration Date	30008	Holds the value for Calibration Date
10	Sensor Life Date	30009	Holds the value for Sensor Life Date
11	Serial No 1st Part	30010	Holds the HEX value Serial No 1st Part
12	Serial No 2nd Part	30011	Holds the HEX value Serial No 2nd Part
13	Serial No 3rd Part	30012	Holds the HEX value Serial No 3rd Part
14	Serial No 4th Part	30013	Holds the HEX value Serial No 4th Part
15	Sensor Serial No 1st Part	30014	Holds the HEX value Sensor Serial No 1st Part
16	Sensor Serial No 2nd Part	30015	Holds the HEX value Sensor Serial No 2nd Part
17	Sensor Serial No 3rd Part	30016	Holds the HEX value Sensor Serial No 3rd Part
18	Sensor Serial No 4th Part	30017	Holds the HEX value Sensor Serial No 4th Part
19	Software Version	30018	Holds the value for Software Version value
COIL REGISTERS			
1	Alarm + Ack	10000	This address is used to Acknowledge Alarm condition.
3	MIN Value clear	10001	This address is used to reset Minimum value
4	MAX Value clear	10002	This address is used to reset Maximum value

DECODING CALIBRATION AND LIFE DATE

To decode calibration date at address 30008

Example: unsigned int value at above address: 13931

Table 14

Parameter	Calculation	Value	Final date
Year	13931 / 520	26.79	2026 (26)
Month	13931 mod 520 / 40	10.275	Oct (10)
Date	13931 mod 520 mod 40	11	11 (11)

So the calibration date is 11 Oct 2026.

Same calculation for Sensor Life Date.

DECODING SERIAL NO

Example: serial number set in device is : 66231022

Convert hex values present at following addressed using hex to ASCII table given below

Table 15

Address	30010	30011	30012	30013
HEX Value	3636H	3233H	3130H	3232H
ASCII Value	66	23	10	22

Same calculation for Sensor Name, Sensor Serial no

HEX to ASCII table

Table 16

DEC	HEX	CHAR	DEC	HEX	CHAR	DEC	HEX	CHAR	DEC	HEX	CHAR
32	20	SPACE	48	30	0	64	40	@	80	50	P
33	21	!	49	31	1	65	41	A	81	51	Q
34	22	"	50	32	2	66	42	B	82	52	R
35	23	#	51	33	3	67	43	C	83	53	S
36	24	$	52	34	4	68	44	D	84	54	T
37	25	%	53	35	5	69	45	E	85	55	U
38	26	&	54	36	6	70	46	F	86	56	V
39	27	'	55	37	7	71	47	G	87	57	W
40	28	(	56	38	8	72	48	H	88	58	X
41	29	)	57	39	9	73	49	I	89	59	Y
42	2A	*	58	3A	:	74	4A	J	90	5A	Z
43	2B	+	59	3B	;	75	4B	K			
44	2C	,	60	3C	<	76	4C	L			
45	2D	_	61	3D	=	77	4D	M			
46	2E	-	62	3E	>	78	4E	N			
47	2F	/	63	3F	?	79	4F	O			

Table 17

CHANNEL STATUS				
SR.NO	**DETECTOR STATUS**	**INPUT REGISTER ADDRESS**	**HEX VALUE**	**BINARY VALUE**
1	NO FAULT / NO ALARM	30001	0x00	00000000
2	SENSOR OPEN INDICATION	30001	0x01	00000001
3	OVER RANGE INDICATION	30001	0x02	00000010
4	WARM UP CONDITION	30001	0x04	00000100
5	CURRENT LOOP OPEN	30001	0x100	100000000
6	INHIBIT MODE	30001	0x200	1000000000

Table 18

CHANNEL STATUS FOR ALARM						
SR.NO	**CONDITIONS**	**LATCH**	**LOGIC**	**DEC VALUE**	**HEX VALUE**	**BINARY VALUE**
1.	AL1 Triggered, AL2(Disabled)/ (OFF STATE)	No & Yes	High	2056	0x808	100000001000
2.	AL1 Ack, AL2(Disabled)/ (OFF STATE)	Yes	High	10248	0x2808	10100000001000
3.	AL1 Ack, AL2 Triggered	Yes	High	14376	0x3828	0011100000101000
4.	AL2 Triggered, AL1(Disabled) / (OFF STATE)	No & Yes	High	4128	0x1020	1000000100000
5.	AL2 Ack, AL1(Disabled)/ (OFF STATE)	Yes	High	20512	0x5020	101000000100000
6.	AL2 Ack, AL1 Triggered	Yes	High	22568	0x5828	101100000101000
7.	AL1 Triggered, AL2(Disabled) / (OFF STATE)	No & Yes	Low	2064	0x810	100000010000
8.	AL1 Ack, AL2(Disabled) / (OFF STATE)	Yes	Low	10256	0x2810	10100000010000
9.	AL1 Ack, AL2 Triggered	Yes	Low	14416	0x3850	11100001010000
10.	AL2 Triggered, AL1(Disabled)/ (OFF STATE)	No & Yes	Low	4160	0x1040	1000001000000

11.	AL2 Ack, AL1(Disabled) / (OFF STATE)	Yes	Low	20544	0x5040	0101000001000000
12.	AL2 Ack, AL1 Triggered	Yes	Low	22608	0x5850	101100001010000
13.	AL1 Triggered and Ack, Al1 (OFF STATE)	Yes	High/ low	8192	0x2000	10000000000000
14.	AL2 Triggered and Ack, AL2 (OFF STATE)	Yes	High/ low	16384	0x4000	100000000000000
15.	AL1 Triggered, AL2 Triggered	No & Yes	High	6184	0x1828	1100000101000
16.	AL1 Triggered, AL2 Triggered and Ack	Yes	High	30760	0x7828	111100000101000
17.	AL1 Triggered, AL2 Triggered and Acked (OFF STATE)	Yes	High/ low	24576	0x6000	110000000000000
18.	AL1 Triggered, AL2 Triggered	No & Yes	Low	6224	0x1850	1100001010000
19.	AL1 Triggered, AL2 Triggered and Ack	Yes	Low	30800	0x7850	111100001010000

2nd Iteration Of Alarm Triggering (Latch: Yes only)

SR.NO	CONDITIONS	Latch	Logic	DEC VALUE	HEX VALUE	BINARY VALUE
1.	AL1 Triggered, AL2 (OFF STATE)	Yes	High	18440	0x4808	100100000001000
2.	AL1 ACK, AL2 (OFF STATE)	Yes	High	26632	0x6808	110100000001000
3.	AL2 Triggered, AL1 (OFF STATE)	Yes	High	12320	0x3020	11000000100000
4.	AL2 ACK, AL1 (OFF STATE)	Yes	High	28704	0x7020	111000000100000
5.	AL1 Triggered, AL2 (OFF STATE)	Yes	Low	18448	0x4810	100100000010000
6.	AL1 ACK, AL2 (OFF STATE)	Yes	Low	26640	0x6810	110100000010000
7.	AL2 Triggered, AL1 (OFF STATE)	Yes	Low	12352	0x3040	11000001000000
8.	AL2 ACK, AL1 (OFF STATE)	Yes	Low	28736	0x7040	111000001000000

Note: "OFF STATE" means alarm setting is enabled but not triggered or it was triggered and brought to off state by lowering gas value

Table 19

SENSOR STATUS				
SR.NO	DETECTOR STATUS	INPUT REGISTER ADDRESS	DEC VALUE	BINARY VALUE
1	CALIBRATION SUCCESS	30002	0	00000000
2	CAL DUE	30002	1	00000001
3	CALIBRATION FAIL	30002	2	00000010
4	PASS & SENSOR REPLACE	30002	4	00000100
5	SENSOR REPLACE WITH LIFE DAYS ARE OVER	30002	8	00001000

13.3 DECIMAL POINT VALUE DESCRPITION

In case of unit as %V/V, %LEL, PPM - %V/V & PPM - %LEL, the actual value of reading or writing for process value, AL1_set pt., AL1_Hyst, AL2_set pt., AL2 Hyst , Min/Max value, output low, output high & Range of unit will depend on the decimal point set in the instrument.

For reading:

If value coming from instrument is ‘x’ then,

Actual value to be read = $\dfrac{\text{‘x’}}{\text{DP Factor}}$

For writing:

If you want to send ‘y’ value to the instrument then,

Please refer following table for DP factor.

= y X DP Factor

Table 20

Sr .No	DECIMAL POINT	DP FACTOR
1.	Two - DP (DP=2)	100
2.	One - DP (DP=1)	10
3.	No - DP (DP=0)	1

13.4 FAULT INDICATION VALUE

Table 21

SR. NO.	FAULT INDICATION	HEX VALUE	DECIMAL VALUE
1.	‘SENSOR OPEN’ INDICATION	7FFF	32767
2.	‘OVER RANGE’ INDICATION	7FFF	32767

13.5 FAULT CONDITIONS

Table 22

SR. NO.	SYMPTOMS	PROBLEMS	SOLUTION
1.	Instrument doesn't turn 'ON'	1) Instrument has failed. 2) Connection Problem.	Check Power Supply connection & make sure that connection are proper as per the connection diagram shown in Operating Manual. OR Contact the factory.
2.	Display shows 'OVER RANGE' / 'SENSOR OPEN' RANGE' & Current Output (1mA, 3.7mA, 21mA, 22mA) / Voltage Output (0V, 1V, 5V, 10V).	1) The gas concentration is more or less than the Selected range. 2) The sensor module is not properly connected. 3) Calibration Problem	Check the Sensor Module is properly interfaced to the unit as per the connection diagram shown in operating manual. OR Contact the factory.
3.	4-20mA / 0-10V / 0-5V / 0-1V Output is not proper.	1) Check the output range 2) Electronic module has Failed. 3) 4-20mA / 0-10V / 0-5V / 0-1V Output setting	1) Check O/P range and current O/P 2) Contact the factory.
4.	Magnetic keys not Working.	1) Magnetic key/ wand problems 2) Switch is damaged.	1) Contact the factory.

13.6 IMPORTANT NOTES

When 'Gas Concentration' crosses the Set Point limits for the Relays during the process, it is shown by the activation of the Relays with their respective LED indications i.e. 'AL1' & 'AL2'.

a) i) If Latch is given for the Relays, then Relays will be activated & latched when 'Gas Concentration' crosses the Set Point limits for the Relays during the process & LED indications for the Relays will also be activated. Relays will stay in that condition even if the 'Gas Concentration 'comes below the Set Point. This condition of the Relays is called as 'LATCH'. Latched Relays will be unlatched or acknowledged by pressing the '

' key. Latching is not applicable to LED indications. LED indications for the Relays gets activated as per the 'Gas Concentration'. When the 'Gas Concentration' crosses the Set Point limits for the Relays, it will get activated & will automatically deactivate when the 'Gas Concentration' comes within the Set Point limits.

The relay and LED indications when alarm is in latch condition are as follow:

- If Process Value (PV) crosses set point and not acknowledged: Relay 'ON', LED 'Blinks'.
- If PV crosses set point and acknowledge: Relay 'OFF', 'LED ON'.
- If PV is within set point and not acknowledge: Relay 'ON', 'LED 'Blinks'.

- If PV is within set point and acknowledge: Relay 'OFF', 'LED OFF'.

a)ii) If Latch is not given for the Relays then the Relays & their LED indications will get activated when 'Gas Concentration' crosses the Set Point limits & will get automatically deactivated when 'Gas Concentration' comes within the Set Point limits for the Relays.

a)iii) For "latch: disable, Acknowledgement: enable setting", When the Relays & their LED indications will get activated when 'Gas Concentration' crosses the Set Point limits. At that condition if you press Acknowledgement key then only Relays will turn off. If you don't press Acknowledgement key then the Relays & their LED indications will turn off automatically when gas concentration comes below set points.

The relay and LED indications when alarm is in Non-Latch condition are as follow.

- If Process Value (PV) crosses set point: Relay 'ON', LED Blinks:
- If PV is within set point: Relay 'OFF', 'LED OFF'.

b) While ordering the 'GT-2500' Specify the Logic (HIGH / LOW) for the Set Points for the Relays, Latch / Unlatch condition for the Relays.

c) In 'Operator Setting Mode' if no key is pressed within two minute then the Unit will automatically return to the 'Normal Working Mode' except Calibration & Bump test mode time out 10 minutes & 20 minutes respectively.

d) To set or select any parameter or Go to the next parameter with or without doing any setting in any Programming mode, Press " 

" key.

e) If device enters any of the operator setting mode or calibration mode & bump test mode, if password is correct or when device is in start-up routine, then current output will be set current in Inhibit mode.

f) If the gas concentration crossed gas range as per display will show 'OVER RANGE' and current output will get as Up Scale / Down Scale

g) System response for 5 seconds at the time of system check.

  g.1) On display "SYSTEM CHECK" will be flashing.

  g.2) All L.E.D. will be ON.

  g.3) All Relays will be ON.

  g.4) The Current / Voltage output will be 12mA / 5V/ 2.5V/ 500mV.

h) Do not use 'Test Mode' / 'System Check' when the unit is completely installed on the site. For the operation take permission from your senior authority; otherwise it may be very dangerous.

i) Relation between inhibit Current/ Voltage & Gas value for Oxygen & Nitrogen sensor module

Table 23

For 4-20mA output current range		
GAS WITH RANGE	**INHIBIT CURRENT**	**GAS VALUE FOR INHIBIT CURRENT**
O2 (25 %V/V)	17.4mA	20.90 %V/V
O2 (100 %V/V)	7.34mA	20.90 %V/V
N2 (100 %V/V)	16.656mA	79.10 %V/V
For 0-10 output voltage range		
GAS WITH RANGE	**INHIBIT VOLTAGE**	**GAS VALUE FOR INHIBIT CURRENT**
O2 (25 %V/V)	8.36V	20.90 %V/V
O2 (100 %V/V)	2.09V	20.90 %V/V
N2 (100 %V/V)	7.91V	79.10 %V/V
For 0-5 output voltage range		
GAS WITH RANGE	**INHIBIT VOLTAGE**	**GAS VALUE FOR INHIBIT CURRENT**
O2 (25 %V/V)	4.18V	20.90 %V/V
O2 (100 %V/V)	1.045V	20.90 %V/V
N2 (100 %V/V)	3.955V	79.10 %V/V
For 0-1 output voltage range		
GAS WITH RANGE	**INHIBIT VOLTAGE**	**GAS VALUE FOR INHIBIT CURRENT**
O2 (25 %V/V)	836mV	20.90 %V/V
O2 (100 %V/V)	209mV	20.90 %V/V
N2 (100 %V/V)	791mV	79.10 %V/V

14 ORDERING INFORMATION

QSEN SERIES GT-2500 ORDERING INFORMATION

Table 24

1) Qsen Series-GT-2500-0-0

ENCLOSURE
QsenL (JB-FLP-IIC, Flameproof)
QsenS (JB-90, Flameproof)
QsenX (JB-150, 24VDC Input, Flameproof)
QsenXi (JB-150, 230VAC Input, Flameproof)
QsenP (WP-P, Weatherproof)
QsenO (WP, Weatherproof)

MODEL NO.

RS-485
0) NO
1) YES

HART
0) NO
1) YES

SENSOR MODULE ORDERING INFORMATION

Table 25

2) SSM-FLP-O2-EC-08-03

SENSMART SWAPPABLE MODULE, FLAMEPROOF

GAS NAME

SENSOR TECHNOLOGY

GAS RANGE

GAS UNIT

NOTE:

- Before ordering please refer the above ordering information or contact below address for assistance

15 REVISION HISTORY

Table 26

Sr. No	VERSION NO	REVISION	EFFECTIVE DATE	REMARK
1.	V2.00 JUL 2021	R0	07/07/2021	NEW RELEASE
2.	V2.00 JUL 2021	R1	10/07/2021	SOFTWARE UPDATE
3.	V2.04 JUL 2021	R2	16/08/2021	HARDWARE & SOFTWARE UPDATE
4.	V2.00 OCT 2021	R4	09/10/2021	SOFTWARE UPDATE
5.	V2.03 DEC 2021	R5	24/12/2021	SOFTWARE UPDATE
6.	V2.03 DEC 2021	R6	24/12/2021	CONTROLLER CHANGED
7.	V2.04 NOV 2022 V3.01 NOV 2022	R7	18/11/2022	1. MODBUS ADDRESS (40000 & 40053)
8.	V2.04 NOV 2022 V3.01 NOV 2022	R8	20/01/2023	CODE MENU NOTE UPDATE
9.	V2.05 FEB 2023 V3.02 FEB 2023	R9	09/03/2023	1) SOFTWARE UPDATE 2) NEW FEATUERS ADDED
10.	V2.06 APR 2023 V3.03 APR 2023	R10	02/06/2023	ENCLOSURE UPDATE FLP-JB-90 (Flameproof) & WP(Weatherproof) PID SOFTWARE UPDATE
11.	V2.07 AUG 2023 V3.04 AUG 2023	R11	18/08/2023	SOFTWARE UPDATE
12.	V2.08 SEP 2023 V3.05 NOV 2023	R12	26/12/2023	GT-2500-WP DETAILS UPDATED
13.	V2.08 SEP 2023 V3.05 NOV 2023	R13	29/05/2024	MANUAL UPDATED
14.	V2.09 JUL 2024 V3.06 JUL 2024	R14	27/07/2024	CO2 SENSOR SOFTWARE UPDATE

16 MISCELLANEOUS

Table 27

SR. NO	VERSION NO	REVISION	EFFECTIVE DATE	TOTAL TABLE NO.	TOTAL FIGURE NO.
1.	V2.00 JUL 2021	R0	07/07/2021	22	10
2.	V2.00 JUL 2021	R1	10/07/2021	22	10
3.	V2.04 JUL 2021	R2	16/08/2021	21	10
4.	V2.00 OCT 2021	R4	09/10/2021	21	10
5.	V2.03 DEC 2021	R5	24/12/2021	21	10
6.	V2.03 DEC 2021 V3.00 MAY 2022	R6	24/12/2021 27/09/2022	21	10
7.	V2.04 NOV 2022 V3.01 NOV 2022	R7	18/11/2022	21	20
8.	V2.04 NOV 2022 V3.01 NOV 2022	R8	20/01/2023	21	20
9.	V2.05 FEB 2023 V3.02 FEB 2023	R9	09/03/2023	25	10
10.	V2.06 APR 2023 V3.03 APR 2023	R10	02/06/2023	25	26
11.	V2.07 AUG 2023 V3.04 AUG 2023	R11	18/08/2023	25	26
12.	V2.08 SEP 2023 V3.05 NOV 2023	R12	26/12/2023	27	33
13.	V2.08 SEP 2023 V3.05 NOV 2023	R13	29/05/2024	27	33
14.	V2.09 JUL 2024 V3.06 JUL 2024	R14	27/07/2024	27	33

THE END

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